Analytic Combinatorics and Large Language Models

An Expository Introduction

RACA Notes

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Preface

This book is an extended expository introduction to the research territory where analytic combinatorics meets large language models. It is written for a student who knows calculus, linear algebra, and a little probability, has perhaps seen complex numbers, and wants to understand both halves of the subject from scratch: the symbolic method and singularity analysis of Flajolet and Sedgewick on one side, and the probabilistic, formal-language, and statistical-mechanics views of neural language models on the other.

The organizing claim of the book is that these two traditions, despite rarely citing each other, are talking about the same mathematical objects: partition functions are generating functions, critical temperatures are radii of convergence, phase transitions are singularities, rational-series theorems from automata are dominant-pole theorems from analytic combinatorics. The research frontier consists largely of building bridges that the two communities have, until recently, avoided building. The final chapters make those bridges explicit and give a student a plausible starting point.

Part I develops the classical machinery of analytic combinatorics from zero, up to the $n^{-3/2}\rho^{-n}$ asymptotic for unambiguous context-free languages. Part II turns to weighted formal languages, probabilistic grammars, tightness, and the entropy-rate identity $h = \log(1/R)$. Part III introduces autoregressive language models and the formal-language theory of transformers, then surveys the weighted-finite-automaton approximation pipeline. Part IV is devoted to Boltzmann sampling, Gibbs-measure views of temperature decoding, and phase transitions. Part V covers Zipf's law, motif statistics, and the ten open problems we currently see at the frontier.

Exercises appear in selected chapters but are not the point. The point is to get a reader who starts knowing none of this to the point where they can open an arXiv paper from 2025 on, say, phase transitions in Pythia or WFA extraction from a Mistral model, and see what is going on.

Part I

Analytic Combinatorics from Scratch

Chapter 1

What is a Generating Function?

A sequence of numbers $(a_n)_{n\geq 0}$ is the natural output of a counting problem: how many binary strings of length n, how many trees with n nodes, how many ways to parse a sentence of length n according to a given grammar. Studying the sequence directly forces us to deal with each index separately, and recurrences — while precise — often resist closed-form extraction. The central idea of generating function theory is to encode the entire sequence as a single algebraic object,

$$A(z) \;=\; \sum_{n\geq 0} a_n z^n,$$

and then let the algebra of formal power series do the bookkeeping. This is not mere notational convenience. Operations on sequences that would otherwise require index gymnastics — shifting, convolving, differencing — correspond to clean algebraic operations on $A(z)$. Recurrences, which are linear or polynomial relations among the a_n, become *functional equations* for $A(z)$, and those equations can often be solved in closed form. The solution then exposes structure invisible in the raw sequence.

Definition 1.1. The set of *formal power series* over $\mathbb{Q}$, denoted $\mathbb{Q}[[z]]$, consists of all expressions $A(z) = \sum_{n\geq 0} a_n z^n$ with $a_n \in \mathbb{Q}$. Two series are added term by term: $(A+B)(z) = \sum_n (a_n + b_n) z^n$. Their product is defined by

$$(A \cdot B)(z) \;=\; \sum_{n\geq 0} c_n z^n, \qquad c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

The product formula follows from collecting all pairs of terms $a_k z^k$ and $b_m z^m$ whose exponents add to n: $z^k \cdot z^m = z^n$ requires $m = n - k$, so the coefficient of z^n in the product is $a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0$.

At this stage z is a *formal indeterminate* — it carries no numerical value, and questions of convergence are entirely irrelevant. The set $\mathbb{Q}[[z]]$ with these operations behaves like the rational numbers do under addition and multiplication: one can add, subtract, and multiply freely. (Algebraists call $\mathbb{Q}[[z]]$ a *commutative ring*; this terminology is not needed for what follows.)

Definition 1.2. The *coefficient extraction operator* $[z^n]$ picks out the n-th coefficient of a power series:

$$[z^n]A(z) \;:=\; a_n.$$

It is linear ($[z^n](A+B) = [z^n]A + [z^n]B$) and respects products via the convolution formula $[z^n](A \cdot B) = \sum_{k=0}^{n} a_k b_{n-k}$.

One further algebraic fact is useful: any series $A(z)$ with $a_0 \neq 0$ has a multiplicative inverse $A(z)^{-1}$ in $\mathbb{Q}[[z]]$, constructed coefficient by coefficient. The idea is to find a series $D(z) = \sum d_n z^n$ satisfying $A(z) \cdot D(z) = 1$. The constant term gives $a_0 d_0 = 1$, so $d_0 = 1/a_0$. The z^1 equation gives $a_0 d_1 + a_1 d_0 = 0$, so $d_1 = -a_1 d_0/a_0$. In general, $d_n = -(a_1 d_{n-1} + \cdots + a_n d_0)/a_0$; each coefficient is determined by the preceding ones. When $a_0 = 0$, division by a_0 fails and no inverse exists.

Two operations on power series appear repeatedly. The *shift* by z sends $A(z)$ to $zA(z) = \sum_{n\geq 1} a_{n-1} z^n$, so $[z^n](zA(z)) = a_{n-1}$. The *formal derivative* is $A'(z) = \sum_{n\geq 1} n a_n z^{n-1}$, defined term by term with no analytic content; it satisfies the product rule $(f \cdot g)' = f' \cdot g + f \cdot g'$ by direct verification on the coefficient formulas.

The method: from recurrences to functional equations

The central technique of this chapter is a three-step recipe for converting a recurrence into a closed-form generating function.

1. **Define** the generating function $A(z) = \sum_{n\geq 0} a_n z^n$.

2. **Multiply** every term of the recurrence $a_n =$ (expression in $a_{n-1}, a_{n-2}, \ldots$) by z^n and **sum** over all valid n. Recognize each resulting sum as an expression involving $A(z)$.

3. **Solve** the resulting equation for $A(z)$.

Step 2 is the key move. Why does it work? Because every occurrence of a_{n-k} in the recurrence, after multiplication by z^n and summation, becomes a shifted version of $A(z)$. For instance, $\sum_n a_{n-1} z^n = zA(z)$, and $\sum_n a_{n-2} z^n = z^2 A(z)$, and in general multiplying by z^k in front of $A(z)$ shifts the index by k. The recurrence, which was an infinite family of equations (one for each n), collapses into a single equation for the single unknown $A(z)$.

We now demonstrate this on three progressively harder examples.

Binary strings

Let b_n denote the number of binary strings of length n. Every such string is a sequence of n bits, each chosen from $\{0, 1\}$, so $b_n = 2^n$. We can derive this from the recurrence and simultaneously demonstrate the recipe. Any binary string of length $n \geq 1$ is obtained by appending one of two symbols to a string of length $n - 1$, giving $b_n = 2b_{n-1}$ for $n \geq 1$, with $b_0 = 1$.

Following the recipe: define $B(z) = \sum_{n\geq 0} b_n z^n$, multiply both sides of the recurrence by z^n, and sum over $n \geq 1$:

$$\sum_{n\geq 1} b_n z^n \;=\; 2z \sum_{n\geq 1} b_{n-1} z^{n-1}.$$

The left side is $B(z) - b_0 = B(z) - 1$. For the right side, substitute $m = n - 1$ in the sum: $\sum_{n\geq 1} b_{n-1} z^{n-1} = \sum_{m\geq 0} b_m z^m = B(z)$, so the right side is $2zB(z)$. Thus

$$B(z) - 1 \;=\; 2zB(z), \qquad \text{so} \qquad B(z) \;=\; \frac{1}{1-2z}.$$

This could equally be read as the geometric series $\sum_{n\geq 0} (2z)^n = \sum_{n\geq 0} 2^n z^n$, confirming $[z^n]B(z) = 2^n$. The functional equation route is more instructive: it works even when no closed-form formula for b_n is obvious beforehand.

Fibonacci numbers

The Fibonacci sequence is defined by $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$. Setting $F(z) = \sum_{n\geq 0} F_n z^n$ and converting the recurrence: multiply $F_n = F_{n-1} + F_{n-2}$ by z^n and sum over $n \geq 2$,

$$\sum_{n\geq 2} F_n z^n \;=\; z \sum_{n\geq 2} F_{n-1} z^{n-1} \;+\; z^2 \sum_{n\geq 2} F_{n-2} z^{n-2}.$$

The left side is $F(z) - F_1 z - F_0 = F(z) - z$. The two sums on the right are z times $\sum_{m\geq 1} F_m z^m = F(z) - F_0 = F(z)$ and z^2 times $\sum_{m\geq 0} F_m z^m = F(z)$, respectively. Collecting terms,

$$F(z) - z \;=\; zF(z) + z^2 F(z),$$

which gives immediately

$$F(z) \;=\; \frac{z}{1 - z - z^2}.$$

To extract F_n explicitly, we factor the denominator and use partial fractions. Define the *golden ratio* $\varphi = (1 + \sqrt{5})/2 \approx 1.618$ and its conjugate $\hat{\varphi} = (1 - \sqrt{5})/2 \approx -0.618$. These are the two roots of $x^2 - x - 1 = 0$ (check: $\varphi^2 = \varphi + 1$ and $\hat{\varphi}^2 = \hat{\varphi} + 1$, both by direct computation). We claim

$$1 - z - z^2 \;=\; (1 - \varphi z)(1 - \hat{\varphi} z).$$

To verify, expand the right side: $(1 - \varphi z)(1 - \hat{\varphi} z) = 1 - (\varphi + \hat{\varphi})z + \varphi\hat{\varphi}\, z^2$. Now $\varphi + \hat{\varphi} = 1$ and $\varphi\hat{\varphi} = (1 + \sqrt{5})(1 - \sqrt{5})/4 = (1 - 5)/4 = -1$, so the product is $1 - z - z^2$ as claimed. The roots of $1 - z - z^2 = 0$ are therefore $z = 1/\varphi \approx 0.618$ and $z = 1/\hat{\varphi} = -\varphi \approx -1.618$.

Now we decompose $F(z)$ into partial fractions. We seek constants A and B such that

$$F(z) \;=\; \frac{z}{(1 - \varphi z)(1 - \hat{\varphi} z)} \;=\; \frac{A}{1 - \varphi z} + \frac{B}{1 - \hat{\varphi} z}.$$

Multiply both sides by $(1 - \varphi z)(1 - \hat{\varphi} z)$:

$$z \;=\; A(1 - \hat{\varphi} z) + B(1 - \varphi z).$$

Comparing constant terms (z^0): $0 = A + B$, so $B = -A$. Comparing coefficients of z: $1 = -A\hat{\varphi} - B\varphi = -A\hat{\varphi} + A\varphi = A(\varphi - \hat{\varphi})$. Since $\varphi - \hat{\varphi} = \sqrt{5}$, we get $A = 1/\sqrt{5}$ and $B = -1/\sqrt{5}$. Therefore

$$F(z) \;=\; \frac{1}{\sqrt{5}} \cdot \frac{1}{1 - \varphi z} \;-\; \frac{1}{\sqrt{5}} \cdot \frac{1}{1 - \hat{\varphi} z}.$$

Each factor is a geometric series, exactly as in the binary-string example: $1/(1 - \varphi z) = \sum_{n\geq 0} \varphi^n z^n$ and $1/(1 - \hat{\varphi} z) = \sum_{n\geq 0} \hat{\varphi}^n z^n$. Reading off the coefficient of z^n gives Binet's formula,

$$F_n \;=\; \frac{\varphi^n - \hat{\varphi}^n}{\sqrt{5}}.$$

The key observation is that this closed form was *forced* by the location of the poles of $F(z)$. The dominant term as $n \to \infty$ comes from the pole closest to the origin, which is at $z = 1/\varphi$, yielding $F_n \sim \varphi^n/\sqrt{5}$. This is not a coincidence: singularities of generating functions control asymptotics, a theme developed fully in Chapters 2 and 4. The standard reference for the entire theory is [18].

Catalan numbers

The Catalan numbers C_n count full binary trees (every internal node has exactly two children) with n internal nodes; equivalently, they count triangulations of a convex $(n+2)$-gon, Dyck paths of length $2n$, and well-formed parenthesizations of length n, among many other combinatorial families. The sequence begins $1, 1, 2, 5, 14, 42, \ldots$ and satisfies the recurrence

$$C_n \;=\; \sum_{k=0}^{n-1} C_k\, C_{n-1-k}, \qquad C_0 = 1.$$

This recurrence reflects a structural decomposition: a binary tree with $n \geq 1$ internal nodes consists of a root together with a left subtree on k internal nodes and a right subtree on $n-1-k$ nodes, for some $0 \leq k \leq n-1$.

Following the same recipe as before: define $C(z) = \sum_{n\geq 0} C_n z^n$, multiply both sides of the recurrence by z^n, and sum over $n \geq 1$. The left side gives $\sum_{n\geq 1} C_n z^n = C(z) - C_0 = C(z) - 1$. For the right side, observe that $\sum_{k=0}^{n-1} C_k C_{n-1-k}$ is the convolution $[z^{n-1}](C(z)^2)$ (the coefficient of z^{n-1} in $C(z)^2$). So

$$\sum_{n\geq 1}\left(\sum_{k=0}^{n-1} C_k C_{n-1-k}\right) z^n \;=\; z \sum_{n\geq 1} [z^{n-1}](C(z)^2)\, z^{n-1} \;=\; z\, C(z)^2.$$

(The first equality factors $z^n = z \cdot z^{n-1}$ and pulls out the z. For the second, substitute $m = n-1$: the sum $\sum_{m\geq 0}[z^m](C^2) \cdot z^m$ simply reconstructs the series $C(z)^2$ from its coefficients — a tautology, since any power series *is* the sum of its terms.) Equating left and right sides:

$$C(z) - 1 \;=\; z\, C(z)^2, \qquad \text{i.e.,} \qquad C(z) \;=\; 1 + z\, C(z)^2.$$

This is our first *quadratic* functional equation. Rearrange it as $z\, C(z)^2 - C(z) + 1 = 0$, which is a quadratic equation in the unknown $C(z)$ with coefficients that happen to involve z. Applying the quadratic formula (with $a = z$, $b = -1$, $c = 1$):

$$C(z) \;=\; \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \cdot z \cdot 1}}{2z} \;=\; \frac{1 \pm \sqrt{1-4z}}{2z}.$$

The sign must be chosen so that $C(z)$ is a valid formal power series with $C_0 = 1$. To check: expand the numerator $1 + \sqrt{1-4z} = 1 + (1 - 2z - \cdots) = 2 - 2z - \cdots$, so $(1 + \sqrt{1-4z})/(2z) = (2 - 2z - \cdots)/(2z) = 1/z - 1 - \cdots$, which contains a z^{-1} term and is not a power series. The minus sign gives $1 - \sqrt{1-4z} = 1 - (1 - 2z - \cdots) = 2z + \cdots$, so $(1 - \sqrt{1-4z})/(2z) = (2z + \cdots)/(2z) = 1 + \cdots$, which begins with $C_0 = 1$ as required. We take the minus sign:

$$C(z) \;=\; \frac{1 - \sqrt{1-4z}}{2z}.$$

To extract coefficients, we need the *generalized binomial theorem.* For any real number α and non-negative integer n, define the *generalized binomial coefficient*

$$\binom{\alpha}{n} \;=\; \frac{\alpha(\alpha-1)(\alpha-2)\cdots(\alpha-n+1)}{n!},$$

a product of n terms in the numerator (an empty product for $n = 0$, giving $\binom{\alpha}{0} = 1$). When α is a positive integer this reduces to the usual binomial coefficient; when α is not a non-negative integer the product never terminates, and the resulting series $(1+w)^\alpha = \sum_{n\geq 0} \binom{\alpha}{n} w^n$ is infinite. This identity is valid as a formal power series (and converges for $|w| < 1$).

Apply this with $\alpha = 1/2$ and $w = -4z$:

$$\sqrt{1-4z} \;=\; (1-4z)^{1/2} \;=\; \sum_{n\geq 0} \binom{1/2}{n}(-4z)^n.$$

We need the explicit form of $\binom{1/2}{n}$ for $n \geq 1$. Writing out the falling factorial:

$$\binom{1/2}{n} = \frac{\frac{1}{2}(\frac{1}{2}-1)(\frac{1}{2}-2)\cdots(\frac{1}{2}-n+1)}{n!} = \frac{\frac{1}{2}\cdot(-\frac{1}{2})\cdot(-\frac{3}{2})\cdots(\frac{3-2n}{2})}{n!}.$$

The numerator has n factors, each with a 2 in the denominator, contributing $1/2^n$ overall. The remaining numerator (ignoring the 2^n) is $1\cdot(-1)\cdot(-3)\cdots(3-2n)$. The first factor is $+1$; the remaining $n-1$ factors are all negative, contributing a sign of $(-1)^{n-1}$. Their absolute values are $1, 1, 3, 5, \ldots, 2n-3$, whose product is the *double factorial* $(2n-3)!! = 1\cdot 3\cdot 5\cdots(2n-3)$. To convert this to ordinary factorials, note that $(2m)! = (1\cdot 3\cdot 5\cdots(2m-1))\cdot(2\cdot 4\cdot 6\cdots 2m) = (2m-1)!!\cdot 2^m m!$, so $(2m-1)!! = (2m)!/(2^m m!)$; setting $m = n-1$ gives $(2n-3)!! = (2n-2)!/(2^{n-1}(n-1)!)$. Assembling everything:

$$\binom{1/2}{n} = \frac{(-1)^{n-1}(2n-2)!}{2^n\cdot n!\cdot 2^{n-1}(n-1)!} = \frac{(-1)^{n-1}}{2^{2n-1}\cdot n}\binom{2n-2}{n-1} \quad \text{for } n \geq 1,$$

where the last step uses $\binom{2n-2}{n-1} = (2n-2)!/((n-1)!(n-1)!)$ and $n! = n\cdot(n-1)!$. (Verify for small n: $\binom{1/2}{1} = 1/2$, $\binom{1/2}{2} = -1/8$, $\binom{1/2}{3} = 1/16$.)

Now substitute into $C(z) = (1-\sqrt{1-4z})/(2z)$:

$$C(z) = \frac{1}{2z}\Big(1 - \sum_{n\geq 0}\binom{1/2}{n}(-4z)^n\Big) = \frac{-1}{2z}\sum_{n\geq 1}\binom{1/2}{n}(-4z)^n,$$

since the $n=0$ term of the sum is $\binom{1/2}{0}(-4z)^0 = 1$, which cancels the leading 1. Dividing by $2z$ shifts the power of z down by one, so for $n \geq 0$:

$$[z^n]C(z) = \frac{-1}{2}\binom{1/2}{n+1}(-4)^{n+1} = \frac{(-4)^{n+1}}{-2}\cdot\frac{(-1)^n}{2^{2n+1}(n+1)}\binom{2n}{n}.$$

The sign: $(-4)^{n+1} = (-1)^{n+1}\cdot 4^{n+1}$, and $(-1)^{n+1}\cdot(-1)^n = (-1)^{2n+1} = -1$, so the factor $(-4)^{n+1}\cdot(-1)^n/(-2) = (-1)\cdot 4^{n+1}/(-2) = 4^{n+1}/2$. The powers of 2: $4^{n+1}/2 = 2^{2n+2}/2 = 2^{2n+1}$, which cancels the 2^{2n+1} in the denominator. What remains is

$$[z^n]C(z) \;=\; \frac{1}{n+1}\binom{2n}{n},$$

the standard formula for the n-th Catalan number. The appearance of $\sqrt{1-4z}$ is significant: $C(z)$ is not a rational function but an *algebraic* one, satisfying a polynomial equation over $\mathbb{Q}(z)$. Algebraic generating functions have branch-point singularities rather than poles. The branch point of $C(z)$ at $z = 1/4$ is a square-root singularity, and from the local expansion $C(z) \sim 2 - 2\sqrt{1-4z}$ near $z = 1/4$, one can show (see Chapter 5) that

$$C_n \;\sim\; \frac{4^n}{n^{3/2}\sqrt{\pi}},$$

a formula whose $n^{-3/2}$ subexponential factor is a direct signature of the square-root branch point. This pattern is universal: any combinatorial class whose generating function is algebraic with a square-root singularity as its dominant singularity will exhibit $n^{-3/2}$ subexponential decay in its coefficients. Unambiguous context-free grammars always produce algebraic generating functions, and this has concrete implications for the statistical structure of natural language sentences — a connection we will revisit when discussing the relationship between pushdown automata and the token distributions produced by language models.

Ordinary versus exponential generating functions

The generating functions above are *ordinary generating functions* (OGFs): the sequence element a_n is literally the coefficient of z^n. For *unlabeled* combinatorial structures, where we only care about shape and not about which specific elements occupy which positions, OGFs are the natural choice.

But many problems in combinatorics — and in the analysis of language models — involve *labeled* structures, where the n atoms in a structure carry distinct labels from $\{1, \ldots, n\}$ and the assignment of labels matters. The canonical example is permutations: the number of permutations of n distinct elements is $n!$. Combining two labeled structures requires choosing which labels go where. Suppose we have a labeled A-structure on some subset of k labels, and a labeled B-structure on the remaining $n-k$ labels. The number of ways to choose the subset is $\binom{n}{k}$, so the total count of combined structures of size n is $\sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k}$. This is called *binomial convolution*, and it differs from the ordinary convolution $\sum_k a_k b_{n-k}$ by the binomial coefficient.

Definition 1.3. The *exponential generating function* (EGF) of a sequence $(a_n)_{n\geq 0}$ is

$$\hat{A}(z) \;=\; \sum_{n\geq 0} a_n \frac{z^n}{n!}.$$

The key algebraic fact is that binomial convolution of sequences corresponds to ordinary multiplication of EGFs. To see this, multiply $\hat{A}(z) = \sum a_n z^n/n!$ by $\hat{B}(z) = \sum b_n z^n/n!$. The coefficient of z^n in the product is

$$[z^n](\hat{A}\cdot\hat{B}) \;=\; \sum_{k=0}^{n} \frac{a_k}{k!}\cdot\frac{b_{n-k}}{(n-k)!} \;=\; \frac{1}{n!}\sum_{k=0}^{n}\binom{n}{k} a_k\, b_{n-k},$$

since $\binom{n}{k} = n!/(k!(n-k)!)$. So $n!\cdot[z^n](\hat{A}\cdot\hat{B})$ is exactly the binomial convolution — the right combinatorial count for labeled structures. This is the algebraic reason that labeled structures prefer EGFs.

We write $[z^n/n!]\hat{A}(z)$ as shorthand for $n!\cdot[z^n]\hat{A}(z) = a_n$, i.e., the coefficient of $z^n/n!$ in the EGF.

Example 1.4. Consider the sequence $a_n = n!$, counting permutations. Its OGF is $\sum_{n\geq 0} n!\, z^n$. The coefficients $n!$ grow much faster than any geometric sequence C^n, which means this series is only useful as a formal object; it does not converge to an analytic function for any $z \neq 0$. (Chapter 2 will make the notion of convergence precise via the *radius of convergence*.) Its EGF, on the other hand, is

$$\hat{S}(z) \;=\; \sum_{n\geq 0} n!\cdot\frac{z^n}{n!} \;=\; \sum_{n\geq 0} z^n \;=\; \frac{1}{1-z},$$

which is a clean rational function. The coefficient $[z^n/n!]\,\hat{S}(z) = 1$ tells us there is, up to labeling, one "shape" of permutation at each size; multiplying by $n!$ recovers the actual count $n!$.

The distinction between OGF and EGF is not merely aesthetic: using an OGF for a labeled structure or an EGF for an unlabeled one leads to incorrect functional equations. Part of the skill of analytic combinatorics is recognizing which generating function type matches the combinatorial product being modeled. The *symbolic method*, developed in Chapter 3, makes this assignment systematic: the constructions SEQ, SET, CYC, and MSET each translate directly into GF operations, and the labeled/unlabeled distinction determines whether the translation uses OGF or EGF conventions.

Road map

This chapter has introduced generating functions purely as formal algebraic objects, and the examples have already shown their power: a recurrence becomes a functional equation, and the equation's solution encodes both exact counts and, implicitly, asymptotic behavior. The next chapter treats z as a complex variable and develops the analytic theory that lets us extract asymptotics from singularities — the principle that the growth rate of a_n is governed by the location of the nearest singularity of $A(z)$ in $\mathbb{C}$, and the subexponential factor is governed by the type of that singularity. Chapter 3 introduces the symbolic method, turning combinatorial grammar rules directly into GF equations without any manual index manipulation. Chapters 4 and 5 then apply singularity analysis to rational and algebraic generating functions respectively, recovering the φ^n growth of Fibonacci numbers and the $4^n n^{-3/2}$ growth of Catalan numbers from first principles. Throughout, we will draw connections to the statistical structure of sequences produced by large language models: such models define probability distributions over discrete sequences, and generating functions provide the right language for analyzing those distributions at scale.

Exercise 1.5. Find the ordinary generating function for the sequence $a_n = 3^n$ (ternary strings of length n) by both methods: (a) summing the geometric series directly, and (b) applying the recurrence-to-GF recipe to the recurrence $a_n = 3a_{n-1}$, $a_0 = 1$.

Exercise 1.6. Use the generating function method to solve the recurrence $a_n = 2a_{n-1} + 1$ for $n \geq 1$, with $a_0 = 0$. (Hint: multiply by z^n and sum. You will need to handle $\sum_{n\geq 1} z^n = z/(1-z)$ for the constant term on the right side. Use partial fractions on the resulting $A(z)$ to find a closed form for a_n.)

Exercise 1.7. Verify the generalized binomial coefficient formula $\binom{1/2}{n} = (-1)^{n-1}/(2^{2n-1}n) \cdot \binom{2n-2}{n-1}$ for $n = 1, 2, 3, 4$ by computing both sides directly.

Further Reading

- [18] — The definitive reference for analytic combinatorics. Part A (Chapters I–II) covers formal power series, OGFs, EGFs, and the translation dictionary between combinatorial constructions and generating-function operations with full proofs and extensive examples.
- [41] — Focuses on algorithms for random generation and enumeration of combinatorial structures; includes generating-function derivations as part of the algorithmic analysis, with a hands-on computational style that complements the more theoretical approach of the present chapter.
- **H. S. Wilf**, *generatingfunctionology* (2nd ed., Academic Press, 1994; freely available online) — A short, readable introduction to generating functions aimed at undergraduates. Covers OGFs and EGFs through worked problems; an excellent companion for anyone finding the present chapter's pace too quick.
- **R. P. Stanley**, *Enumerative Combinatorics*, Vol. 1 (Cambridge, 1997) — Chapter 1 gives a thorough algebraic treatment of formal power series and the ring $\mathbb{Q}[[z]]$; Chapter 4 covers rational and algebraic generating functions in depth appropriate for graduate study.
- **D. E. Knuth**, *The Art of Computer Programming*, Vol. 1, §1.2.9 — A classic account of generating functions as a tool for solving recurrences, with worked examples drawn from algorithm analysis; especially useful for readers interested in the connection to computer science.

Chapter 2

Just Enough Complex Analysis

Chapter 1 treated z as a formal symbol, manipulated according to algebraic rules with no thought of convergence or magnitude. The real power of analytic combinatorics comes from promoting z to a genuine complex variable and extracting the asymptotic growth of the coefficients $[z^n]A(z)$ from the analytic structure of the function A. This chapter develops the minimum complex analysis required to make that move rigorous and computational. The reader who has seen complex numbers but not complex functions will find everything needed here; the reader who has taken a full complex analysis course may skim and use this chapter as a reference for notation. The standard treatment of all topics below is [18], Part A.

2.1 Complex Numbers and Convergence

Recall that $\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\}$ with the convention $i^2 = -1$. The *modulus* of $z = x + iy$ is $|z| = \sqrt{x^2 + y^2}$, and the *argument* $\arg z$ is the angle $\theta \in (-\pi, \pi]$ such that $x = |z|\cos\theta$, $y = |z|\sin\theta$. Euler's formula

$$e^{i\theta} = \cos\theta + i\sin\theta$$

allows every nonzero complex number to be written in *polar form* $z = re^{i\theta}$ with $r = |z| > 0$ and $\theta = \arg z$. Multiplication becomes transparent in polar form: $r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 r_2\, e^{i(\theta_1+\theta_2)}$, so moduli multiply and arguments add.

Convergence in $\mathbb{C}$ is measured by the modulus: a sequence (z_k) converges to L if $|z_k - L| \to 0$. A series $\sum_k w_k$ *converges absolutely* if $\sum_k |w_k| < \infty$, and absolute convergence implies convergence. The geometric series identity

$$\sum_{k=0}^{N} w^k = \frac{1 - w^{N+1}}{1 - w}, \quad w \neq 1,$$

shows that $\sum_{k=0}^{N} w^k \to 1/(1-w)$ as $N \to \infty$ whenever $|w| < 1$, since $|w|^{N+1} \to 0$. We will use this repeatedly.

2.2 Power Series and Radius of Convergence

Let $A(z) = \sum_{n\geq 0} a_n z^n$ be a formal power series with coefficients $a_n \in \mathbb{C}$. Treating z as a complex variable, the set of z for which this series converges absolutely is a disk.

Theorem 2.1 (Cauchy–Hadamard)**.** *The power series $A(z) = \sum_{n\geq 0} a_n z^n$ converges absolutely for every z with $|z| < R$ and diverges for every z with $|z| > R$, where the* radius of convergence *R is given by*

$$\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n}.$$

Proof sketch. Fix z with $|z| < R$. Then $|z| < 1/\limsup |a_n|^{1/n}$, so there exists $\delta > 0$ with $|z| \leq (1-\delta)/\limsup |a_n|^{1/n}$. By the definition of lim sup, for all sufficiently large n we have $|a_n|^{1/n} \leq (1/R) + \epsilon$ for any $\epsilon > 0$; choosing ϵ small enough that $(1/R + \epsilon)|z| < 1$, we get $|a_n z^n| = (|a_n|^{1/n}|z|)^n \leq r^n$ for some $r < 1$ and all large n. The series $\sum |a_n z^n|$ is then bounded above by a convergent geometric series, so $\sum a_n z^n$ converges absolutely.

Conversely, if $|z| > R$, then $\limsup |a_n|^{1/n}|z| > 1$, so $|a_n z^n| > 1$ for infinitely many n, and the terms do not tend to zero. The series diverges. □

The combinatorial import of Cauchy–Hadamard is immediate: the radius of convergence captures the *exponential growth rate* of (a_n). If $R > 0$ is finite, then a_n grows (roughly) like R^{-n}, in the sense that $\limsup |a_n|^{1/n} = R^{-1}$ (and for the well-behaved sequences arising from combinatorics, the ordinary limit exists and equals R^{-1}). More precisely, $a_n = R^{-n} \cdot \theta(n)$ where $\theta(n)$ is *subexponential*: $\limsup |\theta(n)|^{1/n} = 1$. Determining the fine structure of $\theta(n)$ is precisely what asymptotic analysis does.

Example 2.2. For $B(z) = 1/(1-2z) = \sum_{n\geq 0} 2^n z^n$, we have $b_n = 2^n$ and $|b_n|^{1/n} = 2$ for all n, so $R = 1/2$. The singularity of B at $z = 1/2$ is directly visible as a pole, and the exponential growth rate $2^n = (1/2)^{-n}$ agrees with R^{-n}.

For the Fibonacci generating function $F(z) = z/(1-z-z^2)$, the poles are at the roots of $1 - z - z^2 = 0$, namely $z = (-1 \pm \sqrt{5})/2$. The root of smallest modulus is $1/\varphi$ where $\varphi = (1+\sqrt{5})/2 \approx 1.618$ is the golden ratio, so $R = 1/\varphi$ and $f_n \sim C\varphi^n$ for some explicit constant C.

2.3 Analytic Functions

A function $f : U \to \mathbb{C}$ defined on an open set $U \subseteq \mathbb{C}$ is *holomorphic* on U (also called *analytic*) if it is complex-differentiable at every point of U. A fundamental and non-trivial theorem of complex analysis states that the following conditions are equivalent for a function on an open set: (1) f is complex-differentiable (the limit $\lim_{h\to 0}(f(z+h) - f(z))/h$ exists in $\mathbb{C}$); (2) f is infinitely differentiable and satisfies the Cauchy–Riemann equations $\partial_x u = \partial_y v$, $\partial_y u = -\partial_x v$ where $f = u + iv$; (3) f is locally representable by a convergent power series around every point. We will take this equivalence as given.

The intuition behind this equivalence is worth pausing on, even without a proof. Real differentiability requires only that a function have a well-defined slope when you approach a point along the real line—from the left or from the right. Complex differentiability demands something far more stringent: the limit $(f(z+h) - f(z))/h$ must yield the *same* value no matter what direction h approaches zero in the complex plane—horizontally, vertically, diagonally, or along any curve whatsoever. This is an overdetermined system: two partial derivatives of the real and imaginary parts must simultaneously satisfy the Cauchy–Riemann equations, which is already a strong constraint, and having this hold at every point is stronger still. The result is that a function with even one complex derivative everywhere is automatically infinitely smooth, and its values in any open disk are completely determined by a convergent power series—a rigidity that has no real analogue. Real-differentiable functions can have kinks and corners in their higher derivatives; complex-differentiable functions cannot, because the complex constraint leaves no room for local irregularity.

The key example: if $A(z) = \sum_{n\geq 0} a_n z^n$ has radius of convergence $R > 0$, then A is holomorphic on the open disk $|z| < R$. This follows because power series can be differentiated term by term within their disk of convergence, and the resulting series again converges. Conversely, a holomorphic function on $|z| < R$ is completely determined by its power series at the origin. The formal generating functions of Chapter 1, as soon as their coefficients grow at most geometrically, are genuine holomorphic functions on a nontrivial disk.

Definition 2.3. A *singularity* of a function f is a point $z_0 \in \mathbb{C}$ at which f fails to be holomorphic: there is no convergent power series representation of f in any neighborhood of z_0. A *dominant singularity* is a singularity of smallest modulus $|z_0|$.

2.4 The Geometric Series as Prototype

The function $f(z) = 1/(1-z)$ is the model example. For $|z| < 1$ the series $\sum_{n\geq 0} z^n$ converges absolutely, and multiplying by $(1-z)$ collapses the partial sums to $1 - z^{N+1} \to 1$, giving

$$\frac{1}{1-z} = \sum_{n\geq 0} z^n, \quad |z| < 1.$$

The function $1/(1-z)$ is holomorphic on all of $\mathbb{C} \setminus \{1\}$: the formula $1/(1-z)$ makes perfect sense and defines a smooth function at, say, $z = 2$ or $z = -3$, even though the series $\sum z^n$ diverges there. In the language of complex analysis, $1/(1-z)$ is the *analytic continuation* of the series beyond its disk of convergence — the unique holomorphic function that agrees with the series where the series converges, but exists on a larger domain. The series itself, however, is unable to see past the boundary $|z| = 1$. The point $z = 1$ is a *simple pole*: $f(z) = 1/(1-z)$ blows up like $(1-z)^{-1}$ near $z = 1$, which is the simplest possible singularity.

The lesson is structural: the singularity at $z = 1$ is what *prevents* the radius of convergence from exceeding 1, because the series cannot represent a function that blows up. In general, the radius of convergence of a power series at a point z_0 equals the distance from z_0 to the nearest singularity of the function that the series represents. Every analytic-combinatorics computation ultimately reduces to locating the nearest singularity of a generating function.

2.5 Cauchy's Integral Formula for Coefficients

The coefficients of a power series can be recovered by a contour integral. Let f be holomorphic on $|z| < R$ and let $0 < r < R$. Then

$$a_n = [z^n]f(z) = \frac{1}{2\pi i} \oint_{|z|=r} \frac{f(z)}{z^{n+1}}\, dz,$$

where the contour integral runs counterclockwise around the circle of radius r. This follows from Cauchy's integral formula for derivatives: for a holomorphic f on an open disk,

$$f^{(n)}(0) = \frac{n!}{2\pi i} \oint_{|z|=r} \frac{f(z)}{z^{n+1}}\, dz,$$

combined with Taylor's theorem $f(z) = \sum_n a_n z^n$ implying $f^{(n)}(0) = n!\, a_n$. Dividing both sides by $n!$ gives the coefficient formula.

The crucial observation for asymptotics is that the contour $|z| = r$ can be *deformed outward*, provided it remains in the region where f is holomorphic. Suppose f is holomorphic on a disk of radius r (with all singularities outside $|z| = r$). Write $M(r) = \max_{|z|=r} |f(z)|$. Then the standard *ML bound* on contour integrals gives

$$|a_n| = \frac{1}{2\pi}\left|\oint_{|z|=r} \frac{f(z)}{z^{n+1}}\, dz\right| \leq \frac{1}{2\pi} \cdot 2\pi r \cdot \frac{M(r)}{r^{n+1}} = \frac{M(r)}{r^n}.$$

(The factor $2\pi r$ is the length of the circle; $|f(z)/z^{n+1}| \leq M(r)/r^{n+1}$ on the contour.) This is an *exponential* upper bound on $|a_n|$: the decay rate r^{-n} improves as r increases, while $M(r)$ is a constant (depending on r but not on n). The contour can be pushed outward as far as possible

without crossing a singularity of f; the singularity forms an impassable barrier. Taking r just below the distance to the nearest singularity ρ gives $|a_n| \le M(\rho-\epsilon)/(\rho-\epsilon)^n$ for any $\epsilon > 0$, which is essentially ρ^{-n} times a constant. The dominant contribution to a_n therefore comes from the behavior of f near its nearest singularity.

This is the geometric heart of singularity analysis: singularities are not obstacles to computation but rather the *source* of asymptotic information.

2.6 Classifying Singularities

The type of a singularity determines the subexponential correction to the exponential growth rate. For a generating function with non-negative real coefficients — which is the case for every counting problem — a classical result (the Pringsheim–Vivanti theorem) guarantees that the point $z = R$ on the positive real axis is always a singularity. We write $\rho = R > 0$ for this dominant singularity. (Generating functions with complex or alternating-sign coefficients can have their dominant singularity elsewhere in $\mathbb{C}$, but this case rarely arises in combinatorics.) The three types that appear most frequently are the following.

Poles. Consider $f(z) = 1/(1-z/\rho)^k$ for a positive integer k. We build up the coefficient formula from first principles. For $k = 1$, this is the geometric series: $1/(1-w) = \sum_{n\ge0} w^n$ with $w = z/\rho$, giving $[z^n]f = \rho^{-n}$. For $k = 2$, differentiate the geometric series: $\frac{d}{dw}\frac{1}{1-w} = \frac{1}{(1-w)^2} = \sum_{n\ge1} n\,w^{n-1} = \sum_{n\ge0}(n+1)w^n$, so $[z^n]\frac{1}{(1-z/\rho)^2} = (n+1)\rho^{-n}$. Differentiating again: $\frac{1}{(1-w)^3} = \frac{1}{2}\sum_{n\ge0}(n+2)(n+1)w^n$, so $[z^n] = \binom{n+2}{2}\rho^{-n}$. In general, differentiating $k-1$ times:

$$\frac{1}{(1-z/\rho)^k} = \sum_{n\ge0}\binom{n+k-1}{k-1}\rho^{-n}z^n.$$

(This can also be derived from the generalized binomial theorem of Chapter 1 with $\alpha = -k$.)

The coefficient $\binom{n+k-1}{k-1}$ is a polynomial of degree $k-1$ in n. To see this, write it out:

$$\binom{n+k-1}{k-1} = \frac{(n+k-1)(n+k-2)\cdots(n+1)}{(k-1)!}.$$

The numerator is a product of $k-1$ factors, each of the form $n+(\text{constant})$, so it is a polynomial of degree $k-1$ in n with leading term n^{k-1}. Dividing by the constant $(k-1)!$:

$$\binom{n+k-1}{k-1} \sim \frac{n^{k-1}}{(k-1)!} \quad \text{as } n\to\infty.$$

A simple pole ($k = 1$) contributes a pure exponential ρ^{-n}; a double pole ($k = 2$) contributes $(n+1)\rho^{-n} \sim n\rho^{-n}$; a triple pole ($k = 3$) contributes $\binom{n+2}{2}\rho^{-n} \sim \frac{n^2}{2}\rho^{-n}$; and so on. Most rational generating functions have poles, and they arise naturally whenever a linear recurrence has a unique dominant root.

Square-root branch points. If $f(z) = \sqrt{1-z/\rho} = (1-z/\rho)^{1/2}$, then the generalized binomial theorem gives $f(z) = \sum_{n\ge0}\binom{1/2}{n}(-z/\rho)^n$. In Chapter 1 we derived $\binom{1/2}{n} = (-1)^{n-1}/(2^{2n-1}n)\binom{2n-2}{n-1}$, so $[z^n]f = \binom{1/2}{n}(-1)^n\rho^{-n} = -\rho^{-n}/(2^{2n-1}n)\binom{2n-2}{n-1}$.

It remains to estimate $\binom{2n-2}{n-1}$ for large n. This is a central binomial coefficient (up to a shift), and its asymptotics follow from Stirling's approximation $m! \approx \sqrt{2\pi m}\,(m/e)^m$. Write $m = n-1$:

$$\binom{2m}{m} = \frac{(2m)!}{(m!)^2} \approx \frac{\sqrt{2\pi(2m)}\,(2m/e)^{2m}}{(\sqrt{2\pi m}\,(m/e)^m)^2} = \frac{\sqrt{4\pi m}\cdot 2^{2m}m^{2m}/e^{2m}}{2\pi m\cdot m^{2m}/e^{2m}} = \frac{2^{2m}}{\sqrt{\pi m}}.$$

So $\binom{2n-2}{n-1} \sim 4^{n-1}/\sqrt{\pi(n-1)} \sim 4^{n-1}/\sqrt{\pi n}$ for large n. Substituting and noting $4^{n-1}/2^{2n-1} = 2^{2n-2}/2^{2n-1} = 1/2$:

$$[z^n]\sqrt{1-z/\rho} = -\frac{1}{2\sqrt{\pi}}\, n^{-3/2}\, \rho^{-n}\big(1+O(1/n)\big).$$

The exponent $-3/2$ in the polynomial correction $n^{-3/2}$ is the *fingerprint* of a square-root singularity. This type arises pervasively in combinatorics because the kernel method and quadratic equations produce square roots; trees, lattice paths, and many self-referential structures lead to generating functions with $\sqrt{1-z/\rho}$ as a factor. We will return to this in Chapter 5.

Logarithmic singularities. If $f(z) = \log(1/(1-z/\rho))$, then $f(z) = \sum_{n\geq 1} \rho^{-n} z^n/n$, so $[z^n]f = \rho^{-n}/n$. The correction factor is $1/n$: it decays to zero (unlike the constant correction at a simple pole) but more slowly than the $n^{-3/2}$ of a square-root singularity. Logarithms appear, for instance, in the generating functions for labelled structures involving cycles, and in the analysis of certain algorithms. They are less common than poles or branch points but arise naturally whenever the symbolic construction includes SET or CYC operators.

Essential singularities. Functions such as $e^{1/(1-z)}$ have an essential singularity at $z = 1$: they blow up in a highly irregular, non-power-law fashion that cannot be captured by any finite Laurent expansion. The elementary transfer-theorem machinery does not apply at essential singularities, and special methods are required. Such singularities are rare in the combinatorial generating functions considered in this book, and we will not develop the necessary tools here.

2.7 The Big Picture

The moral of this chapter is simple to state and powerful to apply. If $A(z)$ is a generating function whose nearest singularity to the origin lies at $|z| = \rho$, then a_n grows exponentially like ρ^{-n}. That much is determined purely by the *location* of the singularity. The *type* of the singularity—pole of order k, square-root branch point, logarithm—controls the subexponential factor n^α for some α determined by the local singular behavior. Flajolet and Odlyzko's transfer theorem [17], developed in Chapter 4, makes this dictionary precise and gives explicit asymptotic expansions to any desired order of precision [18].

All the complex analysis we use in this book reduces to three steps: (i) find a useful closed form for the generating function $A(z)$, using the recurrence method of Chapter 1 or the symbolic method of Chapter 3; (ii) locate the singularities on the boundary of the disk of convergence; (iii) read the asymptotics off their type using the classification above. The deeper machinery of complex analysis—contour deformation through branch cuts, analytic continuation to Riemann surfaces, the residue theorem in full generality, the saddle-point method—belongs to a dedicated complex-analysis course and will appear in this book only when no simpler approach suffices. The examples of Chapter 3 will make the three-step program concrete before we prove the transfer theorem in Chapter 4.

Exercise 2.4. Find the radius of convergence of $\sum_{n\geq 1} z^n/n^2$. What is the dominant singularity? (Hint: use Cauchy–Hadamard. Note that $|1/n^2|^{1/n} \to 1$.)

Exercise 2.5. Show that $1/(1-z)^3 = \sum_{n\geq 0} \binom{n+2}{2} z^n$ by applying the generalized binomial theorem (Chapter 1) with $\alpha = -3$ and $w = -z$. The singularity at $z = 1$ is a pole of order 3; verify that $\binom{n+2}{2} = (n+2)(n+1)/2 \sim n^2/2$, consistent with the catalogue entry n^{k-1} for $k = 3$.

Exercise 2.6. The generating function $\sum_{n\geq 0} n!\, z^n$ has radius of convergence $R = 0$ (verify this using Cauchy–Hadamard). What does this say about the location of singularities? Can you make sense of the "function" this series represents? (This is an example where the formal power series of Chapter 1 is a valid algebraic object but has no analytic avatar.)

Further Reading

- [18], Part B — Chapters IV–VI develop the complex-analytic foundations used throughout this book: analytic continuation, the transfer theorem, and singularity analysis in full generality. Reading Part B alongside the present chapter is strongly recommended.
- [17] — The original paper introducing the transfer theorem and Δ-domains; more concise than the textbook treatment and useful for seeing the theorem in its original form.
- **L. V. Ahlfors**, *Complex Analysis* (3rd ed., McGraw-Hill, 1979) — The standard graduate-level reference for classical complex analysis. Chapters 1–4 cover all the material in the present chapter (convergence, analyticity, Cauchy's theorem, singularity classification) with full proofs; Chapter 5 introduces residues and contour integration.
- **E. C. Titchmarsh**, *The Theory of Functions* (2nd ed., Oxford, 1939) — A more classical treatment, particularly strong on power series (Chapter 1) and contour integration and residues (Chapter 3); useful as a second reference for the Cauchy–Hadamard theorem and the classification of isolated singularities.
- **P. Henrici**, *Applied and Computational Complex Analysis*, Vol. 1 (Wiley, 1974) — Covers power series and analytic functions with an emphasis on effective computation and coefficient extraction; bridges the gap between pure complex analysis and the algorithmic side of generating functions.

Chapter 3

The Symbolic Method

The central ambition of analytic combinatorics is to pass mechanically from a structural description of a class of combinatorial objects to an equation satisfied by its generating function, and thence—via the complex-analytic machinery of Chapter 2—to precise asymptotic counts. The *symbolic method* is the first half of that program. It is a formal calculus in which set-theoretic operations on combinatorial classes correspond, under a precise dictionary, to algebraic operations on power series. No coefficient computation by hand is required; the structure of an object class *is* its generating function equation. The standard reference for everything in this chapter is [18], Part A.

3.1 Combinatorial Classes

Definition 3.1. A *combinatorial class* is a pair $(\mathcal{C}, |\cdot|)$ where $\mathcal{C}$ is a (typically countably infinite) set and $|\cdot| : \mathcal{C} \to \mathbb{Z}_{\geq 0}$ is a *size function* such that the level set

$$\mathcal{C}_n \;=\; \{\, \gamma \in \mathcal{C} : |\gamma| = n \,\}$$

is finite for every $n \geq 0$. Write $c_n = |\mathcal{C}_n|$. The *ordinary generating function* (OGF) of $\mathcal{C}$ is

$$C(z) \;=\; \sum_{\gamma \in \mathcal{C}} z^{|\gamma|} \;=\; \sum_{n \geq 0} c_n\, z^n.$$

The finiteness condition on $\mathcal{C}_n$ guarantees that $C(z)$ is a well-defined formal power series; whether it converges anywhere other than $z = 0$ is a separate analytic question taken up in Chapter 2. Three archetypal classes will recur throughout: the class $\mathcal{W}$ of binary words over $\{a, b\}$ with $|w|$ equal to the word length; the class $\mathcal{T}$ of plane trees (also called ordered or planar trees) with $|t|$ equal to the number of nodes; and the class $\mathfrak{S}$ of permutations with $|\sigma|$ equal to the length. For permutations one uses instead an *exponential generating function* (EGF), discussed briefly in the subsection on labeled classes below; the unlabeled classes $\mathcal{W}$ and $\mathcal{T}$ are handled by ordinary generating functions.

Two distinguished classes serve as universal building blocks.

Definition 3.2. The *neutral class* $\mathcal{E} = \{\epsilon\}$ contains a single object of size 0; its OGF is $E(z) = 1$. The *atomic class* $\mathcal{Z} = \{\bullet\}$ contains a single object of size 1; its OGF is $Z(z) = z$.

Every combinatorial class is, in the symbolic framework, built from $\mathcal{E}$ and $\mathcal{Z}$ by iterating a small family of constructions. Making this precise is the content of the next section.

3.2 The Translation Dictionary for OGFs

Proposition 3.3 (Symbolic Transfer, Unlabeled Case)**.** *Let $\mathcal{A}$ and $\mathcal{B}$ be combinatorial classes with OGFs $A(z)$ and $B(z)$ respectively. The following constructions on classes translate into the indicated operations on OGFs.*

1. ***Disjoint union.*** *If $\mathcal{A}+\mathcal{B}$ denotes the disjoint union of $\mathcal{A}$ and $\mathcal{B}$ (with size inherited from each summand), then*
$$\mathcal{A}+\mathcal{B} \longleftrightarrow A(z)+B(z).$$
2. ***Cartesian product.*** *Define the size of a pair by $|(a,b)| = |a|+|b|$. Then*
$$\mathcal{A}\times\mathcal{B} \longleftrightarrow A(z)\,B(z).$$
3. ***Sequence.*** *Provided $\mathcal{A}$ contains no object of size 0,*
$$\mathrm{SEQ}(\mathcal{A}) \longleftrightarrow \frac{1}{1-A(z)}.$$

Proof sketch. For the disjoint union, the sum defining the OGF splits over the two components:
$$\sum_{\gamma\in\mathcal{A}\sqcup\mathcal{B}} z^{|\gamma|} = \sum_{\alpha\in\mathcal{A}} z^{|\alpha|} + \sum_{\beta\in\mathcal{B}} z^{|\beta|} = A(z)+B(z).$$
For the Cartesian product, the additive size convention gives
$$\sum_{(a,b)\,\in\,\mathcal{A}\times\mathcal{B}} z^{|(a,b)|} = \sum_{a\in\mathcal{A}}\sum_{b\in\mathcal{B}} z^{|a|+|b|} = \left(\sum_{a\in\mathcal{A}} z^{|a|}\right)\left(\sum_{b\in\mathcal{B}} z^{|b|}\right) = A(z)\,B(z).$$
This is precisely the Cauchy convolution of coefficient sequences, matching the observation in Chapter 1 that $[z^n]\,A(z)\,B(z) = \sum_{k=0}^{n} a_k\,b_{n-k}$.

For sequences, by definition $\mathrm{SEQ}(\mathcal{A}) = \mathcal{E}+\mathcal{A}+\mathcal{A}^2+\mathcal{A}^3+\cdots$, where $\mathcal{A}^k$ is the k-fold Cartesian product and the sum is disjoint union. The absence of size-0 objects in $\mathcal{A}$ ensures that each $\mathcal{A}^k$ contributes only to $[z^n]$ for $n\ge k$, so the formal sum is locally finite and defines a valid combinatorial class. Applying the first two rules,
$$\mathrm{SEQ}(\mathcal{A}) \longleftrightarrow 1+A(z)+A(z)^2+A(z)^3+\cdots = \frac{1}{1-A(z)},$$
the last equality holding as an identity of formal power series (since $A(z)$ has zero constant term by hypothesis). □

Remark 3.4. The SEQ constructor is often written with a subscript constraint: $\mathrm{SEQ}_{\ge k}(\mathcal{A})$ denotes sequences of length at least k, with OGF $A(z)^k/(1-A(z))$. This follows because $\mathrm{SEQ}_{\ge k}(\mathcal{A}) = \mathcal{A}^k\times\mathrm{SEQ}(\mathcal{A})$, so the OGF is $A(z)^k\cdot 1/(1-A(z))$. In particular $\mathrm{SEQ}_{\ge 1}(\mathcal{A})$ has OGF $A(z)/(1-A(z))$.

Marking variables and bivariate generating functions

A single extra variable u allows one to track a statistic simultaneously with size. Suppose $\chi:\mathcal{C}\to\mathbb{Z}_{\ge 0}$ is a combinatorial parameter. Define the *bivariate OGF*
$$C(z,u) = \sum_{\gamma\in\mathcal{C}} z^{|\gamma|}\,u^{\chi(\gamma)} = \sum_{n,k\ge 0} c_{n,k}\,z^n u^k,$$
where $c_{n,k} = |\{\gamma : |\gamma| = n,\ \chi(\gamma)=k\}|$. The univariate OGF is recovered as $C(z,1)$. The mean of χ over objects of size n is
$$\mathbf{E}[\chi \mid |\gamma| = n] = \frac{[z^n]\,\partial_u C(z,u)\big|_{u=1}}{[z^n]\,C(z,1)}.$$
To see why: differentiating $u^{\chi(\gamma)}$ with respect to u and setting $u=1$ replaces each $u^{\chi(\gamma)}$ by $\chi(\gamma)$, so $[z^n]\,\partial_u C(z,u)|_{u=1} = \sum_{|\gamma|=n}\chi(\gamma)$. Dividing by $c_n = [z^n]\,C(z,1)$ gives the mean.

The *pointing* (or *derived*) class $\mathcal{C}^\bullet$ marks one distinguished atom in each structure. If objects of size n each contain exactly n atoms, then $C^\bullet(z) = z\,C'(z)$ where $C' = dC/dz$, because pointing multiplies the number of structures of size n by n: the GF of pointed objects is $\sum_{n\ge 0} n\,c_n\,z^n = z\sum_{n\ge 0} n\,c_n\,z^{n-1} = z\,C'(z)$.

Labeled classes and EGFs

For labeled structures—where the n atoms carry distinct labels from $\{1, \ldots, n\}$—the correct accounting replaces OGFs with *exponential generating functions* (EGFs): $\hat{C}(z) = \sum_{n\geq 0} c_n\, z^n/n!$. The translation rules for disjoint union and sequence are formally identical in shape. The *labeled product* of classes $\hat{\mathcal{A}}$ and $\hat{\mathcal{B}}$ yields OGF $\hat{A}(z)\,\hat{B}(z)$; the $1/n!$ normalization of the EGF absorbs the $\binom{n}{k}$ ways to partition a label set of size n between the two factors. Beyond sequences, labeled structures admit two further constructions: $\mathrm{SET}(\mathcal{A})$ (unordered collections, with repetition forbidden) translates to $\exp \hat{A}(z)$, and $\mathrm{CYC}(\mathcal{A})$ (cyclic sequences) translates to $\log(1/(1-\hat{A}(z)))$. Here is the derivation of the SET rule, which is short and illuminates the whole labeled framework.

A set of k labeled $\mathcal{A}$-structures is formed by (i) partitioning the label set $[n]$ into k blocks of sizes $n_1, \ldots, n_k$ with $n_1 + \cdots + n_k = n$, (ii) placing one $\mathcal{A}$-structure on each block, and (iii) forgetting the order of the k components (since a set is unordered). Steps (i) and (ii) together contribute

$$\binom{n}{n_1, \ldots, n_k} \cdot a_{n_1} \cdots a_{n_k}$$

ways to build an ordered k-tuple of $\mathcal{A}$-structures on labels $[n]$. Step (iii) divides by $k!$ to forget the ordering. Summing over all $k \geq 0$ and all compositions of n, the EGF of $\mathrm{SET}(\mathcal{A})$ is

$$\widehat{\mathrm{SET}(\mathcal{A})}(z) \;=\; \sum_{k\geq 0} \frac{\hat{A}(z)^k}{k!} \;=\; \exp\bigl(\hat{A}(z)\bigr).$$

The $k!$ in the denominator is the key: it is exactly the number of ways to permute k indistinguishable components, canceling the overcounting from treating an unordered set as an ordered tuple. The labeled product rule (the EGF of an ordered k-tuple is $\hat{A}(z)^k$) combined with division by $k!$ is precisely the exponential series, which is why EGFs and the exponential function are inseparable in the labeled world.

For $\mathrm{CYC}(\mathcal{A})$, the same argument applies with one change: cyclic sequences have k rotations rather than $k!$ permutations as symmetries, so the EGF of k-cycles is $\hat{A}(z)^k/k$ instead of $\hat{A}(z)^k/k!$. Summing over $k \geq 1$ gives

$$\widehat{\mathrm{CYC}(\mathcal{A})}(z) \;=\; \sum_{k\geq 1} \frac{\hat{A}(z)^k}{k} \;=\; \log\frac{1}{1-\hat{A}(z)}.$$

The full derivation, including careful bookkeeping for the $k = 0$ case and the formal treatment via species, is in [18], Chapter II; see also [**?**] for the species-theoretic account.

3.3 Worked Examples

Example 3.5 (Binary words)**.** Let $\mathcal{A} = \{a\}$ and $\mathcal{B} = \{b\}$, each a copy of $\mathcal{Z}$. A binary word is an element of $\mathrm{SEQ}(\mathcal{A} + \mathcal{B})$. By Proposition 3.3,

$$W(z) \;=\; \frac{1}{1-(z+z)} \;=\; \frac{1}{1-2z} \;=\; \sum_{n\geq 0} 2^n z^n.$$

There are 2^n binary words of length n. This is the simplest instance of the method: a one-line specification yields a one-line OGF.

Example 3.6 (Plane trees and Catalan numbers)**.** A plane tree $t \in \mathcal{T}$ consists of a root node (contributing one unit of size) together with an ordered, possibly empty sequence of plane trees (the subtrees hanging from the root). The self-referential description translates immediately:

$$\mathcal{T} \;=\; \mathcal{Z} \times \mathrm{SEQ}(\mathcal{T}) \quad\Longrightarrow\quad T(z) \;=\; \frac{z}{1-T(z)}.$$

This is a functional equation for T, not a closed form—but extracting the closed form is straightforward. Rearranging gives $T(z)(1-T(z)) = z$, or

$$T(z)^2 - T(z) + z \;=\; 0.$$

The quadratic formula yields two branches; the branch with $T(0) = 0$ (no trees of size 0) is

$$T(z) \;=\; \frac{1-\sqrt{1-4z}}{2}.$$

Expanding the square root via the generalized binomial theorem (Chapter 1), $(1-4z)^{1/2} = \sum_{n\geq 0}\binom{1/2}{n}(-4z)^n$, so

$$[z^n]\,\sqrt{1-4z} \;=\; \binom{1/2}{n}(-4)^n \;=\; \frac{-2}{n}\binom{2(n-1)}{n-1} \quad (n \geq 1).$$

Since $T(z) = \frac{1}{2}(1-\sqrt{1-4z})$, we get

$$[z^n]\,T(z) \;=\; \frac{1}{n}\binom{2(n-1)}{n-1} \;=\; C_{n-1},$$

the $(n-1)$-st Catalan number. Crucially, the symbolic specification $\mathcal{T} = \mathcal{Z}\times\mathrm{SEQ}(\mathcal{T})$ *determines* the appearance of the square-root singularity: any class satisfying a quadratic equation of this shape will have a square-root branch point at its radius of convergence, with the $n^{-3/2}$ asymptotic law that Chapter 4 will make precise.

Example 3.7 (Integer compositions)**.** A *composition* of n is an ordered tuple of positive integers summing to n. A positive integer of value k is naturally modeled as a sequence of exactly k atoms, so the class of positive integers is $\mathcal{I} = \mathrm{SEQ}_{\geq 1}(\mathcal{Z})$, with OGF

$$I(z) \;=\; \frac{z}{1-z}.$$

A composition is then a nonempty sequence of positive integers: $\mathcal{K} = \mathrm{SEQ}_{\geq 1}(\mathcal{I})$, giving

$$K(z) \;=\; \frac{I(z)}{1-I(z)} \;=\; \frac{z/(1-z)}{1-z/(1-z)} \;=\; \frac{z}{1-2z}.$$

Thus $[z^n]\,K(z) = 2^{n-1}$ for $n \geq 1$: there are 2^{n-1} compositions of n. This is again a one-paragraph derivation; a direct combinatorial proof (choosing which of the $n-1$ gaps between n ones to include as part breaks) gives the same answer, but the generating-function route required no such insight.

Example 3.8 (Bivariate marking: number of a's in a binary word)**.** Let $\chi(w)$ count the number of occurrences of the letter a in a binary word w. Mark each a by u and each b by 1 to obtain the bivariate OGF:

$$W(z,u) \;=\; \frac{1}{1-zu-z} \;=\; \frac{1}{1-z(u+1)}.$$

Expanding,

$$[z^n u^k]\,W(z,u) \;=\; [u^k]\,(u+1)^n \;=\; \binom{n}{k}.$$

So among the 2^n binary words of length n, exactly $\binom{n}{k}$ contain exactly k letters a. Setting $u = 1$ recovers $W(z,1) = 1/(1-2z)$. The mean number of a's in a word of length n is

$$\frac{[z^n]\,\partial_u W(z,u)\big|_{u=1}}{[z^n]\,W(z,1)} \;=\; \frac{n\cdot 2^{n-1}}{2^n} \;=\; \frac{n}{2},$$

as expected by symmetry. The bivariate framework will become essential in Chapter 8 when we study parameters of parse trees arising from probabilistic context-free grammars.

3.4 Why the Symbolic Method Matters

The examples above share a common pattern: write down what an object *is* (in terms of simpler objects), apply the dictionary in Proposition 3.3, and obtain a functional equation for the generating function. This equation is typically polynomial or algebraic in the GF, and sometimes transcendental (involving exp or log in the labeled setting). The algebraic case—arising whenever the specification uses only $+$, $\times$, and SEQ—is of special importance: a GF satisfying a polynomial equation with algebraic-function coefficients is itself algebraic, and algebraic functions have branch-point singularities of square-root type (or more generally, p/q type for integers p and q). As Chapter 4 will show, the location and nature of the dominant singularity determines the leading asymptotic behavior of the coefficients; for the algebraic case, the result is always a term of the form $\kappa\rho^{-n}n^{-3/2}$ for explicit constants κ and ρ.

Chapter 5 makes this connection explicit through context-free grammars. A context-free grammar over a terminal alphabet is, from the symbolic-method perspective, exactly a system of simultaneous equations of the form

$$Y_i(z) \;=\; \Phi_i(z, Y_1(z), \ldots, Y_k(z)), \qquad i = 1, \ldots, k,$$

where each Φ_i is a polynomial (the polynomial is read off from the productions of the grammar). Such systems define algebraic functions by the implicit function theorem over formal power series, and the theory of algebraic singularities then gives the $n^{-3/2}$ universal exponent as a consequence of the irreducibility of the grammar. Every unambiguous context-free language has $2^{\Theta(n)}$ or $n^{-3/2}\rho^{-n}$ asymptotics; the symbolic method is the reason this dichotomy exists.

The connection to language models, foreshadowed for Part III, runs as follows. A generative language model over a finite vocabulary can be viewed as a probability distribution on finite strings. When the model is constrained to generate strings that belong to a combinatorial class $\mathcal{C}$—for instance, well-formed programs, grammatical sentences, or structured data formats—the generating function of $\mathcal{C}$ encodes the normalization constant and partition function of the model. Sampling from the model efficiently, computing the probability of a given string, and estimating the entropy rate all reduce, in the structured setting, to analytic properties of $C(z)$. The symbolic method is thus not merely a counting device; it is the algebraic skeleton onto which probabilistic and information-theoretic content is attached [18].

Exercise 3.9. Let $\mathcal{P}$ be the class of nonempty strings over $\{(,)\}$ that form a balanced parenthesization (i.e., Dyck prefixes of zero net height at the end). Write a symbolic specification for $\mathcal{P}$ using SEQ and a suitable decomposition at the first return to height zero, and derive the OGF. Verify that $[z^{2n}]\,P(z) = C_n$, the n-th Catalan number.

Exercise 3.10. A *unary-binary tree* is either a leaf (one node) or an internal node with exactly one child or exactly two children. Write a symbolic equation for the class $\mathcal{U}$ of unary-binary trees (with size = number of nodes), derive the OGF as a root of a quadratic, and identify the radius of convergence.

Exercise 3.11. Using the labeled setting, let $\mathcal{S}$ be the class of permutations, with size equal to the number of elements. The EGF is $\hat{S}(z) = \sum_n z^n = 1/(1-z)$. Now let $\mathcal{D}$ be the class of derangements. Using $\mathrm{SET}(\mathrm{CYC}_{\geq 2}(\mathcal{Z}))$ (cycles of length at least 2), derive the EGF of derangements and confirm that the probability a random permutation of $[n]$ is a derangement tends to e^{-1} as $n \to \infty$.

Further Reading

- [18], Chapters I–II — The definitive treatment of the symbolic method, covering both the unlabeled (OGF) and labeled (EGF) translation dictionaries, with proofs, an extensive

worked-example catalogue, and the full derivation of the SET, CYC, and MSET rules omitted from the present chapter.

- **P. Flajolet and M. Soria**, "General combinatorial schemas: Gaussian limit distributions and exponential tails," *Discrete Mathematics* 114 (1993), 159–180 — Shows how bivariate generating functions and the symbolic method together yield limit laws (Gaussian distributions of parameters) for broad families of combinatorial structures; a readable illustration of how the bivariate marking machinery of this chapter extends to probability.

- **R. P. Stanley**, *Enumerative Combinatorics*, Vol. 2 (Cambridge, 1999) — Chapter 5 covers trees and the Lagrange inversion formula; Chapter 6 develops algebraic, D-finite, and noncommutative generating functions. Together they extend the symbolic-method viewpoint to more advanced territory.

- [41] — Develops algorithms for random generation and enumeration of combinatorial structures (set partitions, permutations, labeled trees); the algorithmic perspective complements the symbolic-method viewpoint of this chapter.

- **F. Bergeron, G. Labelle, and P. Leroux**, *Combinatorial Species and Tree-Like Structures* (Cambridge, 1998) — The standard reference for the theory of species, the categorical underpinning of the symbolic method; essential for readers who want to understand why the translation dictionary works at the level of bijections rather than just coefficient identities.

Chapter 4

Singularity Analysis

Chapter 2 established the basic relationship between generating functions and asymptotics: if $A(z) = \sum_{n\geq 0} a_n z^n$ has radius of convergence R, then the sequence a_n grows exponentially at rate $1/R$. That is the coarsest possible information about the coefficients. This chapter refines it considerably. The central insight, due to Flajolet and Odlyzko [17], is that the *type* of the dominant singularity of A on its circle of convergence determines not only the exponential growth rate ρ^{-n} (where $\rho = R$) but also the subexponential correction factor and the multiplicative constant. The full asymptotic takes the form $a_n \sim Cn^{\alpha-1}\rho^{-n}$, and reading off C, α, and ρ is a matter of inspecting the local behavior of A near its singularity. This is the Flajolet–Odlyzko transfer theorem, the workhorse of analytic combinatorics [18].

4.1 Exponential Growth Rate

The Cauchy–Hadamard formula gives $1/R = \limsup_{n\to\infty} |a_n|^{1/n}$. Rephrasing: if $\rho = R$ is the radius of convergence, then a_n grows no faster than $(\rho^{-1} + \varepsilon)^n$ for any $\varepsilon > 0$, and ρ^{-n} is the sharpest exponential bound. More precisely, $|a_n| \leq (\rho^{-1}+\varepsilon)^n$ eventually, and for any $C > 1$ the bound $|a_n| = \rho^{-n} \cdot o(C^n)$ holds. The factor ρ^{-n} is called the *exponential part* of the asymptotic. What ρ does *not* tell us is any polynomial or logarithmic correction: two sequences $n^{1/2}\rho^{-n}$ and $n^{-3/2}\rho^{-n}$ have the same exponential growth rate but vastly different behaviors for finite n. The polynomial correction is what singularity analysis supplies.

4.2 Dominant Singularities

Definition 4.1. A *dominant singularity* of a power series $A(z)$ with radius of convergence ρ is a point ζ with $|\zeta| = \rho$ at which A fails to extend analytically.

Every power series has at least one dominant singularity, since A cannot be continued to an entire function unless it converges everywhere. The analytic behavior of A near each dominant singularity contributes an oscillatory term to a_n. When there are multiple dominant singularities $\zeta_1, \ldots, \zeta_k$ on the circle $|z| = \rho$, the coefficient a_n is a superposition of terms $c_j\zeta_j^{-n}$, and since $|\zeta_j^{-n}| = \rho^{-n}$ for all j, these terms interfere, producing oscillations rather than clean growth.

A simple illustration: the generating function $1/(1 - z^2)$ has radius of convergence 1, with two dominant singularities at $z = 1$ and $z = -1$. Partial fractions give

$$\frac{1}{1-z^2} = \frac{1}{2}\left(\frac{1}{1-z} + \frac{1}{1+z}\right),$$

so the coefficient of z^n is $\frac{1}{2}(1 + (-1)^n)$, which is 1 for even n and 0 for odd n. The coefficient sequence oscillates with period 2 rather than growing monotonically, which is the characteristic

signature of two dominant singularities with opposite signs. No single real exponential $C\rho^{-n}$ approximates these coefficients uniformly.

For the remainder of this chapter, we restrict to the *aperiodic case*: A has a unique dominant singularity, located at a positive real point $z = \rho$. This is the generic situation for combinatorial classes defined by positive structural rules, and it is where the transfer theorem applies cleanly.

4.3 Rational Generating Functions

When $A(z) = P(z)/Q(z)$ is rational with $\deg P < \deg Q$, the partial fraction decomposition gives an exact formula for every coefficient. If Q has zeros $z_1, \ldots, z_r$ with multiplicities $m_1, \ldots, m_r$, then

$$a_n = \sum_{i=1}^{r} \mathrm{Pol}_i(n) \cdot z_i^{-n},$$

where Pol_i is a polynomial of degree $m_i - 1$ in n. The dominant contribution comes from the zero of Q of smallest modulus, say $z_1 = \rho$ with multiplicity m. All other terms are $O(|z_i|^{-n}) = O((\rho + \delta)^{-n})$ for some $\delta > 0$, which is exponentially smaller. Consequently

$$a_n \sim C n^{m-1} \rho^{-n}$$

as $n \to \infty$, for an explicit constant C determined by the Laurent expansion of A at $z = \rho$.

Example 4.2 (Fibonacci numbers)**.** Let $A(z) = z/(1 - z - z^2)$, the generating function for the Fibonacci sequence established in Chapter 1. The denominator $1 - z - z^2$ factors as $-(z - 1/\varphi)(z + \varphi)$, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio. The two poles are at $z = 1/\varphi \approx 0.618$ and $z = -\varphi \approx -1.618$. Since $1/\varphi < \varphi$, the dominant pole is $z = 1/\varphi$, and it is simple, so $m = 1$.

The partial fraction decomposition was already carried out in Chapter 1 using the factorization $1 - z - z^2 = (1 - \varphi z)(1 - \hat{\varphi} z)$, where $\hat{\varphi} = (1 - \sqrt{5})/2$. The result is $A(z) = (1/\sqrt{5})/(1 - \varphi z) - (1/\sqrt{5})/(1 - \hat{\varphi} z)$, giving $C = 1/\sqrt{5}$. The transfer from the rational case then gives

$$F_n \sim \frac{\varphi^n}{\sqrt{5}},$$

which is Binet's formula to leading order. (The subleading term from the pole at $-\varphi$ is $(-1)^{n+1}/(\sqrt{5}\,\varphi^n)$, which decays exponentially and contributes less than $1/2$ in absolute value for all $n \geq 1$, so the round-to-nearest interpretation recovers the exact formula.)

4.4 Algebraic Singularities and the Transfer Theorem

Not every generating function arising from combinatorics is rational. The Catalan generating function $C(z) = (1 - \sqrt{1 - 4z})/(2z)$ has a square-root branch point, which is neither a pole nor a removable singularity. Binary trees, general planar maps, and most classes defined by functional equations of the form $T(z) = z \cdot \phi(T(z))$ give rise to algebraic singularities of this type [13]. The appropriate tool is the Flajolet–Odlyzko transfer theorem.

First we need a geometric notion. Intuitively, the challenge near a singularity ρ is that A may fail to be analytic exactly at ρ while remaining analytic on a region extending slightly beyond the circle $|z| = \rho$ in all other directions. The Δ-domain formalizes this.

Definition 4.3. For $\rho > 0$, $\phi \in (0, \pi/2)$, and $\eta > \rho$, the Δ-*domain at* ρ is the set

$$\Delta(\rho, \phi, \eta) = \{z \in \mathbb{C} : |z| < \eta,\ z \neq \rho,\ |\arg(z - \rho)| > \phi\}.$$

We say A is Δ-*analytic at* ρ if it is analytic in some $\Delta(\rho, \phi, \eta)$.

Concretely, a Δ-domain is a disk of radius $\eta > \rho$ centered at the origin, with a small wedge-shaped slit removed around the positive real ray at and beyond ρ. The function A need not extend to ρ itself, but it must be analytic in the slit disk. One may picture shining a narrow flashlight from the origin outward: the beam extends slightly past ρ in all directions except along the slit, where ρ itself blocks the view.

Theorem 4.4 (Flajolet–Odlyzko Transfer Theorem, [17])**.** *Let $\rho > 0$ and suppose $A(z)$ is Δ-analytic at ρ. Suppose further that as $z \to \rho$ within the Δ-domain,*

$$A(z) = C\left(1 - \frac{z}{\rho}\right)^{-\alpha}(1 + o(1))$$

for some $\alpha \in \mathbb{R} \setminus \mathbb{Z}_{\leq 0}$ and $C \neq 0$. Then

$$[z^n]A(z) \sim \frac{C}{\Gamma(\alpha)}\, n^{\alpha-1}\, \rho^{-n}$$

as $n \to \infty$.

The proof is a beautiful application of contour deformation from Ch2, and we sketch it here to give the reader confidence that the formula is not magic. Start with the Cauchy coefficient formula from Chapter 2:

$$a_n = \frac{1}{2\pi i}\oint_{|z|=r} \frac{A(z)}{z^{n+1}}\, dz,$$

where the contour is a small circle of radius $r < \rho$ centered at the origin. By the contour deformation principle (Chapter 2), we may deform this circle outward to any path that stays within the region where A is holomorphic. Because A is Δ-analytic at ρ, we can push the contour past the circle $|z| = \rho$ everywhere *except* near the singularity, wrapping it tightly around the branch cut or pole at $z = \rho$; this is the *Hankel contour*. The Hankel contour consists of a large outer arc plus two segments running along opposite sides of the branch cut connecting ρ to infinity. By the ML bound from Chapter 2, the contribution from the large outer arc is at most (length) $\times$ (maximum of $|A(z)/z^{n+1}|$ on the arc), and since $|z^{n+1}|$ grows exponentially in n on the arc, this contribution decays exponentially and is negligible. The two segments hugging the singularity are what survive, and on them the integrand is governed entirely by the local behavior $A(z) \approx C(1 - z/\rho)^{-\alpha}$. Substituting $z = \rho(1 - t/n)$ and rescaling, the integral over the two segments reduces to an Euler-type integral that evaluates to $\int_0^\infty t^{\alpha-1}e^{-t}\, dt = \Gamma(\alpha)$, with the ρ^{-n} and $n^{\alpha-1}$ factors emerging from the change of variables. Dividing out $\Gamma(\alpha)$ produces the stated formula. Full details appear in [18], Chapter VI.

What matters for applications is the structural decomposition of the conclusion:

1. The exponential factor ρ^{-n} is determined entirely by the *location* of the singularity, not its type. Moving the singularity closer to or farther from the origin changes the exponential growth rate.

2. The polynomial factor $n^{\alpha-1}$ is determined by the *order of contact* of A with its singularity. A large positive α means A blows up strongly at ρ, and the coefficients grow correspondingly fast. A negative α means A vanishes at ρ, and the coefficients decay polynomially relative to ρ^{-n}.

3. The constant $C/\Gamma(\alpha)$ is fixed by the local behavior of A near ρ; it is independent of n and encodes global information about the combinatorial class through the single number C.

The Δ-analyticity hypothesis is essential. It rules out situations where a second dominant singularity (not at ρ) interferes. In the aperiodic setting, one typically verifies Δ-analyticity by confirming that all other singularities of A lie strictly outside $|z| = \rho$ or can be analytically continued around.

4.5 Catalogue of Exponents

The transfer theorem applies across a range of singularity types. We record the most important cases.

Poles. If $\alpha = k$ is a positive integer, then $(1 - z/\rho)^{-k}$ has a pole of order k at ρ. The gamma function value is $\Gamma(k) = (k-1)!$, so

$$[z^n]\frac{C}{(1-z/\rho)^k} \sim \frac{C}{(k-1)!}\, n^{k-1}\, \rho^{-n}.$$

For $k = 1$ (simple pole) this gives $a_n \sim C\rho^{-n}$ with no polynomial factor. For $k = 2$ (double pole), $a_n \sim Cn\rho^{-n}$. This is consistent with the partial fraction analysis of Section 3.

Square-root singularity. If $A(z) \sim C\sqrt{1 - z/\rho}$ near ρ, then $\alpha = -1/2$ and A vanishes like a square root. Since $\Gamma(-1/2) = -2\sqrt{\pi}$,

$$a_n \sim \frac{C}{-2\sqrt{\pi}}\, n^{-3/2}\, \rho^{-n} = -\frac{C}{2\sqrt{\pi}}\, n^{-3/2}\, \rho^{-n}.$$

The $n^{-3/2}$ decay relative to ρ^{-n} is characteristic; the negative sign in $\Gamma(-1/2)$ combines with the sign of C to give a positive coefficient sequence when $C < 0$ (which is the typical case when A approaches 0 at ρ from below).

Inverse square-root singularity. If $A(z) \sim C/\sqrt{1 - z/\rho}$, then $\alpha = 1/2$. Since $\Gamma(1/2) = \sqrt{\pi}$,

$$a_n \sim \frac{C}{\sqrt{\pi}}\, n^{-1/2}\, \rho^{-n}.$$

The $n^{-1/2}$ polynomial prefactor is the signature of an inverse square-root singularity. This arises, for instance, in counting lattice paths with unconstrained endpoint.

Logarithmic singularity. The function $\log(1/(1 - z/\rho))$ is a limiting case outside the strict hypotheses of the theorem as stated (since $\alpha \to 0$ as $z \to \rho$). A direct computation gives $[z^n]\log(1/(1 - z/\rho)) = \rho^{-n}/n$, a $1/n$ decay that sits between the $\alpha \to 0$ and $\alpha \to 1$ cases.

4.6 Worked Example: Catalan Numbers

Example 4.5 (Catalan numbers)**.** In Chapter 1 we derived the generating function for the Catalan numbers $C_n = \binom{2n}{n}/(n+1)$:

$$C(z) = \frac{1 - \sqrt{1 - 4z}}{2z}.$$

The function $\sqrt{1 - 4z}$ has a branch point at $z = 1/4$, which is the dominant singularity of $C(z)$. All other singularities (the apparent pole at $z = 0$ is removable) lie no closer to the origin. Thus $\rho = 1/4$.

To apply the transfer theorem, expand $C(z)$ near $z = 1/4$. Let $u = 1 - 4z$, so $u \to 0$ as $z \to 1/4$. Then

$$C(z) = \frac{1 - \sqrt{u}}{2z} = \frac{1 - \sqrt{u}}{(1-u)/2} = \frac{2(1 - \sqrt{u})}{1 - u}.$$

Expanding $(1 - u)^{-1} = 1 + u + O(u^2)$ and keeping terms through $O(\sqrt{u})$:

$$C(z) = 2(1 - \sqrt{u})(1 + u + O(u^2)) = 2 - 2\sqrt{u} + O(u) = 2 - 2\sqrt{1 - 4z} + O(1 - 4z).$$

Since $\sqrt{1-4z} = \sqrt{4(1/4-z)} = 2\sqrt{1/4-z}$, we can also write $\sqrt{1-4z} = \sqrt{4}\sqrt{1-z/\rho} = 2(1-z/\rho)^{1/2}$, so

$$C(z) = 2 - 4(1-z/\rho)^{1/2} + O(1-z/\rho).$$

The constant term $2 = C(1/4)$ does not contribute to the asymptotics of coefficients. The singular part is $-4(1-z/\rho)^{1/2}$, corresponding to $\alpha = 1/2$ and $C = -4$. However, this is not the form $(1-z/\rho)^{-\alpha}$; rather it is $(1-z/\rho)^{+1/2}$, corresponding to $-\alpha = 1/2$, i.e., $\alpha = -1/2$ in the theorem's notation. We have $C = -4$ and $\alpha = -1/2$.

A naive first attempt at applying the transfer theorem might use $C_{\text{sing}} = -4$, but this would give $C_n \sim 2 \cdot 4^n/(\sqrt{\pi}\, n^{3/2})$, which is off by a factor of 2 from the classical asymptotic. The error lies in the change of variables. The full expansion is $C(z) = 2 - 2\sqrt{1-4z} + O(1-4z)$, and the coefficient of $(1-4z)^{1/2}$ is -2, not -4. Since $1-4z = 4(1/4-z) = 4\rho(1-z/\rho)$, we have $\sqrt{1-4z} = 2\sqrt{\rho}\sqrt{1-z/\rho} = \sqrt{1-z/\rho}$ (because $\rho = 1/4$, so $\sqrt{4\rho} = 1$). Therefore $C(z) \sim 2 - 2(1-z/\rho)^{1/2}$, giving $C_{\text{sing}} = -2$ and $\alpha = -1/2$. Applying the transfer theorem:

$$C_n \sim \frac{-2}{\Gamma(-1/2)}\, n^{-3/2}\, 4^n = \frac{-2}{-2\sqrt{\pi}}\, n^{-3/2}\, 4^n = \frac{4^n}{\sqrt{\pi}\, n^{3/2}}.$$

This agrees exactly with the asymptotic derived from Stirling's formula applied to the exact formula $C_n = \binom{2n}{n}/(n+1)$. The moral is that the change of variables from $(1-4z)$ to $(1-z/\rho)$ must be tracked carefully; a factor of $\sqrt{4\rho}$ is easy to drop. The $n^{-3/2} \cdot 4^n$ form is the prototype of what Chapter 5 will call the $n^{-3/2}$ *law*.

4.7 Summary and Preview

The Flajolet–Odlyzko transfer theorem completes the central algorithmic workflow of analytic combinatorics. Given a combinatorial class $\mathcal{A}$: (i) apply the symbolic method (Chapter 3) to translate the structural decomposition of $\mathcal{A}$ into a functional equation for the generating function $A(z)$; (ii) analyze that equation to locate the dominant singularity ρ and determine the local exponent α; (iii) read off the asymptotic $a_n \sim (C/\Gamma(\alpha))\, n^{\alpha-1}\, \rho^{-n}$ directly from the singularity data. No other computation is required.

The power of this approach lies in what it unifies. Rational GFs yield polynomial-times-exponential asymptotics via poles; algebraic GFs yield the same form with non-integer exponents via branch points. Both cases are handled by the single theorem [17], with the difference in exponent type α being the only distinction.

Chapter 5 will examine the class of context-free grammars in detail. A fundamental theorem of analytic combinatorics states that any unambiguous context-free grammar gives rise to an algebraic generating function, and that in the aperiodic case the dominant singularity is always a square-root branch point, hence $\alpha = -1/2$ universally. This means the coefficient of any unambiguous context-free language satisfies $a_n \sim C \cdot n^{-3/2} \cdot \rho^{-n}$ (with ρ and C depending on the grammar, but the exponent $-3/2$ being universal), a remarkable structural prediction. The same $n^{-3/2}$ law turns out to be intimately connected with the length distribution of sentences generated by language models, a connection we will develop in Parts III–V.

Further Reading

- [17] — The original paper proving the transfer theorem in the form used throughout this chapter; contains the Hankel-contour proof and the full catalogue of singular types, and is more concise than the book-length treatment in [18].
- [18], Chapter VI — The textbook development of singularity analysis, including the Δ-domain framework, a detailed treatment of the Hankel contour argument, and worked examples covering random trees, urns, and maps.

- [13] — A monograph on random trees whose first half is an accessible introduction to the singularity analysis of algebraic and D-finite generating functions; Chapter 2 is especially useful for understanding how the characteristic system of the present chapter extends to systems of equations.
- **A. Odlyzko**, "Asymptotic enumeration methods," in *Handbook of Combinatorics*, Vol. 2 (Elsevier, 1995) — A comprehensive survey of coefficient-extraction techniques from generating functions, covering the transfer theorem, saddle-point methods, and the Darboux method; excellent for seeing the transfer theorem in context alongside other asymptotic tools.
- [3] — Proves universal limit laws for coefficients of algebraic generating functions beyond the leading term; relevant for understanding why the $n^{-3/2}$ exponent governs not just asymptotics but also distributional limits of combinatorial parameters.

Chapter 5

Algebraic Generating Functions and the $n^{-3/2}$ Law

Chapter 4 equipped us with the Flajolet–Odlyzko transfer theorem: once we know the singular behaviour of a generating function near its dominant singularity, we can read off the asymptotics of its coefficients with minimal effort. This chapter deploys that theorem to prove one of the most striking results in analytic combinatorics. For an unambiguous context-free grammar satisfying mild irreducibility conditions, the number of distinct words of length n satisfies

$$c_n \sim C\, n^{-3/2}\, \rho^{-n}$$

for a grammar-specific constant $C > 0$ and exponential rate $\rho > 0$. The crucial point is the universality of the exponent $-3/2$: it does not depend on the grammar, only on the fact that the language is context-free. The rest of Part I is devoted to understanding why this exponent is forced.

5.1 Unambiguous Context-Free Grammars

A *context-free grammar* $\mathcal{G} = (V, \Sigma, S, \mathcal{P})$ consists of a finite set V of nonterminal symbols, a finite alphabet Σ of terminal symbols disjoint from V, a designated start symbol $S \in V$, and a finite set $\mathcal{P}$ of productions of the form $A \to \alpha$ where $A \in V$ and $\alpha \in (V \cup \Sigma)^*$. The language $L(\mathcal{G}) \subseteq \Sigma^*$ is the set of all terminal strings derivable from S by successive application of productions. A grammar is *unambiguous* if every string $w \in L(\mathcal{G})$ admits a unique leftmost derivation, or equivalently a unique parse tree.

Ambiguity is the enemy of enumeration: if a string can be parsed in multiple ways, a naive translation of the grammar into generating-function equations overcounts. Unambiguity ensures that the generating function for $L(\mathcal{G})$, counting words by length, can be extracted directly from the grammar's combinatorial structure. Part II (Chapters 6–9) develops the theory of context-free languages in depth; here we need only the fact that unambiguous grammars translate productions into polynomial equations on generating functions.

5.2 From Productions to Polynomial Equations

The translation from grammar to generating function proceeds production by production. Each production $A \to \alpha$ contributes a monomial: a terminal symbol a contributes a factor of z, and a nonterminal B contributes a factor of $B(z)$, the ordinary generating function counting strings derivable from B by length. Summing over all productions with left-hand side A yields a polynomial expression for $A(z)$.

Example 5.1 ($a^n b^n$)**.** Consider the grammar with a single nonterminal S and productions $S \to a\,S\,b \mid \varepsilon$. This generates the language $\{a^n b^n : n \geq 0\}$. The production $S \to a\,S\,b$ contributes $z \cdot S(z) \cdot z = z^2 S(z)$; the production $S \to \varepsilon$ contributes 1. The functional equation is

$$S(z) = 1 + z^2 S(z),$$

with solution $S(z) = 1/(1 - z^2)$. This is rational, so the transfer theorem gives $s_n = 1$ for even $n \geq 0$ and $s_n = 0$ otherwise — a trivial but clean verification.

Example 5.2 (Plane trees)**.** Plane trees (rooted, ordered trees) satisfy the structural decomposition $T \to \bullet \cdot \mathrm{SEQ}(T)$: a tree is a root followed by an ordered sequence of subtrees. This translates to

$$T(z) = \frac{z}{1 - T(z)},$$

or equivalently the polynomial equation

$$T(z)^2 - T(z) + z = 0.$$

The quadratic formula gives two branches; the branch analytic at $z = 0$ is

$$T(z) = \frac{1 - \sqrt{1 - 4z}}{2}.$$

The coefficients are $[z^n]T(z) = C_{n-1}$, the $(n-1)$-th Catalan number, for $n \geq 1$.

The general principle is now clear. Given an unambiguous grammar in Chomsky normal form (productions of the form $A \to BC$ or $A \to a$), each production contributes a monomial that is at most quadratic in the nonterminal generating functions. Summing over all productions for each nonterminal yields a polynomial system. The system can be solved or, failing a closed form, analysed via elimination theory: successively eliminate nonterminals until a single polynomial equation $P(z, S(z)) = 0$ remains for the start symbol.

5.3 The Chomsky–Schützenberger Theorem

The passage from grammar to polynomial equation is the combinatorial content of the following fundamental result.

Theorem 5.3 (Chomsky–Schützenberger, 1963)**.** *Let L be a context-free language generated by an unambiguous grammar, and let c_n denote the number of words of length n in L. Then the ordinary generating function*

$$L(z) = \sum_{n \geq 0} c_n\, z^n$$

is algebraic over $\mathbb{Q}(z)$: there exists a nonzero polynomial $P \in \mathbb{Q}[z, y]$ such that $P(z, L(z)) = 0$.

The proof proceeds by induction on the size of the grammar [11]. Convert the grammar to Chomsky normal form; for each nonterminal A, write $A(z) = \sum_{\text{productions } A \to \alpha} \phi(\alpha, z)$ where ϕ encodes the monomial. The resulting polynomial system over $\mathbb{Q}[z]$ has a unique power-series solution (by a bootstrapping argument on coefficients), and elimination theory applied to a polynomial system over $\mathbb{Q}(z)$ always yields an algebraic equation for the start symbol. The algebraicity is a purely structural consequence of the grammar's finiteness.

Two points deserve emphasis. First, Theorem 5.3 is a counting result: it concerns the number of distinct words of each length, not the structure of individual derivations. Second, the theorem by itself says nothing about the growth rate of c_n; that requires additional analytic input. The asymptotic chain is explicit: algebraicity comes from Chomsky–Schützenberger; the asymptotic form of the coefficients then follows from the Flajolet–Odlyzko transfer theorem [17], via a structural analysis of the singularities of algebraic functions. The bridge between the two is Puiseux's theorem.

5.4 Puiseux's Theorem and the Structure of Singularities

Algebraic functions do not have arbitrary singularity types. Every singularity of an algebraic function is a branch point of rational order, and the local expansion takes a precise form.

Theorem 5.4 (Puiseux)**.** *Let $F(z)$ be algebraic, with $P(z, F(z)) = 0$ for some nonzero polynomial $P \in \mathbb{C}[z, y]$. At any singularity ρ of F, there exists a positive integer d (the branching order) and an integer k_0 such that F admits a local Puiseux expansion*

$$F(z) = \sum_{k \geq k_0} a_k \, (1 - z/\rho)^{k/d}$$

convergent in a slit neighbourhood of ρ.

To understand why such an expansion exists, consider what happens geometrically near a branch point ρ. The function $F(z)$ is multi-valued there: just as $\sqrt{z}$ takes two values for each nonzero z, an algebraic function satisfying a degree-d polynomial equation can take up to d values near ρ. Puiseux's theorem says these values are not chaotic — they are organized as a power series in the single variable $w = (z - \rho)^{1/d}$. Concretely, near the branch point the function "sees" a d-sheeted cover of the complex plane, and on each sheet it has a perfectly well-behaved Laurent series, but written in w rather than $z - \rho$. For the generating functions arising from quadratic equations (such as the Catalan generating function $T(z)$), the polynomial P has degree 2 in F, so generically $d = 2$: the branching is of square-root type, and the two sheets are exchanged by analytic continuation around ρ. The proof uses Newton's polygon method, a visual algorithm that reads d directly from the Newton diagram of $P(z, F) = 0$ without solving the equation; see [18] Chapter VII for an explicit account.

This is a classical consequence of Newton's polygon method and the preparation theorem; see [18] for a combinatorial treatment. The expansion is a power series in the fractional variable $(1 - z/\rho)^{1/d}$, so singularities are always branch points, never essential singularities or poles of irrational order.

For asymptotic coefficient extraction, what matters is the leading exponent $\alpha = k_0/d$ of the singular part. If $k_0 \geq 0$, the function is bounded near ρ and the dominant singular contribution comes from the smallest $k \geq 0$ with $k/d \notin \mathbb{Z}$ (the non-analytic part). The key question is: what is the generic value of this leading fractional exponent for generating functions arising from context-free grammars?

5.5 Why Square Roots Are Generic

The answer is that $d = 2$ and $k_0 = 1$ generically, i.e., the dominant singularity is a square-root branch point. To see why, consider the implicit function setup. The generating function $L(z)$ satisfies $P(z, L(z)) = 0$. The implicit function theorem guarantees that $L(z)$ is analytic at z_0 as long as $P_F(z_0, L(z_0)) := \partial P/\partial F\big|_{(z_0, L(z_0))} \neq 0$. A singularity at ρ therefore requires

$$P(\rho, L(\rho)) = 0 \quad \text{and} \quad P_F(\rho, L(\rho)) = 0.$$

This is the *characteristic system.* Now Taylor-expand P around the point $(\rho, L(\rho))$. Writing $\Delta z = z - \rho$ and $\Delta F = F(z) - L(\rho)$, we have

$$0 = P(z, F(z)) \approx P_z \, \Delta z + P_F \, \Delta F + \frac{1}{2} P_{FF} \, (\Delta F)^2 + \cdots$$

Since $P_F = 0$ at the singularity, the linear term in ΔF vanishes, and to leading order:

$$P_z \, \Delta z + \frac{1}{2} P_{FF} \, (\Delta F)^2 \approx 0.$$

Solving for ΔF:

$$\Delta F \sim \sqrt{-\frac{2P_z}{P_{FF}}} \cdot \sqrt{\Delta z} = \sqrt{-\frac{2P_z}{P_{FF}}} \cdot \sqrt{\rho} \cdot \sqrt{1 - z/\rho} \cdot (-1).$$

Provided $P_{FF}(\rho, L(\rho)) \neq 0$ (the genericity condition: the characteristic system does not degenerate further), the leading singular behaviour of F near ρ is exactly a square root. In terms of the Puiseux expansion, $d = 2$ and the expansion begins at $k = 1$, giving $(1 - z/\rho)^{1/2}$.

This mechanism is the geometric reason for universality. The characteristic system imposes two equations on two unknowns $(\rho, L(\rho))$; generically this determines a point, and at that point the second-order analysis above forces a square root. Higher-order degeneracies (e.g., $P_{FF} = 0$ as well) require an additional coincidence and are non-generic.

For context-free languages, this genericity is made precise by the Drmota–Lalley–Woods theorem, extended and quantified by Banderier and Drmota [13, 3]: for an irreducible, strongly connected, and aperiodic grammar (conditions ensuring the generating function has a single dominant singularity on its circle of convergence, and that the grammar cannot be decomposed into non-interacting parts), the dominant singularity of $L(z)$ is a square-root branch point. Near that singularity,

$$L(z) = g(z) - h(z)\sqrt{1 - z/\rho},$$

where g and h are analytic at ρ and $h(\rho) \neq 0$. The function $g(\rho)$ is the finite limit of $L(z)$ as $z \to \rho$; the square-root term carries all the singular asymptotic information.

5.6 Applying the Transfer Theorem

We now have everything needed to apply the Flajolet–Odlyzko transfer theorem [17]. Write the singular expansion as

$$L(z) = g(z) - h(z)\sqrt{1 - z/\rho} = g(z) - h(z)\,(1 - z/\rho)^{1/2}.$$

The analytic part $g(z)$ contributes coefficients that are exponentially smaller than those from the branch-cut term (its singularities are further from the origin, by the irreducibility assumption). The transfer theorem applies to the singular factor $(1 - z/\rho)^{1/2}$, which corresponds to $\alpha = -1/2$ in the Chapter 4 convention $(1 - z/\rho)^{-\alpha}$, giving

$$[z^n](1 - z/\rho)^{1/2} \sim \frac{-1/2}{\Gamma(-1/2)} \cdot \frac{\rho^{-n}}{n^{3/2}} \cdot (-1) = \frac{\rho^{-n}}{2\sqrt{\pi}\, n^{3/2}},$$

using $\Gamma(-1/2) = -2\sqrt{\pi}$ and $\binom{1/2}{n}(-1)^{n-1} \sim (2\sqrt{\pi})^{-1} n^{-3/2}$ for large n. Multiplying by $-h(\rho)$ from the expansion of L:

$$c_n = [z^n]L(z) \sim \frac{h(\rho)}{2\sqrt{\pi}}\, n^{-3/2}\, \rho^{-n}.$$

This is the $n^{-3/2}$ law. We state it as a theorem.

Theorem 5.5 ($n^{-3/2}$ law for context-free languages)**.** *Let L be an unambiguous context-free language generated by an irreducible, strongly connected, and aperiodic grammar. Then there exist constants $C > 0$ and $\rho > 0$ such that the number of words of length n in L satisfies*

$$c_n \sim C\, n^{-3/2}\, \rho^{-n} \quad \text{as } n \to \infty.$$

The exponential rate ρ is grammar-specific; the polynomial exponent $-3/2$ is universal across all grammars satisfying the stated conditions.

The universality is the content of the theorem: ρ encodes the particular language, but the exponent $-3/2$ does not. It is forced by the combination of algebraicity (from the Chomsky–Schützenberger theorem) and the genericity of square-root singularities (from the characteristic system analysis).

5.7 Worked Example: Plane Trees Revisited

Example 5.6 (Plane trees, continued from Example 5.2)**.** Recall the equation $T(z)^2 - T(z) + z = 0$, so $P(z, F) = F^2 - F + z$. The characteristic system is

$$P(z, F) = 0 \quad \text{and} \quad P_F(z, F) = 2F - 1 = 0.$$

From $P_F = 0$ we get $F = 1/2$. Substituting into $P = 0$:

$$\frac{1}{4} - \frac{1}{2} + \rho = 0 \implies \rho = \frac{1}{4}.$$

The local expansion follows from the quadratic formula: since $P_{FF} = 2$ and $P_z = 1$, the formula $\Delta F \approx \sqrt{-P_z/(\frac{1}{2}P_{FF})} \cdot \sqrt{\Delta z}$ gives

$$T(z) = \frac{1}{2} - \frac{1}{2}\sqrt{1 - 4z}.$$

Here $g(z) = 1/2$ and $h(z) = 1/2$, so $h(\rho) = 1/2$. The transfer theorem yields

$$[z^n]T(z) \sim \frac{1/2}{2\sqrt{\pi}}\, n^{-3/2}\, 4^n = \frac{4^n}{4\sqrt{\pi}\, n^{3/2}}.$$

Since $[z^n]T(z) = C_{n-1} = \binom{2n-2}{n-1}/n$, one can verify directly via Stirling's approximation that $C_{n-1} \sim 4^{n-1}/(\sqrt{\pi}(n-1)^{3/2}) \cdot 2 \sim 4^n/(4\sqrt{\pi}n^{3/2})$, confirming the transfer theorem computation to leading order.

The example is instructive for a second reason. The Catalan numbers are among the most thoroughly studied sequences in combinatorics, and their $n^{-3/2}$ polynomial prefactor is well known. Theorem 5.5 says that this is not a coincidence special to plane trees: it is the inevitable consequence of any generating function satisfying a polynomial equation with a square-root dominant singularity.

5.8 Implications for the Rest of the Book

The $n^{-3/2}$ law provides one of analytic combinatorics' most compact predictions, and it has empirical content. Suppose one observes a combinatorial process producing strings of various lengths, and one fits the word-count histogram to the model $c_n \approx C\rho^{-n}n^{\beta}$. Then:

- $\beta = -3/2$ is consistent with an underlying unambiguous context-free structure.
- $\beta \neq -3/2$ is evidence against such a structure. For instance, $\beta = -1/2$ would suggest a different type of algebraic singularity (a branching order still equal to 2 but with the singular part entering at the $k = 0$ level), while rational generating functions produce purely exponential growth with no polynomial correction, corresponding to $\beta = 0$ in the logarithmic regime.

This turns the transfer theorem into a hypothesis-testing instrument: measure the polynomial exponent β from data, compare to $-3/2$, and draw structural inferences. Part V of this book takes this seriously in the context of large language models. A central open question (Gap 5 of Part V) is whether LLMs, which have been shown to be capable of learning context-free grammars under appropriate training conditions, produce output-length distributions whose polynomial prefactor is consistent with the $n^{-3/2}$ signature. If the answer is yes, it suggests the internal representations learned by LLMs preserve the algebraic character of the target grammar. If not, the deviation itself is informative.

The structural message of Theorem 5.5 also connects to the broader programme of [18]: analytic combinatorics is a machine that converts combinatorial structure into asymptotic information, and the most powerful lever in the machine is the identification of dominant singularity type. Square-root singularities are not special to context-free languages; they arise wherever a self-referential equation fails the implicit function theorem at first order and not at second. But context-free languages are one of the most natural and important settings in which this occurs, which is why the $-3/2$ exponent has become something of a hallmark.

Remark 5.7. The irreducibility and aperiodicity conditions in Theorem 5.5 are not vacuous. A periodic grammar (for instance, one generating only strings of even length) produces $c_n = 0$ for odd n, breaking the simple asymptotic formula. The correct treatment involves a periodicised version of the transfer theorem; see [18]. For aperiodic grammars, the formula holds as stated.

Summary of Part I

We now have the complete analytic-combinatorics toolkit assembled over Chapters 1–5. The workflow is four steps: (i) encode a combinatorial class via the symbolic method (Chapter 3) to obtain a functional equation for its generating function; (ii) identify the dominant singularity of that generating function, using the implicit-function characteristic system or direct computation; (iii) extract the local singular form, invoking Puiseux's theorem when the generating function is algebraic; (iv) apply the Flajolet–Odlyzko transfer theorem (Chapter 4) to convert the singular form into coefficient asymptotics. The present chapter has shown that for unambiguous context-free languages this entire pipeline collapses to a single universal output: the $n^{-3/2}$ polynomial prefactor is a theorem, not a coincidence.

The remainder of the book applies this machinery beyond the purely combinatorial setting. Part II develops the formal language theory that contextualises Theorem 5.5. Parts III and IV extend the framework to probabilistic generative models, where the generating function becomes a probability-generating function and the asymptotics must be reinterpreted accordingly. Part V confronts the theory with neural language models, asking which analytic-combinatoric signatures survive the transition from symbolic to statistical generation. The tools are in place; the questions are open.

Further Reading

- [11] — The original paper proving that the generating function of any unambiguous context-free language is algebraic; essential reading for understanding the historical context and the original inductive proof that underpins Theorem 5.3.
- [13] — A monograph devoted to random trees and context-free structures; Chapters 2–3 cover Puiseux expansions, the characteristic system, and the square-root universality theorem in a self-contained and detailed way, making it the most direct complement to the present chapter.
- [3] — Establishes a general framework ("algebraic systems") subsuming many classes of context-free generating functions, and proves universal distributional limit laws; the paper is the state of the art on the $n^{-3/2}$ universality beyond leading-order asymptotics.
- [18], Chapter VII — Covers algebraic functions and the Newton polygon method for computing Puiseux expansions algorithmically; complements the sketch given in Theorem 5.4 with explicit procedures for finding d and the coefficients a_k.
- **M. Drmota, É. Fusy, M. Kang, V. Kraus, and J. Rué**, "Asymptotic study of subcritical graph classes," *SIAM Journal on Discrete Mathematics* 25 (2011), 1615–1651

— Illustrates how the characteristic-system approach of this chapter extends to more complex algebraic systems (subcritical graph families), where the same $n^{-3/2}$ universality holds under analogous irreducibility conditions.

Part II

Formal Languages and Probabilistic Sources

Chapter 6

Formal Languages and the Chomsky Hierarchy

Part I developed analytic combinatorics in a setting-agnostic way: a combinatorial class is simply a set of objects equipped with a notion of size, and the transfer theorems translate structural decompositions into functional equations for generating functions. This chapter grounds that framework in a concrete and historically fundamental family of classes, namely the *formal languages*—subsets of the free monoid Σ^* generated by grammars or recognized by machines. The structural organisation of such languages is the Chomsky hierarchy, a four-level stratification whose levels correspond, through the analytic-combinatorial lens, to progressively richer families of generating functions. The punchline, already foreshadowed in Chapter 5, is that regular languages have rational generating functions, unambiguous context-free languages have algebraic generating functions, and moving further up the hierarchy introduces genuinely transcendental and even exotic analytic behavior. Standard references for automata theory and formal languages are [23] and [51].

6.1 Alphabets, Strings, and Languages

Definition 6.1. An *alphabet* Σ is a finite nonempty set whose elements are called *symbols* or *letters*. A *string* over Σ is a finite sequence $w = \sigma_1\sigma_2\cdots\sigma_k$ with each $\sigma_i \in \Sigma$. The *length* of w is $|w| = k$. The unique string of length zero is the *empty string* ε.

The set Σ^* consists of all finite strings over Σ including ε; it is the free monoid generated by Σ under the operation of *concatenation* uv (juxtaposition). The subset $\Sigma^n \subset \Sigma^*$ collects strings of exactly length n, so $|\Sigma^n| = |\Sigma|^n$. We write $\Sigma^+ = \Sigma^* \setminus \{\varepsilon\}$ for the set of nonempty strings. A *language* over Σ is any subset $L \subseteq \Sigma^*$. The combinatorial quantity of primary interest is the sequence $(|L \cap \Sigma^n|)_{n\geq 0}$, which counts the words of each length; the generating function $L(z) = \sum_{n\geq 0} |L \cap \Sigma^n| z^n$ will be the central object of study.

6.2 Deterministic Finite Automata and Regular Languages

Definition 6.2. A *deterministic finite automaton* (DFA) is a tuple $M = (Q, \Sigma, \delta, q_0, F)$ where Q is a finite set of *states*, $\delta : Q \times \Sigma \to Q$ is the *transition function*, $q_0 \in Q$ is the *start state*, and $F \subseteq Q$ is the set of *accepting states*.

The transition function extends to strings via $\hat{\delta}(q, \varepsilon) = q$ and $\hat{\delta}(q, w\sigma) = \delta(\hat{\delta}(q, w), \sigma)$ for $\sigma \in \Sigma$. The language *accepted* by M is $L(M) = \{w \in \Sigma^* : \hat{\delta}(q_0, w) \in F\}$.

Example 6.3. Let $\Sigma = \{a, b\}$ and $L = \{w \in \Sigma^* : w \text{ contains an even number of } a\text{'s}\}$. The DFA with $Q = \{q_0, q_1\}$, start state q_0, $F = \{q_0\}$, and transitions $\delta(q_i, a) = q_{1-i}$ and $\delta(q_i, b) = q_i$ accepts exactly L. The two states track the parity of the a-count.

A *nondeterministic finite automaton* (NFA) relaxes δ to a relation $Q \times \Sigma \to 2^Q$ and allows ε-transitions. The subset construction converts any NFA with n states into a DFA with at most 2^n states accepting the same language, so the two models are equivalent in expressive power.

Definition 6.4. A language L is *regular* if $L = L(M)$ for some DFA M.

6.3 The Myhill–Nerode Theorem

The minimal DFA recognizing a language L is characterised by an intrinsic equivalence relation on strings. Define $x \sim_L y$ if and only if for every $z \in \Sigma^*$, $xz \in L \iff yz \in L$: two strings are equivalent when they have identical sets of future continuations leading to acceptance. This is the *Myhill–Nerode equivalence* of L.

Theorem 6.5 (Myhill–Nerode)**.** *A language L is regular if and only if $\sim_L$ has finitely many equivalence classes. When L is regular, the number of classes equals the number of states in the minimal DFA for L.*

The proof in the forward direction is straightforward: two strings reaching the same state of a DFA are $\sim_L$-equivalent, so the number of classes is at most $|Q|$. Conversely, if there are only finitely many classes, one constructs a DFA whose states are the equivalence classes, and verifies it accepts L.

The theorem immediately shows that the language $\{a^n b^n : n \geq 0\}$ is not regular: the strings $\varepsilon, a, aa, aaa, \ldots$ are pairwise $\sim_L$-inequivalent, since $a^i z \in L \iff z = b^i$ while $a^j z \in L \iff z = b^j$, so distinct powers of a have distinct future sets, giving infinitely many classes.

6.4 Regular Expressions

Regular expressions over Σ are built by the following inductive syntax: the constants $\emptyset$ (empty language), ε (language $\{\varepsilon\}$), and σ (language $\{\sigma\}$) for each $\sigma \in \Sigma$ are regular expressions; if R and S are regular expressions then so are $R + S$ (union), RS (concatenation), and R^* (Kleene star, zero or more concatenations of strings from R). The language denoted by a regular expression is defined inductively from these operations.

Theorem 6.6 (Kleene)**.** *A language is regular if and only if it is denoted by some regular expression.*

The equivalence is constructive in both directions: one converts a DFA to a regular expression by a state-elimination procedure, and one converts a regular expression to an NFA by structural induction. In the analytic-combinatorial vocabulary, the operations $+$, concatenation, and $*$ correspond exactly to disjoint union, the SEQ construction applied to a pair of classes, and SEQ applied to a single class without restriction on length; the close parallel between regular expressions and the symbolic method is visible here and will be developed further through weighted automata in Chapter 7.

6.5 The Pumping Lemma for Regular Languages

Theorem 6.7 (Pumping Lemma, Regular)**.** *If L is a regular language, there exists a constant $p \geq 1$ (the* pumping length*) such that for every $w \in L$ with $|w| \geq p$, there is a decomposition $w = xyz$ satisfying: (i) $|xy| \leq p$, (ii) $|y| \geq 1$, and (iii) $xy^k z \in L$ for every $k \geq 0$.*

The proof is a direct application of the pigeonhole principle: let $p = |Q|$ where Q is the state set of a DFA for L. Reading w from q_0, the machine visits $|w| + 1 \geq p + 1$ states, so by pigeonhole some state q is visited twice within the first $p + 1$ steps; take y to be the substring read between the two visits to q. Since the machine loops on q when reading y, repeating or deleting y does not change the final state, so $xy^k z \in L(M) = L$ for all $k \geq 0$.

To see that $L = \{a^n b^n : n \geq 0\}$ is not regular, suppose it were regular with pumping length p. Take $w = a^p b^p \in L$. The decomposition satisfies $|xy| \leq p$, so $y = a^j$ for some $j \geq 1$; but then $xy^2 z = a^{p+j} b^p \notin L$, a contradiction.

6.6 Context-Free Grammars and Pushdown Automata

Definition 6.8. A *context-free grammar* (CFG) is a tuple $G = (V, \Sigma, P, S)$ where V is a finite set of *nonterminal* symbols, Σ is a finite alphabet of *terminal* symbols (with $V \cap \Sigma = \emptyset$), $P \subseteq V \times (V \cup \Sigma)^*$ is a finite set of *productions* written $A \to \alpha$, and $S \in V$ is the *start symbol*.

A *derivation step* $\alpha A \beta \Rightarrow \alpha\gamma\beta$ applies a production $A \to \gamma$ to replace one occurrence of A. The reflexive-transitive closure $\Rightarrow^*$ defines reachability. The *language generated* by G is $L(G) = \{w \in \Sigma^* : S \Rightarrow^* w\}$. A language is *context-free* (CFL) if it equals $L(G)$ for some CFG G.

Example 6.9. The grammar $S \to aSb \mid \varepsilon$ has a single nonterminal S, terminals $\{a, b\}$, and two productions. One verifies that $L(G) = \{a^n b^n : n \geq 0\}$: the only derivation of length n produces $a^n b^n$, giving a generating function $L(z) = \sum_{n \geq 0} z^{2n} = 1/(1 - z^2)$, which is rational here by a coincidence specific to this grammar.

Example 6.10. The grammar $S \to SS \mid (S) \mid \varepsilon$ generates the *Dyck language* of balanced parentheses. The functional equation $S(z) = 1 + z^2 S(z)^2$ (using z as a size marker for each matched pair) yields $S(z) = (1 - \sqrt{1 - 4z^2})/(2z^2)$, which is algebraic and not rational.

A grammar is *ambiguous* if some string $w \in L(G)$ admits two distinct leftmost derivations (leftmost means always rewriting the leftmost nonterminal). Ambiguity is not merely a syntactic inconvenience: *inherently ambiguous* languages exist for which no unambiguous grammar exists, and for such languages the symbolic method via the transfer matrix / Chomsky–Schützenberger approach fails to apply directly. The algebraic generating function correspondence from Chapter 5 requires an unambiguous grammar.

A *pushdown automaton* (PDA) is a nondeterministic finite automaton augmented with a stack of unbounded depth; its transition function may push or pop symbols from the stack. The context-free languages are exactly the languages accepted by PDAs, in the same sense that DFAs capture the regular languages.

Theorem 6.11 (Pumping Lemma, Context-Free)**.** *If L is a CFL with pumping length p, then every $w \in L$ with $|w| \geq p$ decomposes as $w = uvxyz$ with $|vxy| \leq p$, $|vy| \geq 1$, and $uv^k xy^k z \in L$ for all $k \geq 0$.*

The proof uses the parse tree of w: if $|w|$ exceeds $p = |\Sigma|^{|V|+1}$, the tree has height greater than $|V|$, so some nonterminal A appears twice along a root-to-leaf path, giving a pumping structure in the tree. The language $\{a^n b^n c^n : n \geq 0\}$ is not context-free: taking $w = a^p b^p c^p$ and any decomposition $w = uvxyz$ with $|vxy| \leq p$, the substring vxy can span at most two of the three symbol types, so pumping distorts their equal counts.

6.7 The Chomsky Hierarchy

Formal grammars are classified by the form of their productions. This yields a strict four-level containment:

1. **Type 3 (Regular):** Productions of the form $A \to a$ or $A \to aB$ (right-linear grammars). Equivalent to finite automata. The class is closed under union, intersection, complement, concatenation, and Kleene star.

2. **Type 2 (Context-Free):** Productions of the form $A \to \alpha$ for $\alpha \in (V \cup \Sigma)^*$, with no restriction on α. Equivalent to pushdown automata. The class is closed under union, concatenation, and Kleene star, but *not* under intersection or complement: the intersection of two CFLs need not be context-free, as the example $\{a^n b^n c^k\} \cap \{a^k b^n c^n\} = \{a^n b^n c^n\}$ demonstrates.

3. **Type 1 (Context-Sensitive):** Productions of the form $\alpha A \beta \to \alpha\gamma\beta$ where $\gamma \neq \varepsilon$ (non-contracting). Equivalent to linear-bounded automata (Turing machines whose tape length is bounded by the input length). The language $\{a^n b^n c^n : n \geq 0\}$ is a canonical example, context-sensitive but not context-free.

4. **Type 0 (Recursively Enumerable):** Unrestricted grammars; productions may have any form. Equivalent to Turing machines. The *halting language* $\{(\langle M \rangle, w) : M \text{ halts on } w\}$ is in this class but is not recursive (decidable), placing it strictly above the context-sensitive class.

The inclusions Reg $\subsetneq$ CFL $\subsetneq$ CSL $\subsetneq$ RE are all strict, with canonical separating examples at each boundary. The regular and context-free levels are by far the most analytically tractable, and they account for nearly all structures arising in the study of language models.

6.8 The Analytic-Combinatorial Payoff

The structural information encoded in a formal language is reflected faithfully in the analytic type of its counting generating function $L(z) = \sum_{n \geq 0} |L \cap \Sigma^n| z^n$.

Theorem 6.12. *If L is a regular language, then $L(z)$ is a rational function of z.*

The proof sketch is given in full in Chapter 7, but the idea is transparent: a DFA with state set Q defines a linear system. For each state q, let $f_q(z) = \sum_{n \geq 0} |\{w \in \Sigma^n : \hat{\delta}(q_0, w) = q\}| z^n$ count the strings that land in q after being read from the start state. The transitions $\delta(q, \sigma) = q'$ yield equations $f_{q'}(z) = \sum_{\sigma : \delta(q,\sigma) = q'} z \cdot f_q(z)$ (plus the contribution of ε at the start state), a finite linear system over $\mathbb{Q}(z)$ whose solution is a vector of rational functions. The generating function $L(z) = \sum_{q \in F} f_q(z)$ is therefore rational.

Theorem 6.13 (Chomsky–Schützenberger, Chapter 5)**.** *If L is generated by an unambiguous CFG, then $L(z)$ is algebraic over $\mathbb{Q}(z)$.*

This is the content of Chapter 5's main theorem. The grammar's productions translate into a polynomial system for the generating functions of each nonterminal, and under the unambiguity hypothesis the solution is an algebraic function. Singularity analysis then yields the asymptotic form of $|L \cap \Sigma^n|$.

At level 1, the correspondence breaks down in a more severe way. Context-sensitive languages can exhibit generating functions with a *natural boundary* on their circle of convergence—a circle beyond which no analytic continuation exists, so that the singularity analysis developed in Part I simply does not apply. This is not a pathological edge case: for certain context-sensitive languages the coefficients grow faster than any geometrically bounded sequence, or oscillate in ways incompatible with the Darboux–transfer-matrix framework. The hierarchy thus presents a clean stratification:

$$\begin{aligned} \text{regular} &\longleftrightarrow \text{rational } L(z), \\ \text{unambiguous CFL} &\longleftrightarrow \text{algebraic } L(z), \\ \text{context-sensitive} &\longleftrightarrow \text{possibly transcendental or exotic } L(z). \end{aligned}$$

For the remainder of this book we work almost exclusively at the regular and context-free levels of the hierarchy (Types 3 and 2, respectively). The context-sensitive level appears in occasional warnings: if a model class tacitly encodes context-sensitive computations, one should not expect the clean rational or algebraic asymptotics that the transfer theorems deliver. Understanding precisely where a language sits in the hierarchy is therefore not merely a theoretical exercise—it determines which analytic tools are available.

Chapter 7 develops the theory of weighted finite automata, the probabilistic refinement of level 3 that assigns real-valued weights to transitions and thereby models probability distributions over Σ^*. This is the direct precursor to the n-gram language models of Part III, and it is where the connection between regular languages and rational generating functions becomes computationally actionable.

Further Reading

- [23] — the definitive classical treatment of automata theory, formal languages, and the Chomsky hierarchy, with careful proofs of all major results including the Myhill–Nerode theorem and both pumping lemmas.
- [51] — a modern, accessible introduction to the theory of computation covering DFAs, NFAs, CFGs, pushdown automata, and undecidability; the standard graduate textbook for the material of this chapter.
- [18] — Part A develops the symbolic method and the connection between grammar structure and generating function type; the bridge between formal languages and analytic combinatorics that underlies this chapter's payoff section.
- [6] — treats rational and recognizable power series over noncommutative alphabets; the algebraic foundation for the transfer from regular languages to rational generating functions developed in Chapter 7.

Chapter 7

Weighted Finite Automata and Rational Power Series

Chapter 6 established finite automata as devices that assign a binary verdict—accept or reject—to every string over a finite alphabet. The present chapter replaces that verdict with a weight drawn from an algebraic structure called a semiring. Each string w receives a weight $f(w) \in K$, and the resulting object is a *weighted finite automaton* (WFA). When $K = \mathbb{R}_{\geq 0}$ and the weights are normalized to sum to one, the WFA becomes a probabilistic generative model over Σ^*. The central theorem of the chapter is that the length-indexed generating function of any such model is automatically a rational function of z, bringing the full machinery of Part I—partial fractions, dominant-pole asymptotics, entropy rates via spectral radii—to bear on probabilistic language models. This rationality result is not an approximation or a limiting statement; it is an exact algebraic consequence of the matrix structure of WFAs, and it is the engine driving the WFA-approximation program developed in Chapter 12.

7.1 Semirings

A *semiring* $(K, \oplus, \otimes, \mathbf{0}, \mathbf{1})$ is a set K equipped with two binary operations satisfying the following axioms: $(K, \oplus, \mathbf{0})$ is a commutative monoid; $(K, \otimes, \mathbf{1})$ is a monoid; $\otimes$ distributes over $\oplus$ from both sides; and $\mathbf{0} \otimes a = a \otimes \mathbf{0} = \mathbf{0}$ for all $a \in K$. The crucial omission relative to a ring is that $\oplus$ need not have inverses. The name "semi" reflects exactly this missing structure.

Two semirings dominate the present chapter. The *Boolean semiring* $(\{0, 1\}, \vee, \wedge, 0, 1)$ reduces the theory to classical unweighted automata: transition weights are either present or absent, matrix multiplication becomes the Boolean matrix product, and the WFA accepts a string iff the computed weight is 1. The *nonnegative real semiring* $(\mathbb{R}_{\geq 0}, +, \times, 0, 1)$ produces the probabilistic theory. One may also work over $\mathbb{R}$ or $\mathbb{C}$ (discarding the nonnegativity constraint), over the tropical semiring $(\mathbb{R} \cup \{+\infty\}, \min, +, +\infty, 0)$ for optimization problems, or over the formal power series ring $\mathbb{R}[[z]]$ itself. The general theory of WFAs over arbitrary semirings is developed in [14]; for present purposes the nonnegative reals suffice.

7.2 Weighted Finite Automata

Definition 7.1. A *weighted finite automaton* (WFA) over alphabet Σ with n states and weights in semiring K is a triple $\mathcal{A} = (\alpha, \{A_\sigma\}_{\sigma \in \Sigma}, \beta)$ where:

- $\alpha \in K^{1 \times n}$ is a row vector of initial weights,
- $\beta \in K^{n \times 1}$ is a column vector of final weights,
- $A_\sigma \in K^{n \times n}$ is a transition matrix for each letter $\sigma \in \Sigma$.

The weight assigned to a string $w = \sigma_1\sigma_2\cdots\sigma_k$ is

$$f_{\mathcal{A}}(w) = \alpha \cdot A_{\sigma_1}A_{\sigma_2}\cdots A_{\sigma_k} \cdot \beta \in K. \tag{7.1}$$

The empty string ε receives weight $f_{\mathcal{A}}(\varepsilon) = \alpha \cdot \beta$.

The computation (7.1) has a concrete state-sum interpretation. Define the weight of a path $q_0 \xrightarrow{\sigma_1} q_1 \xrightarrow{\sigma_2} \cdots \xrightarrow{\sigma_k} q_k$ as $\alpha_{q_0} \cdot (A_{\sigma_1})_{q_0,q_1} \cdots (A_{\sigma_k})_{q_{k-1},q_k} \cdot \beta_{q_k}$. Then $f_{\mathcal{A}}(w)$ is the $\oplus$-sum of this quantity over all paths labeled by w, which is precisely the $(1,1)$ entry of the matrix product in (7.1) once one accounts for the vector boundary conditions. In the Boolean semiring this collapses to: $f_{\mathcal{A}}(w) = 1$ iff at least one accepting path exists, recovering the classical Myhill-Nerode picture from Chapter 6.

Example 7.2 (Fibonacci WFA)**.** Let $\Sigma = \{a\}$, $n = 2$, $K = \mathbb{R}$. Set

$$\alpha = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad A_a = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Then $f_{\mathcal{A}}(a^k) = \alpha A_a^k \beta$. One verifies directly: $k = 1$: $\alpha A_a \beta = (1,0)\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = (1,0)\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 = F_1$. $k = 2$: $\alpha A_a^2 \beta = (1,0)\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = (1,0)\begin{pmatrix} 1 \\ 2 \end{pmatrix} = 1 = F_2$. $k = 3$: $A_a^3 = A_a^2 \cdot A_a = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$, so $\alpha A_a^3 \beta = (1,0)\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = 2 = F_3$. In general $f_{\mathcal{A}}(a^k) = F_k$, the k-th Fibonacci number (with $F_0 = 0$, $F_1 = 1$). The two-state automaton encodes the linear recurrence $F_k = F_{k-1} + F_{k-2}$ in the companion matrix A_a.

7.3 Stochastic WFAs and Probabilistic Language Models

Specialize now to $K = \mathbb{R}_{\geq 0}$. A WFA is *stochastic* if α is a probability distribution over states (i.e., $\sum_q \alpha_q = 1$, $\alpha_q \geq 0$) and if the following row-stochasticity condition holds: for each state q,

$$\sum_{\sigma\in\Sigma}\sum_{q'}(A_\sigma)_{q,q'} + \beta_q = 1.$$

The interpretation is that from state q, the automaton either emits a letter σ and transitions to state q' (with probability $(A_\sigma)_{q,q'}$) or emits the end-of-sequence symbol EOS and halts (with probability β_q). Under this condition $f_{\mathcal{A}}(w)$ is exactly the probability that the automaton generates the string w and then halts, and the weights sum to one over all $w \in \Sigma^*$.

Stochastic WFAs generalize *hidden Markov models* (HMMs). In a standard HMM there is a Markov chain over hidden states and at each step a symbol is emitted from a state-dependent emission distribution; one may always write such a model as a stochastic WFA by absorbing the emission distributions into the transition matrices A_σ. The converse does not hold without further structural conditions: a stochastic WFA can encode transitions that depend simultaneously on both the current and next state in ways that are not directly factored as emission times transition. The representational difference is analytically subtle but matters for expressiveness arguments in Section 6.

7.4 The Generating Function is Rational

Fix a WFA $\mathcal{A} = (\alpha, \{A_\sigma\}, \beta)$ over $K = \mathbb{R}$ (or $\mathbb{C}$). Define the *aggregate transition matrix*

$$A := \sum_{\sigma\in\Sigma} A_\sigma \in \mathbb{R}^{n\times n}.$$

The length-indexed generating function of $\mathcal{A}$ is

$$F(z) := \sum_{w \in \Sigma^*} f_{\mathcal{A}}(w)\, z^{|w|}.$$

Group terms by length:

$$F(z) = \sum_{k \geq 0} z^k \sum_{\substack{w \in \Sigma^* \\ |w|=k}} f_{\mathcal{A}}(w) = \sum_{k \geq 0} z^k \sum_{\substack{\sigma_1,\ldots,\sigma_k \\ \in \Sigma}} \alpha A_{\sigma_1} \cdots A_{\sigma_k} \beta.$$

The key step: the multilinear expansion of a product of sums gives

$$\sum_{\sigma_1,\ldots,\sigma_k \in \Sigma} A_{\sigma_1} \cdots A_{\sigma_k} = \Big(\sum_{\sigma \in \Sigma} A_\sigma\Big)^k = A^k,$$

because the A_σ are matrices over a ring (not merely a semiring), so ordinary matrix algebra applies. Therefore

$$F(z) = \sum_{k \geq 0} z^k\, \alpha A^k \beta = \alpha \Big(\sum_{k \geq 0} (zA)^k\Big) \beta.$$

The matrix power series $\sum_{k \geq 0} (zA)^k$ converges to $(I - zA)^{-1}$ whenever $|z| < 1/\rho(A)$, where $\rho(A)$ denotes the spectral radius of A. This yields:

Theorem 7.3 (Rationality of WFA Generating Functions)**.** *For any WFA $\mathcal{A} = (\alpha, \{A_\sigma\}, \beta)$ over $\mathbb{R}$ (or $\mathbb{C}$), the length-indexed generating function is rational:*

$$F(z) = \alpha (I - zA)^{-1} \beta,$$

where $A = \sum_{\sigma \in \Sigma} A_\sigma$. This rational function has poles only at the reciprocals of the nonzero eigenvalues of A, and the power series converges for $|z| < 1/\rho(A)$.

The numerator polynomial arises from Cramer's rule applied to $(I - zA)^{-1}$: each entry is a ratio of degree-$(n-1)$ and degree-n polynomials in z, so $F(z) = P(z)/Q(z)$ with $\deg P \leq n-1$ and $Q(z) = \det(I - zA)$ the characteristic polynomial of A (with variable substitution $\lambda \mapsto 1/z$).

The connection to Part I is immediate. Chapter 4 established that the coefficients of a rational generating function $P(z)/Q(z)$ satisfy a linear recurrence with characteristic polynomial Q, and the asymptotic behavior is governed by the pole of smallest modulus (i.e., the reciprocal of the dominant eigenvalue). Precisely, if λ_1 is the eigenvalue of A with largest absolute value and multiplicity m, then

$$[z^n]F(z) \sim C \lambda_1^n n^{m-1}$$

for some constant C depending on the eigenvector structure and the boundary vectors α, β.

For stochastic WFAs the row-stochasticity condition $A\mathbf{1} + \beta = \mathbf{1}$ with $\beta \geq 0$ implies $A\mathbf{1} \leq \mathbf{1}$, so A is substochastic and has spectral radius $\rho(A) \leq 1$. When the automaton defines a proper probability distribution—a condition analyzed in depth in Chapter 8—one has $\rho(A) < 1$, the generating function $F(z)$ has no pole on the unit circle, and the string probabilities decay geometrically in length.

Example 7.4 (Fibonacci asymptotics via Theorem 7.3)**.** Returning to Example 7.2, the aggregate matrix is $A = A_a = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$. Its characteristic polynomial is $\lambda^2 - \lambda - 1 = 0$, with roots $\varphi = (1 + \sqrt{5})/2$ (the golden ratio) and $\hat{\varphi} = (1 - \sqrt{5})/2$. The generating function is

$$F(z) = \alpha (I - zA)^{-1} \beta = \frac{z}{1 - z - z^2},$$

with poles at $z = 1/\varphi$ and $z = 1/\hat{\varphi}$. The dominant pole is $1/\varphi$, and partial fractions (exactly as in Chapter 1) give $[z^n]F(z) = F_n \sim \varphi^n/\sqrt{5}$. Theorem 7.3 recovers this from the WFA structure without any ad hoc computation.

7.5 The Kleene–Schützenberger Theorem

The classical Kleene theorem characterizes the languages accepted by finite automata as exactly the regular languages—those definable by regular expressions built from union, concatenation, and Kleene star. Schützenberger's generalization lifts this to the weighted setting. A formal power series $S = \sum_{w \in \Sigma^*} c_w \cdot w$ over a semiring K is called *recognizable* if it equals $f_{\mathcal{A}}$ for some WFA $\mathcal{A}$, and *rational* if it belongs to the smallest family of series containing all polynomials (finite sums $\sum_w c_w \cdot w$) and closed under sum, product, and the operation $S^* := \sum_{k \geq 0} S^k$ (defined when the constant term of S is $\mathbf{0}$).

Theorem 7.5 (Kleene–Schützenberger, informal)**.** *A formal power series $S \in K\langle\langle \Sigma^* \rangle\rangle$ is recognizable if and only if it is rational.*

This result, proved in full generality in [6], is the noncommutative analogue of Kleene's theorem. The "noncommutative" qualifier is essential: the series lives over the free monoid Σ^* where word concatenation is not commutative, so S^* must be interpreted with care and the star is not the geometric series in a scalar variable but rather the fixed-point equation $S^* = \varepsilon + S \cdot S^*$. The connection to Theorem 7.3 is that when one commutes all the variables—replacing each word w by its length $z^{|w|}$—the rational series maps to a rational function of z, recovering the scalar generating function.

7.6 The Icard Hierarchy and What WFAs Cannot Express

Theorem 7.3 implies a sharp limitation: the generating function of any WFA-definable distribution is rational, hence its Taylor coefficients satisfy a linear recurrence. But Chapter 5 introduced distributions whose generating functions are genuinely algebraic—satisfying a polynomial equation $P(z, F(z)) = 0$ but not a linear equation $Q(z)F(z) = R(z)$—and whose coefficients obey the $n^{-3/2}$ subexponential law characteristic of algebraic singularities. No WFA can reproduce such behavior.

This observation is formalized by the *probabilistic Chomsky hierarchy* of [25]. At the finite-state level sit the *rational stochastic languages*: probability distributions over Σ^* computable by some stochastic WFA. At the context-free level sit the *algebraic stochastic languages*: distributions definable by probabilistic context-free grammars (PCFGs). Icard's non-collapse theorem states that these two classes are strictly distinct: there exist PCFG-definable distributions whose generating functions are algebraic but not rational, and no stochastic WFA matches them. The proof proceeds by exhibiting a PCFG for balanced-parentheses strings, whose length-distribution generating function satisfies $F = z + F^2$ (a quadratic, not a rational function), and arguing that no WFA can replicate this distribution.

Complementarily, [52] showed that at the context-free level, adding arbitrary positive weights to a CFG (producing a general weighted CFG) does not increase the class of conditional parse distributions beyond what a properly normalized PCFG can represent. This means the additional flexibility of arbitrary weights buys nothing at the context-free level—the representational power is captured by the probabilistic normalization alone.

Together, these results carve out the following picture. WFAs occupy a well-defined and analytically tractable niche: their distributions are exactly the rational-stochastic languages, their generating functions are rational, and the full arsenal of Part I applies. PCFGs occupy the algebraic level above, and Chapters 9–11 will develop their theory. The generating functions there satisfy polynomial equations, the Flajolet–Odlyzko transfer theorems for algebraic singularities apply, and the asymptotic coefficients reflect the tree-counting structure of context-free derivations. The general reference for the WFA theory is [14]; the generating function perspective is detailed in [18].

7.7 Significance for Language Models

A large language model is neither a WFA nor a PCFG. It is a high-dimensional, nonlinear function—typically a transformer with billions of parameters—and its associated distribution over Σ^* has no obvious algebraic characterization. Nevertheless, the WFA framework provides two concrete analytical tools.

First, if an LLM's next-token distribution can be *approximated* by a WFA—in the sense that there exists a stochastic WFA $\mathcal{A}$ such that $\|f_{\mathcal{A}} - f_{\mathrm{LLM}}\|$ is small in some appropriate norm—then the LLM's generating function is approximately rational, and the Flajolet–Sedgewick machinery yields approximate asymptotics. The quality of the approximation is controlled by the number of states n: larger n allows more complex rational approximations. Chapter 12 develops this WFA-approximation approach and gives conditions under which the approximation error vanishes as $n \to \infty$.

Second, the Icard hierarchy provides sharp obstructions. If empirical measurements reveal that a language model's length distribution follows a $n^{-3/2}$ law (as Chapter 5 suggests for certain grammatical structures), this is a signature of algebraic, not rational, behavior. No WFA approximation—regardless of the number of states—can reproduce this law exactly, because all rational functions have exponential or polynomial-times-exponential coefficient asymptotics, never a pure power law with negative fractional exponent. The algebraic structure must be modeled at the PCFG level or above. This is the content of Icard's non-collapse theorem applied to empirical data, and it provides a principled test for whether a finite-state approximation is adequate for a given modeling task.

The generating function $F(z) = \alpha(I - zA)^{-1}\beta$ of Theorem 7.3 will reappear throughout Part III. Its dominant pole at $z = 1/\rho(A)$ determines the exponential decay rate of string-length probabilities (when $\mathcal{A}$ is stochastic and irreducible, $\Pr(|W| = n) \sim C\rho(A)^n$). The entropy rate of the source, discussed in Chapter 9, is a separate quantity computed from the per-symbol conditional distributions, not directly from $\rho(A)$. Chapter 8 addresses the foundational question that must precede all asymptotic analysis: when does a probabilistic grammar define a *proper* probability distribution—one that assigns total mass one to Σ^*—and what algebraic conditions govern this consistency?

Further Reading

- [14] — the comprehensive reference for weighted automata over arbitrary semirings, covering the algebraic foundations, closure properties, and decision problems; the standard source for the general theory presented in this chapter.
- [6] — develops rational series over free monoids from first principles, proving the Kleene–Schützenberger theorem in full generality and connecting recognizable series to matrix representations.
- [25] — establishes the probabilistic Chomsky hierarchy and the non-collapse theorem separating rational stochastic languages from algebraic ones; the direct source for the Icard hierarchy discussion in Section 6.
- [52] — shows that weighted CFGs with arbitrary positive weights are representationally equivalent to properly normalized PCFGs at the context-free level, settling a question about the expressiveness of unnormalized weighting schemes.
- [18] — Chapter V on applications of rational and meromorphic asymptotics provides the analytic-combinatorics counterpart to the WFA rationality theorem, including the transfer theorems that translate pole locations into coefficient asymptotics.

Chapter 8

Probabilistic Grammars and Tightness

Every generating-function manipulation in this book presupposes that the object being analyzed is actually a probability distribution — that the weights sum to 1 and no probability mass escapes to infinity. That assumption is not automatic for generative models defined by recursion. A grammar or a neural model may, through the compounding of local conditional probabilities, assign positive measure to the event that generation never halts, in which case the induced measure on finite strings is strictly defective. This chapter develops the tightness condition carefully in two settings: probabilistic context-free grammars, the classical case worked out by Booth and Thompson in 1973, and autoregressive neural language models, the modern case formalized by Du et al. in 2023. The two stories are the same story, written in different dialects fifty years apart. Each culminates in a condition that is necessary and sufficient for the length probability generating function to satisfy $F(1) = 1$, the measure-theoretic precondition without which the singularity analysis of [18] cannot be applied.

8.1 Probabilistic Context-Free Grammars

A context-free grammar $G = (V, \Sigma, P, S)$ consists of a finite set of nonterminals V, a finite terminal alphabet Σ disjoint from V, a finite set of productions $P \subset V \times (V \cup \Sigma)^*$, and a distinguished start symbol $S \in V$. A *probabilistic context-free grammar* (PCFG) augments this structure with a probability assignment: for each nonterminal $A \in V$, suppose the productions with left-hand side A are $A \to \alpha_1, \ldots, A \to \alpha_k$; one assigns nonnegative weights $p_1, \ldots, p_k$ with $\sum_{i=1}^{k} p_i = 1$. The probability of a leftmost derivation $d = (r_1, r_2, \ldots, r_m)$, where $r_i \in P$ are the productions applied in order, is $\prod_i p(r_i)$, the product of the individual production probabilities. The probability of a terminal string $w \in \Sigma^*$ is then

$$p(w) = \sum_{d : d \Rightarrow^* w} \prod_{r \in d} p(r),$$

summed over all leftmost derivations producing w. A PCFG thus attempts to define a probability distribution on $L(G) \subseteq \Sigma^*$.

The local normalization $\sum_i p_i = 1$ for each nonterminal is a necessary but not sufficient condition for this global distribution to be proper. The difficulty arises from infinitely long derivations. Consider the single-nonterminal grammar with productions $S \to SS$ (probability q) and $S \to a$ (probability $1 - q$), where $q > 1/2$. In a single expansion of S, the expected number of S-nonterminals in the right-hand side is $q \cdot 2 + (1 - q) \cdot 0 = 2q > 1$. The derivation process is therefore a supercritical multi-type Galton–Watson branching process, and by the theory of such processes, there is positive probability that the nonterminal population never

dies out — the derivation never reaches a terminal string. The total probability assigned to finite derivations is strictly less than 1.

8.2 The Booth–Thompson Consistency Criterion

[8] asked exactly when a PCFG defines a proper distribution and introduced the term *consistent* for PCFGs where the answer is affirmative.

Definition 8.1. A PCFG is *consistent* if $\sum_{w \in L(G)} p(w) = 1$, i.e. if the total probability assigned to finite terminal derivations equals 1.

Associate to the PCFG its *first-moment matrix* M, indexed by pairs of nonterminals: the entry $M_{A,B}$ is the expected number of occurrences of nonterminal B in the right-hand side of a single random expansion of A, where the expansion is drawn according to the production probabilities. Formally,

$$M_{A,B} = \sum_{\alpha : A \to \alpha \in P} p(A \to \alpha) \cdot |\alpha|_B,$$

where $|\alpha|_B$ denotes the number of occurrences of B in the sentential form α. The matrix M is precisely the mean matrix of the multi-type Galton–Watson branching process with one type per nonterminal, in which each individual of type A produces offspring according to the distribution over right-hand sides of productions $A \to \alpha$.

Theorem 8.2 (Booth–Thompson 1973)**.** *A PCFG is consistent if and only if the spectral radius* $\rho(M) \leq 1$.

When $\rho(M) < 1$ the branching process is subcritical: the nonterminal population dies out almost surely, and moreover the expected total derivation length is finite. The marginal case $\rho(M) = 1$ (critical branching) still yields an almost-sure extinction — and hence a consistent PCFG — but with infinite expected derivation length and heavy tails on parse-tree size. The supercritical case $\rho(M) > 1$ produces inconsistency: with positive probability the derivation is infinite, and $\sum_w p(w) < 1$.

The generating-function reformulation makes the connection to Part I explicit. Let $F_A(z) = \sum_{n \geq 0} q_{A,n} z^n$ be the probability generating function for the total number of terminal symbols produced by a derivation rooted at nonterminal A, where $q_{A,n}$ is the probability that A generates a string of length exactly n. Consistency of the PCFG is equivalent to $F_S(1) = 1$, i.e. the PGF of the start symbol evaluates to 1 at $z = 1$. This is precisely the condition that F_S represents a genuine probability distribution on $\mathbb{Z}_{\geq 0}$. Any attempt to apply the transfer theorems of [18] — expanding F_S in a Puiseux series around the dominant singularity, reading off tail asymptotics of the length distribution — requires this evaluation; a defective distribution has $F_S(1) < 1$ and the standard radius-of-convergence analysis at $z = 1$ is invalidated.

8.3 Chi's Theorem and Maximum-Likelihood Estimation

[10] proved a fundamental result about PCFGs fitted to data. In computational linguistics the standard procedure is to read a treebank — a corpus of sentences paired with their parse trees — and set each production probability $p(A \to \alpha)$ to the relative frequency of that production among all expansions of A in the corpus. This maximum-likelihood estimator has a closed form, and Chi showed it is automatically consistent.

Theorem 8.3 (Chi 1999)**.** *The maximum-likelihood PCFG estimated from any finite treebank is consistent. Moreover, the parse-tree size (number of internal nodes, number of leaves, tree depth) has finite moments of all orders under this distribution.*

The proof proceeds via multi-type Galton–Watson theory. One shows that the ML estimator always satisfies the system of fixed-point equations $\mathbf{F}(1) = \mathbf{1}$ (where $\mathbf{F}(z)$ is the vector of PGFs, one per nonterminal) and that these equations together with the criticality condition $\rho(M) \leq 1$ are consequences of the ML conditions on a finite corpus. Generating functions appear in Chi's argument as auxiliary computational devices — the $z = 1$ evaluation — rather than as the primary objects of study; the core reasoning is probabilistic. The practical upshot is that for any PCFG fitted by maximum likelihood, consistency is automatic and one may apply singularity analysis to $\mathbf{F}(z)$ without further verification.

8.4 Autoregressive Language Models

An *autoregressive language model* over a finite vocabulary Σ is a family of conditional distributions

$$\{p(\cdot \mid h) : h \in \Sigma^*\},$$

where each $p(\cdot \mid h)$ is a probability distribution on $\Sigma \cup \{\text{EOS}\}$, the vocabulary extended by a distinguished end-of-sequence symbol $\text{EOS} \notin \Sigma$. The generative procedure initializes with the empty history, samples $w_1 \sim p(\cdot \mid \varepsilon)$, then $w_2 \sim p(\cdot \mid w_1)$, and so on, halting as soon as EOS is emitted. The probability assigned to a finite string $w = w_1 w_2 \cdots w_n \in \Sigma^*$ is

$$p(w) = p(\text{EOS} \mid w_1 \cdots w_n) \prod_{t=1}^{n} p(w_t \mid w_1 \cdots w_{t-1}).$$

The same measure-theoretic question arises: does $\sum_{w \in \Sigma^*} p(w) = 1$? In general the answer is no. The local normalization $\sum_{v \in \Sigma \cup \{\text{EOS}\}} p(v \mid h) = 1$ for every history h is again necessary but not sufficient. A model may compound small EOS probabilities across long prefixes in a way that places positive measure on the event that generation continues forever, producing an infinite string. In that case $\sum_{w \in \Sigma^*} p(w) < 1$ and the model is not a proper distribution on Σ^*.

8.5 Du et al.'s Measure-Theoretic Framework

[15] provide a rigorous formulation. Define the extended sample space $\Sigma^\infty = \Sigma^* \cup \Sigma^\omega$, the set of all finite or infinite strings over Σ. Equip Σ^∞ with the cylinder σ-algebra: for each finite string $u \in \Sigma^*$ the cylinder $[u] = \{x \in \Sigma^\infty : x \text{ has prefix } u\}$ is measurable, and the σ-algebra is generated by all such cylinders. The autoregressive conditional probabilities determine a consistent family of finite-dimensional distributions (via the chain rule), and by Carathéodory's extension theorem this family extends uniquely to a probability measure μ on Σ^∞.

Definition 8.4 (Du et al. 2023)**.** An autoregressive language model is *tight* if $\mu(\Sigma^*) = 1$, i.e. if the induced measure assigns probability 1 to the event that generation halts in finite time.

To state the necessary and sufficient condition, define $\widetilde{p}_{\text{EOS}}(t)$ as the conditional probability that generation halts at exactly step t given that it has not halted before step t — formally, $\widetilde{p}_{\text{EOS}}(t) = \mathbb{E}[p(\text{EOS} \mid w_1 \cdots w_{t-1}) \mid \text{not halted before } t]$, where the expectation is over prefixes of length $t - 1$ generated by the model conditioned on survival to step t.

Theorem 8.5 (Du et al. 2023, Theorem 4.7)**.** *An autoregressive language model is tight if and only if*

$$\sum_{t=1}^{\infty} \widetilde{p}_{\text{EOS}}(t) = +\infty.$$

This is a Borel–Cantelli-type criterion: if the conditional halting probabilities form a summable sequence, then generation survives to infinity with positive probability, and the model is non-tight. Divergence of the series, by contrast, forces almost-sure halting.

A more directly usable sufficient condition follows from a comparison argument. If there exists a function $f : \mathbb{Z}_{>0} \to [0, 1]$ such that $p(\text{EOS} \mid x) \geq f(t)$ for every prefix x of length $t - 1$ (uniformly over all prefixes of that length) and $\sum_{t=1}^{\infty} f(t) = \infty$, then the model is tight. In particular, if there exists $\varepsilon > 0$ such that $p(\text{EOS} \mid h) \geq \varepsilon$ for every history $h \in \Sigma^*$, then $\widetilde{p}_{\text{EOS}}(t) \geq \varepsilon$ for all t, the series $\sum_t \varepsilon$ diverges, and tightness holds immediately.

8.6 Examples on Both Sides of the Threshold

The abstract criterion becomes concrete through the examples constructed in [15].

Example 8.6 (Non-tight recurrence)**.** Consider a recurrent architecture whose hidden state $h_t \in \mathbb{R}$ evolves by a one-dimensional ReLU map, with EOS probability $p(\text{EOS} \mid h_t) = 1/(e^{h_t} + 1)$ (note: this is a decreasing function of h_t, so as the hidden state grows the model becomes less likely to stop), and where h_t grows roughly linearly in t. Then $\widetilde{p}_{\text{EOS}}(t) \sim C/(e^t + 1)$ for some constant C, which decays exponentially. The series $\sum_t \widetilde{p}_{\text{EOS}}(t)$ converges, so the model is non-tight: with positive probability the sampler runs forever and the output lies in Σ^ω.

Example 8.7 (Tight but heavy-tailed)**.** A softplus RNN can be constructed so that $\widetilde{p}_{\text{EOS}}(t) \sim 1/(t + 1)$, the harmonic series. The divergence $\sum_t 1/(t + 1) = \infty$ confirms tightness, but only barely: there is no finite upper bound on the expected output length, and the length distribution has a power-law tail. This model generates finite strings almost surely, yet its generating function $F(z) = \sum_n p_n z^n$ has radius of convergence exactly 1, with a singularity at $z = 1$ of logarithmic type.

Theorem 8.8 (Du et al. 2023, Section 5.2)**.** *Every Transformer language model with softmax output layer is tight.*

The proof turns on a structural property of the softmax function. For a finite vocabulary $\Sigma \cup \{\text{EOS}\}$ of size V, and any logit vector $\ell \in \mathbb{R}^V$, the softmax probability assigned to EOS satisfies

$$p(\text{EOS} \mid h) = \frac{e^{\ell_{\text{EOS}}}}{\sum_v e^{\ell_v}} > 0.$$

Since the logit vector is computed by a Transformer with bounded weights (the model has finitely many parameters), the logits are continuous functions of the input history as embedded in a compact representation. Over any bounded set of inputs, the minimum EOS softmax probability is bounded away from zero. Even without compactness of the input space, the strict positivity of softmax over a finite alphabet means there is always some positive EOS probability, and one can show (by a careful bounding argument) that this positivity is uniform enough to satisfy the Proposition 4.3 floor condition. The decisive structural feature is that softmax over a finite set cannot assign exactly zero probability to any token: the exponential function is everywhere positive, and the normalizing constant is finite. Constructing a non-tight model requires engineering EOS probabilities that tend to zero at an exponential rate, which the softmax architecture structurally prevents.

8.7 The Parallel Between the Two Theories

The Booth–Thompson theorem and the Du et al. theorem are instances of the same mathematical phenomenon, and it is worth stating the isomorphism precisely. In the PCFG setting, one has a recursive generative process in which each expansion step introduces new subproblems

(nonterminals) that must themselves terminate; consistency asks whether this branching process almost surely reaches a leaf configuration. In the autoregressive setting, one has a sequential generative process in which each step emits a token or halts; tightness asks whether the sequential process almost surely halts. Both are questions about the total mass assigned to finite outcomes by a process defined through local stochastic rules that are individually normalized.

The Booth–Thompson condition $\rho(M) \leq 1$ is equivalent, via the theory of multi-type branching processes, to the statement $\mathbf{F}(1) = \mathbf{1}$ for the vector of PGFs. The Du et al. condition $\sum_t \widetilde{p}_{\mathrm{EOS}}(t) = \infty$ is equivalent to $\sum_n p_n = 1$ for the length distribution, i.e. $F(1) = 1$ for the length PGF. Both conditions say: the probability generating function of the length (or derivation size) of a random output evaluates to 1 at the boundary point $z = 1$. Both are prerequisites for the singularity analysis program of [18]: if $F(1) < 1$ then F is a defective distribution, the radius of convergence may be > 1, and the standard transfer theorems that read off asymptotics from the behavior of F near its dominant singularity require modification or fail outright.

The two theories also share the same critical boundary behavior. A critical PCFG with $\rho(M) = 1$ and a just-barely tight autoregressive model with $\sum_t \widetilde{p}_{\mathrm{EOS}}(t) = \infty$ diverging like the harmonic series both exhibit infinite expected output length and generating functions with a logarithmic singularity at $z = 1$. This is precisely the regime studied by algebraic-logarithmic transfer lemmas. The subcritical PCFG and the uniformly positive-EOS autoregressive model, by contrast, have $F(z)$ analytic in a disk strictly larger than the unit disk, and the length distribution decays exponentially.

Cotterell et al. [12] situate these definitions within the broader framework of language models as probability measures on Σ^*, connecting the linguistic and probabilistic perspectives. The theorem of Du et al. that every softmax Transformer is tight is, from the vantage point of this book, not a peripheral remark about neural architecture but a foundational result: it is the permission slip granting Transformer language models entry into the class of objects to which the machinery of Part I applies.

The next chapter takes up entropy rate — the average per-token surprise under a stationary distribution — and shows how, for tight models, the entropy rate can be computed from the radius of convergence of a suitably defined generating function, closing the loop between information-theoretic and analytic-combinatorial perspectives.

Further Reading

- [8] — the original paper introducing the consistency criterion for PCFGs via the spectral radius of the first-moment matrix; the primary source for the theorem bearing their names.
- [10] — proves that maximum-likelihood PCFG estimation from finite treebanks always yields a consistent grammar, establishing automatic consistency as a property of the standard estimation procedure.
- [15] — provides the measure-theoretic framework for tightness of autoregressive language models, proves the Borel–Cantelli-type criterion, and shows that every softmax Transformer is tight; the modern counterpart to the Booth–Thompson theorem.
- [12] — situates tightness and consistency within a unified probabilistic framework for language models as measures on Σ^*, connecting the PCFG and autoregressive perspectives.
- [18] — the singularity analysis in Parts B and C provides the analytic machinery presupposed by the tightness condition: consistent and tight models are exactly those whose PGFs satisfy $F(1) = 1$, permitting the Darboux and transfer theorems to apply.

Chapter 9

Entropy Rate and the Identity $h = \log(1/R)$

One of the cleanest bridges between information theory and analytic combinatorics is the identity

$$h = \log_2(1/R), \tag{9.1}$$

where h is the entropy rate of a stationary source and R is the radius of convergence of a suitable counting-type generating function. This chapter develops the identity carefully, warns against a common conflation, and surveys what is known about entropy rates for natural language and LLM outputs.

9.1 Entropy Rate

Let $X_1, X_2, \ldots$ be a stationary stochastic process over a finite alphabet Σ. The *block entropy* at length n is

$$H(X_1^n) := -\sum_{w \in \Sigma^n} p(w) \log_2 p(w),$$

where $p(w) = \Pr(X_1 \cdots X_n = w)$. The *entropy rate* is

$$h := \lim_{n \to \infty} \frac{H(X_1^n)}{n},$$

when the limit exists. Stationarity guarantees existence and the further identity

$$h = \lim_{n \to \infty} H(X_n \mid X_1^{n-1}).$$

The conditional form shows that h measures the irreducible uncertainty per new symbol given arbitrarily long past context.

Informally, h bits per symbol are necessary and sufficient (on average) to losslessly encode a long sample. For a memoryless uniform source on Σ, $h = \log_2 |\Sigma|$, the maximum possible. In general, $0 \le h \le \log_2 |\Sigma|$, with $h = \log_2 |\Sigma|$ if and only if the source is uniform i.i.d.

9.2 Shannon–McMillan–Breiman and Typical Sets

The fundamental limit theorem of source coding is the following.

Theorem 9.1 (Shannon–McMillan–Breiman)**.** *Let $\{X_n\}$ be an ergodic stationary source over Σ with entropy rate h. Then*

$$-\frac{1}{n} \log_2 p(X_1 \cdots X_n) \;\to\; h \quad \textit{almost surely as } n \to \infty.$$

This is the *asymptotic equipartition property* (AEP). The block log-probability concentrates, almost surely, at $-nh$.

Definition 9.2 (Typical set)**.** For $\epsilon > 0$, the *ϵ-typical set* at length n is

$$T_n^{(\epsilon)} := \Big\{ w \in \Sigma^n : \big| -\tfrac{1}{n} \log_2 p(w) - h \big| < \epsilon \Big\}.$$

The AEP immediately yields three properties: (i) $\Pr(X_1^n \in T_n^{(\epsilon)}) \to 1$; (ii) each $w \in T_n^{(\epsilon)}$ satisfies $2^{-n(h+\epsilon)} \le p(w) \le 2^{-n(h-\epsilon)}$; and therefore (iii) the cardinality satisfies

$$(1 - \delta_n) \cdot 2^{n(h-\epsilon)} \;\le\; |T_n^{(\epsilon)}| \;\le\; 2^{n(h+\epsilon)}$$

for any $\delta_n \to 0$. Setting $\epsilon \to 0$ slowly: $|T_n| \approx 2^{nh}$ up to subexponential factors.

9.3 From Typical Sets to Generating Functions

Package the cardinality of the typical set into an ordinary generating function. Let $a_n := |T_n|$ for a fixed (small) ϵ. Consider

$$A(z) = \sum_{n \ge 0} a_n z^n, \qquad a_n \approx 2^{nh}.$$

By the Cauchy–Hadamard formula (Chapter 2), the radius of convergence is

$$\frac{1}{R} = \limsup_{n \to \infty} a_n^{1/n} = 2^h.$$

Therefore $R = 2^{-h}$ and we recover the identity (9.1):

$$\boxed{h = \log_2(1/R).}$$

Remark 9.3. The identity is classical for finite-state sources. If the source is generated by a finite automaton with transition matrix M, then a_n counts paths of length n in the automaton. The transfer-matrix method (Chapter 7) gives $[z^n]A(z) \sim C \cdot \rho(M)^n$, where $\rho(M)$ is the Perron eigenvalue. The radius of convergence is $R = 1/\rho(M)$, and the entropy rate of the corresponding stationary Markov chain satisfies $h = \log_2 \rho(M)$. The identity thus unifies the Perron–Frobenius spectral picture (Chapters 6–7) with the information-theoretic picture.

9.4 A Crucial Distinction

The identity $h = \log_2(1/R)$ applies to a *counting* generating function whose coefficients grow like 2^{nh}. It does not apply to the *length probability* generating function, and confusing the two leads to error.

Define the length probability generating function by

$$P(z) := \sum_{n \ge 0} p_n \, z^n, \qquad p_n = \Pr(|W| = n),$$

for a model that generates strings of random length (such as an autoregressive language model with an end-of-sequence token). Since the p_n are probabilities summing to 1, we have $P(1) = 1$, and therefore the radius of convergence satisfies $R_P \ge 1$. Consequently,

$$\log_2(1/R_P) \le 0,$$

which cannot encode a positive entropy rate. What R_P and the singularity structure of P near $z = 1$ *do* encode is the tail behavior of the length distribution: exponential decay, subexponential decay, polynomial decay, and so on.

Two distinct generating functions thus play two distinct roles.

1. **The counting-type generating function** $A(z) = \sum a_n z^n$, where a_n counts strings of length n in the typical set (or, for finite-state sources, paths of length n in the underlying automaton). Its radius satisfies $R = 2^{-h}$ and encodes the entropy rate.

2. **The length probability generating function** $P(z) = \sum p_n z^n$. Its radius satisfies $R_P \geq 1$ and its singularity type encodes the length-distribution tail.

When you encounter the phrase "the generating function of the LLM" in a paper, you must ask which of these two objects is intended. The two questions one can ask about a source — how large is the typical set, and how are string lengths distributed — correspond to two generating functions with radii of convergence on opposite sides of $z = 1$.

9.5 Classical Entropy-Rate Estimation for Natural Language

Entropy-rate estimation for natural language has a history of more than seventy years, and the problem remains open at the precision needed for modern applications.

Shannon (1951). Shannon [50] estimated the entropy of printed English via a guessing experiment. Human subjects predicted the next character of a text given the preceding characters; the distribution of guesses yields an upper bound on conditional entropy, while the reciprocal of the guessed character's rank in a natural ordering yields a lower bound. For contexts of length $N = 100$, Shannon's estimates placed the entropy rate in the range $[0.6, 1.3]$ bits per character. In the language of Section 3, Shannon was estimating $\log_2(1/R)$ for the counting generating function of English, though he never wrote that function down. The width of his interval reflects both experimental variance and the slow convergence of the conditional entropy to its limit, a theme we return to below.

Kontoyiannis et al. (1998). Kontoyiannis, Algoet, Suhov, and Wyner [28] sharpen the estimation problem using longest-match-length statistics. Let L_n denote the length of the longest prefix of $(X_{n+1}, X_{n+2}, \ldots)$ that has appeared as a substring in $(X_1, \ldots, X_n)$. Their estimator is

$$\hat{h}_n := \frac{\log_2 n}{L_n}.$$

Under a Doeblin-type mixing condition on the source, $\hat{h}_n \to h$ almost surely. The convergence rate is controlled by how cleanly the dominant singularity of $A(z)$ is separated from the remainder of the spectrum on the circle $|z| = R$; a simple isolated pole gives geometric convergence, while a more complex singularity structure degrades the rate.

Szpankowski (2001). The book [56] applies the full Flajolet–Sedgewick toolkit — Mellin transforms, depoissonization, singularity analysis — to these questions in the context of digital data structures and source models. Part V, Chapter 17 of the present text covers this material in depth.

9.6 Modern LLM-Era Estimates

Takahashi and Tanaka-Ishii (2018). Takahashi and Tanaka-Ishii [57] extrapolate neural language-model cross-entropies simultaneously to infinite training data and infinite context length. Their extrapolation procedure fits the observed cross-entropies as a function of both corpus size and context window using a two-parameter model, and takes the joint limit. For English, they obtain an entropy-rate estimate of approximately 1.12 bits per character. In the

present framework, this is an estimate of $\log_2(1/R)$ for the English counting generating function, computed indirectly via the cross-entropy of a neural model serving as a proxy for the true source distribution.

The result is consistent with, and more precise than, Shannon's original interval. The narrowing was achieved not by a more careful experiment but by using a much better predictor, trained on vastly more text with much longer effective context.

Scheibner, Smith, and Bialek (2025). Scheibner, Smith, and Bialek [48] return to Shannon's guessing-game paradigm but replace human subjects with frontier large language models as predictors. The central finding is that conditional entropy continues to decrease with context length out to $N \sim 10^4$ characters, at scales of dependency that Shannon could not access and that substantially exceed the context lengths used in [57]. The running entropy-rate estimate does not settle within the range of context lengths they study.

9.7 Slow Convergence as a Diagnostic for Singularity Type

The identity $h = \log_2(1/R)$ specifies the location of the dominant singularity of $A(z)$ but says nothing about its *type*. The type of the singularity determines the rate at which the conditional entropy $H(X_{n+1} \mid X_1^n)$ converges to h, and this convergence rate is exactly what entropy-estimation experiments measure.

To be precise, recall from Chapter 4 that the coefficient asymptotics of a GF with a dominant singularity at $z = R$ depend on the local behavior of $A(z)$ near R.

- If $A(z)$ has a *simple pole* at R, then $[z^n]A(z) = CR^{-n}(1 + O(\theta^n))$ for some $\theta < 1/R$. The conditional entropy satisfies $H(X_{n+1} \mid X_1^n) - h = O(\theta^n)$: exponentially fast convergence. A few hundred characters of context suffice to pin down h to three decimal places. This is the Markov chain and finite-automaton case.

- If $A(z)$ has an *algebraic branch point* of exponent α, meaning $A(z) \sim C(1 - z/R)^\alpha$ as $z \to R$ with $\alpha \notin \mathbb{Z}$, then by the Transfer Lemma (Chapter 4),

$$[z^n]A(z) \;\sim\; C' \cdot R^{-n} \cdot n^{-\alpha-1}.$$

  The conditional entropy converges with a polynomial residual, $H(X_{n+1} \mid X_1^n) - h \sim C'' n^{-\alpha-1}$.

- If $A(z)$ has a *logarithmic singularity*, $A(z) \sim C(1 - z/R)^\alpha \log(1 - z/R)$, the residual is $(\log n)^{-\beta}$, decaying more slowly than any power law.

- A *confluence of singularities* on the circle $|z| = R$ introduces oscillatory corrections on top of the algebraic or logarithmic envelope.

The Shannon-to-Takahashi-Tanaka-Ishii span of ~ 0.6 bits is not measurement noise; it is a signal. Estimates of h via finite-context predictors converge to the true entropy rate from above as context length increases, and the *rate* at which they converge encodes the singularity type of $A(z)$. If the convergence were geometric, the 1951 and 2018 estimates would agree to many significant figures. They do not. Scheibner et al.'s finding that the estimate continues to drift downward past $N = 10^4$ characters is consistent with a singularity that is algebraic or logarithmic, with a slow polynomial or logarithmic tail on the convergence.

Remark 9.4. This reframes a classical problem in information theory. Decades of work have been devoted to estimating the exponent $h = \log_2(1/R)$. The type of the singularity of $A(z)$ — which controls how fast the estimate converges and thus how long a memory the source requires —

has received almost no systematic attention. In the language of Part V, this is a question about the *nature* of the dominant singularity, not merely its *location*. The entropy-rate convergence curves produced by modern LLMs, evaluated at progressively longer contexts, are, in principle, a fine-resolution probe of that singularity type. Threading together the analytic-combinatorics tools of Part I with the estimation procedures of Part V is a program taken up in Chapter 17.

We have completed the foundations of Part II. The identity $h = \log_2(1/R)$ ties together the entropy rate of a stationary source, the asymptotics of its typical set, and the singularity analysis of its counting generating function. The distinction between the counting GF and the length probability GF is essential and is frequently missed. The convergence rate of entropy estimators to h is a window onto the singularity type of $A(z)$, and that type is what modern LLM experiments are beginning to resolve.

Part III turns to autoregressive language models as mathematical objects: first as probability distributions over Σ^*, then through the lens of formal-language theory applied to transformer architectures, and finally as weighted-finite-automaton approximations whose spectral structure directly instantiates the framework developed in these pages.

Further Reading

- [50] — the original guessing-game experiment estimating the entropy rate of printed English; the historical starting point for all subsequent entropy-rate estimation work.
- [28] — establishes the longest-match-length estimator $\hat{h}_n = (\log_2 n)/L_n$ and proves its almost-sure convergence to the entropy rate under mixing conditions; the rigorous nonparametric approach to entropy-rate estimation.
- [56] — applies Mellin transforms, depoissonization, and singularity analysis to entropy-related combinatorial problems; the reference for the analytic toolkit connecting generating-function singularity type to estimation convergence rates.
- [57] — extrapolates neural language model cross-entropies to infinite data and infinite context, obtaining a precise entropy-rate estimate of approximately 1.12 bits per character for English.
- [48] — uses frontier LLMs as predictors in Shannon's guessing paradigm, finding that conditional entropy continues to decrease at context lengths well beyond those studied in earlier work and providing evidence for a slow-converging singularity in the English counting GF.

Part III

Large Language Models and Their Formal Theory

Chapter 10

Autoregressive Language Models

So far we have treated "distributions over strings" abstractly: a probability measure μ on Σ^*, a weighted automaton, a formal power series with nonnegative coefficients summing to one. This chapter introduces the concrete object that the rest of the book analyzes: a large language model. The exposition is deliberately stripped down. An LLM, for our purposes, is a probability distribution on Σ^*, nothing more. The neural network is the parameter, not the object. What matters is what distribution it induces, and what mathematical tools are available for analyzing that distribution. Once we have this, the singularity analysis of Part I and the entropy theory of Part II can be brought to bear directly.

10.1 Tokens, Vocabulary, and End-of-Sequence

An LLM operates over a finite vocabulary Σ, typically consisting of *tokens* obtained by applying a subword encoding to a large text corpus. The three standard algorithms are byte-pair encoding (BPE), WordPiece, and the unigram language model. Each algorithm merges or segments character sequences according to corpus statistics, producing units that are neither words nor characters but lie somewhere between. Modern vocabularies range from $|\Sigma| \approx 30{,}000$ (as in GPT-2 and the Llama family) to $|\Sigma| \approx 200{,}000$ (as in GPT-4). The vocabulary always includes a distinguished *end-of-sequence token* $\mathrm{EOS} \in \Sigma$. Every sampled string ends with the first occurrence of EOS, at which point sampling halts. Strings not containing EOS in their interior are the only grammatical outputs; EOS appearing mid-string would be treated as a terminal event, truncating everything after it.

Tokenization is not a detail to be abstracted away. It determines what counts as an atom for the combinatorial class of LLM outputs. A byte-pair tokenizer running on English text will produce a granularity different from word-level and different from character-level. Zipf exponents on token frequencies, length distributions measured in tokens, and entropy rates all depend on which tokenizer was applied. Two models with identical weights but different tokenizers define genuinely different distributions on Σ^*. Part V returns to this in Chapter 16, where tokenization artifacts appear as poles in the length generating function.

10.2 The Autoregressive Factorization

An LLM defines, for every history $h \in \Sigma^*$, a conditional distribution $p(\cdot \mid h) \in \Delta(\Sigma)$ over the next token, treating EOS as an ordinary token. Given these conditionals, the probability of generating a specific finite string $w = w_1 w_2 \cdots w_n$ followed immediately by EOS is

$$p(w \cdot \mathrm{EOS}) \;=\; p(\mathrm{EOS} \mid w_1 \cdots w_n) \prod_{t=1}^{n} p(w_t \mid w_1 \cdots w_{t-1}),$$

where the empty product at $t = 0$ is 1 (so the probability of the string consisting of EOS alone is $p(\text{EOS} \mid \varepsilon)$). Sampling proceeds left to right: draw $w_1 \sim p(\cdot \mid \varepsilon)$, then $w_2 \sim p(\cdot \mid w_1)$, and so on, until EOS is sampled. This procedure is called *autoregressive decoding.*

Definition 10.1 (Autoregressive language model)**.** An *autoregressive language model* over Σ is a family $\mathcal{M} = \{p(\cdot \mid h)\}_{h \in \Sigma^*}$ of conditional distributions $p(\cdot \mid h) \in \Delta(\Sigma)$, one for each finite history $h \in \Sigma^*$. The induced distribution on Σ^* is defined by the autoregressive factorization above.

The parameters of the neural network are merely a compact representation of the family $\mathcal{M}$; the mathematical object is the family itself. Two distinct parameter vectors that induce the same family of conditionals define the same language model in our sense. This is the appropriate level of abstraction for the analytic combinatorics program: we care about generating functions of the output distribution, and those are determined by $\mathcal{M}$, not by the network's internal weight matrices.

10.3 Softmax and the Role of Temperature

In every modern LLM, the conditional $p(\cdot \mid h)$ is computed via *softmax* applied to a vector of real-valued *logits* $\ell(h) \in \mathbb{R}^{|\Sigma|}$:

$$p(\sigma \mid h) \;=\; \frac{e^{\ell_\sigma(h)}}{\sum_{\sigma' \in \Sigma} e^{\ell_{\sigma'}(h)}}.$$

Softmax maps any real vector to a strictly positive probability distribution: every token, including every token that might look semantically bizarre in context, receives nonzero probability. This seemingly minor property has structural consequences that will dominate the next two sections.

Temperature scaling. In practice one often samples from the tempered distribution

$$p_T(\sigma \mid h) \;=\; \frac{e^{\ell_\sigma(h)/T}}{\sum_{\sigma'} e^{\ell_{\sigma'}(h)/T}},$$

where $T > 0$ is the *sampling temperature.* At $T = 1$ one recovers the model's native distribution. As $T \to 0$, all mass concentrates on the argmax token (greedy decoding). As $T \to \infty$, $p_T(\cdot \mid h)$ converges to the uniform distribution on Σ. Temperature scaling is a one-parameter deformation of the model's output distribution; it preserves the support (every token remains reachable for any $T > 0$) while reshaping the entropy. Temperature is the central object of Part IV, where its effect on the dominant singularity of the length PGF is analyzed.

10.4 Transformers in One Paragraph

A transformer maps the history h to the logits $\ell(h)$ through a stack of L layers, each consisting of a multi-head self-attention sublayer followed by a position-wise feedforward network, with residual connections and layer normalization at each sublayer [58]. The formal object we care about is the input–output function $h \mapsto \ell(h)$; the internal mechanism is parameterization, not the subject of study. What matters analytically is that this function is (i) computable in polynomial time per step (quadratic in the context length via the attention computation), (ii) differentiable in the parameters, and (iii) structurally forced to produce strictly positive softmax outputs at every step. Property (iii) is what the next section exploits. Nothing about the specific form of self-attention, feedforward width, or normalization scheme enters the tightness argument; the argument requires only that the logit function $h \mapsto \ell(h)$ maps into a bounded region of $\mathbb{R}^{|\Sigma|}$ for fixed parameters.

10.5 Tightness for Transformer Language Models

Chapter 8 defined *tightness*: the autoregressive factorization yields a proper probability measure on Σ^* (i.e., $\sum_{w\in\Sigma^*} p(w\cdot\mathrm{EOS}) = 1$) if and only if the sum $\sum_{t=1}^{\infty} \widetilde{p}_{\mathrm{EOS}(t)}$ diverges, where $\widetilde{p}_{\mathrm{EOS}(t)}$ is the probability that EOS has not yet been emitted by step t. For transformer language models, tightness is automatic by a result of Du et al. [15].

The argument is essentially the following. The logit vector $\ell(h) \in \mathbb{R}^{|\Sigma|}$ depends on the history h through the transformer, but because the transformer's parameters are a fixed finite collection of weight matrices and biases, and because attention weights and layer normalizations are bounded per step, there is a uniform constant $C > 0$ depending only on the parameters such that $|\ell_\sigma(h)| \le C$ for all $h \in \Sigma^*$ and all $\sigma \in \Sigma$. Then

$$p(\mathrm{EOS} \mid h) \;=\; \frac{e^{\ell_{\mathrm{EOS}(h)}}}{\sum\limits_{\sigma\in\Sigma} e^{\ell_\sigma(h)}} \;\ge\; \frac{e^{-C}}{|\Sigma|\, e^{C}} \;=\; \frac{e^{-2C}}{|\Sigma|} \;=:\; \varepsilon \;>\; 0.$$

The EOS probability is bounded below by a uniform constant ε that depends only on the parameter bound C and the vocabulary size. By Proposition 4.3 of Du et al., a uniform lower bound $p(\mathrm{EOS} \mid h) \ge \varepsilon > 0$ for all histories implies tightness: the probability of still generating at time t is at most $(1-\varepsilon)^t$, so $\widetilde{p}_{\mathrm{EOS}(t)\le(1-\varepsilon)^t}$, and the survival sum is bounded by a convergent geometric series, meaning generation terminates almost surely.

Theorem 10.2 (Du et al. 2023, [15])**.** *Every softmax-output transformer language model with bounded parameters is tight: it defines a proper probability measure on Σ^*, assigning zero total mass to the set of infinite strings Σ^ω.*

This theorem is not a curiosity. It is the permission slip for everything that follows. All of the singularity-analysis machinery of Part I presupposes that the object being analyzed is actually a probability distribution, not a defective measure leaking mass to Σ^ω. Tightness is what guarantees that the length PGF $P(z) = \sum_{n\ge0} p_n z^n$ satisfies $P(1) = 1$ and that the counting generating function of the typical set has a well-defined radius of convergence whose reciprocal encodes the entropy rate. Transformers come with this guarantee built in, for free, by the structure of softmax.

Remark 10.3. The tightness bound $\varepsilon = e^{-2C}/|\Sigma|$ is extremely loose for any well-trained model. In practice $p(\mathrm{EOS} \mid h)$ varies enormously with h, approaching 1 at natural sentence boundaries and near 0 mid-token. But for the purpose of the existence theorem, the uniform floor suffices.

Contrast with non-tight recurrent models. Du et al. show by explicit construction that an RNN with ReLU activations can be non-tight: one can tune the network so that the EOS logit decays faster than any polynomial in the sequence length, causing $\sum_t \widetilde{p}_{\mathrm{EOS}(t)}$ to converge (their Example 2.4). Non-tightness is therefore a genuine pathology, not an artifact of poor training. It is avoidable only by architectural choice. The softmax output layer is exactly such a choice; other choices, including hard gates or ReLU activations applied directly to the EOS logit, do not yield the same guarantee. This makes the softmax operation load-bearing in a mathematical sense that goes beyond its role as a differentiable normalization.

10.6 Generating Functions for a Concrete LLM

Chapters 8 and 9 distinguished two generating functions associated with a string-valued distribution. For a concrete LLM both are well-defined objects, and the distinction matters.

Length PGF. Define $p_n = \Pr_{\mathcal{M}}(|W| = n)$, the probability that the model generates a string of exactly n tokens before emitting EOS (not counting the EOS token itself). The length

probability generating function is

$$P(z) \;=\; \sum_{n\geq 0} p_n z^n.$$

For a tight model, $P(1) = 1$. The singularity of P closest to the origin on the positive real axis — whether it lies at $z = 1$, beyond $z = 1$, or is a branch point — controls how the tail p_n decays as $n \to \infty$. This is the natural object when asking: "how long does the LLM generate before emitting EOS?" A model with a simple pole of P at $z_0 > 1$ produces geometrically decaying length tails; a model whose P has a branch point at $z = 1$ produces polynomial tails.

Counting GF. For $\alpha \in (0, 1)$, define the *α-typical set* T_n^α to be the set of strings w of length n with $p(w) \geq 2^{-n(h+\alpha)}$, where h is the entropy rate of $\mathcal{M}$. The counting generating function of the typical set is

$$A(z) \;=\; \sum_{n\geq 0} |T_n^\alpha|\, z^n.$$

By the asymptotic equipartition property, $|T_n^\alpha| \sim 2^{nh}$ for large n, so the radius of convergence of $A(z)$ is $R = 2^{-h}$, giving $h = \log_2(1/R)$. This is the natural object when asking: "how large is the set of likely outputs at length n?" The dominant singularity of A at $z = R$ connects the entropy rate of the model to the analytic structure of the counting series.

The two objects address two different questions, and the literature on each is largely disjoint. The length PGF P belongs to the domain of stopping-time analysis; the counting GF A belongs to the domain of information-theoretic combinatorics. Part V makes the separation explicit through the "Gap 1 vs. Gap 2" dichotomy [12]: Gap 1 concerns the discrepancy between a model's probability and a reference measure at fixed length (related to A), while Gap 2 concerns the probability assigned to strings beyond a certain length (related to P). When someone says "the generating function of an LLM," one must always ask which of these is meant.

10.7 An LLM as a Very-High-Order Markov Source

The autoregressive factorization makes an LLM a Markov chain on the state space Σ^* (the entire set of finite prefixes) with transition kernel $p(\cdot \mid h)$. This is an enormously high-dimensional, indeed uncountably branching, Markov chain, but it is still a Markov chain in the formal sense, and the theory of entropy rate, stationary distributions, and mixing from Part II Chapter 9 applies formally. The entropy rate

$$h(\mathcal{M}) \;=\; \lim_{n\to\infty} \frac{1}{n} H(W_1, \ldots, W_n)$$

exists for any tight autoregressive model by the subadditivity of entropy.

What is structurally new in the LLM setting is the scale of state dependence. A classical k-th order Markov source has transition kernel depending only on the last k tokens; its entropy rate can be estimated from $(k + 1)$-gram statistics, and its generating functions are rational by the transfer matrix theorem of Chapters 6–7. An LLM with full self-attention, by contrast, has transition kernel $p(\cdot \mid h)$ depending on the entire prefix h: the distribution over the next token can depend, in principle, on the token at position one. The long-range attention patterns break the finite-Markov-order structure entirely.

This has a direct consequence for the singularity-analysis program. The transfer matrix theorem guarantees rational GFs for finite-state Markov sources. If an LLM's attention machinery lets it approximate context-free or even more exotic behavior (Chapter 11), then the natural question becomes: does the LLM's output distribution lie in the rational, algebraic, or some wilder class of formal power series, and what does the dominant singularity of its length PGF or counting GF look like? These are precisely the questions of Parts IV and V.

Example 10.4. A unigram model, $p(\sigma \mid h) = q(\sigma)$ for a fixed distribution q, is a zero-order Markov source. Its length PGF is $P(z) = q(\mathrm{EOS})z/(1-(1-q(\mathrm{EOS}))z)$, a rational function with a simple pole at $z = 1/(1-q(\mathrm{EOS})) > 1$. Its counting GF has radius $2^{-H(q\setminus\mathrm{EOS})}$ where $H(q \setminus \mathrm{EOS})$ is the per-token entropy over non-EOS tokens. A full transformer deforms both of these objects in ways that depend on the corpus and the architecture, and the analytic structure of the resulting GFs is the subject of the rest of this book.

Chapter 11 maps the transformer architecture against the formal language hierarchy, and makes precise why vanilla transformers without chain-of-thought are constrained to operate near the rational-stochastic corner of that hierarchy. This matters because it implies that the output GFs of such models are expected to be "close to rational"—they should have a small number of dominant poles, with subdominant algebraic or essential singularities suppressed—and therefore amenable to the weighted finite automaton approximation program developed in Chapter 12. The tightness theorem of this chapter is the necessary prerequisite: it tells us the object whose singularities we are about to study is a genuine probability measure, not a ghost.

Further Reading

- [58] — the original transformer paper introducing scaled dot-product attention, multi-head attention, and positional encoding; the architectural description underlying the tightness argument in Section 5.
- [15] — the primary source for the tightness theorem for softmax transformers and for the measure-theoretic framework distinguishing tight from non-tight autoregressive models.
- [12] — develops the Gap 1 vs. Gap 2 dichotomy and the framework for comparing language models as probability measures on Σ^*; directly relevant to Section 6's separation of the length PGF from the counting GF.
- [25] — the probabilistic Chomsky hierarchy establishes that the rational-stochastic and algebraic-stochastic classes are distinct; the theoretical background for the question of where LLM output distributions fall in this hierarchy.
- [18] — the singularity analysis machinery applied throughout Parts IV and V to the length PGF and counting GF of LLMs; the analytic-combinatorics reference for the program outlined at this chapter's close.

Chapter 11

Formal Language Theory of Transformers

When a transformer LLM assigns probabilities to strings, what kind of formal object is it, really? This question matters for the analytic-combinatorics program because the generating function of the output distribution inherits its analytic type from the underlying formal class. If an LLM is (approximately) a weighted finite automaton, its generating function is rational and Chapter 7's partial-fraction story applies verbatim. If it is strictly more expressive, the generating function could be algebraic, D-finite, or wilder still, with correspondingly richer singular structure and qualitatively different asymptotics for $[z^n]L(z)$. This chapter surveys what is known about the answer, drawing on a body of results from circuit complexity theory, automata-theoretic language theory, and the expressivity literature for neural sequence models.

11.1 Circuit Complexity as a Measuring Stick

The tools that turn out to be most useful for characterizing transformer expressivity come not from automata theory directly but from Boolean circuit complexity. Three classes matter for what follows.

Definition 11.1. AC^0 is the class of languages recognizable by constant-depth, polynomial-size circuits over unbounded-fan-in AND, OR, and NOT gates. TC^0 extends AC^0 by adding threshold gates, which output 1 if and only if at least k of their inputs are 1, for any fixed k. NC^1 is the class recognizable by circuits of logarithmic depth and polynomial size over fan-in-2 gates.

It is widely conjectured, though not proved, that these containments are strict: $\mathsf{AC}^0 \subsetneq \mathsf{TC}^0 \subsetneq \mathsf{NC}^1$. Several classical separation results support this picture. PARITY—the language of binary strings containing an even number of 1's—is not in AC^0 (Håstad's switching lemma) but lies in TC^0 via a single threshold gate that computes the sum of the input bits modulo 2 through counting. General context-free language recognition, and the word problem for the symmetric group S_5, lie in NC^1 but are believed to lie outside TC^0, though this separation is conditional on standard complexity-theoretic assumptions.

For us the critical class is TC^0. The central theme of this chapter is that vanilla transformers—constant depth, bounded precision, no scratchpad—are essentially contained in TC^0, and that this ceiling has direct consequences for the analytic structure of the distributions they can represent.

11.2 Saturated Transformers and the TC^0 Upper Bound

[35] introduced the notion of a *saturated transformer* to make the expressivity analysis tractable. A softmax attention layer with queries Q and keys K computes attention weights $\alpha_{ij} =$

$\exp(q_i^\top k_j)/\sum_\ell \exp(q_i^\top k_\ell)$. In the saturated limit, one takes the hard version: each query attends uniformly and exclusively to those positions j that maximize $q_i^\top k_j$. This "hard attention" is a faithful proxy for the asymptotic behavior of softmax when the norms of queries and keys are held bounded and the model is evaluated on long sequences.

Theorem 11.2 (Merrill, Sabharwal, Smith 2022)**.** *Saturated transformers of constant depth and polynomial embedding dimension can be simulated by constant-depth Boolean threshold circuits of polynomial size. Consequently, any language recognized by such a model lies in* TC^0*.*

The proof constructs the circuit layer by layer. Each attention head computes, for each query position, a maximum over key-query dot products—an operation realizable by threshold gates—and then a uniform average over the selected positions, which is again a threshold computation. The feed-forward sublayers are handled by approximating RELU or GELU activations with polynomially many threshold gates. The depth of the resulting circuit is proportional to the depth of the transformer, hence constant.

The survey [53] systematizes and sharpens this line of work. For essentially every variant of "vanilla" transformer one might study—fixed depth, soft or hard attention, log-precision floating-point arithmetic, no intermediate scratchpad—the conclusion is the same: the class of languages recognizable lies inside TC^0. The survey also clarifies which model-theoretic choices (precision, positional encoding scheme, attention type) affect the exact position within TC^0 versus which merely amount to different constant-factor overheads.

The consequence for generating functions is sharp and important. If an LLM is constrained to TC^0 and therefore cannot uniformly recognize context-free languages, it is extremely unlikely to produce a distribution whose generating function satisfies a nontrivial polynomial equation over $\mathbb{Q}(z)$—that is, to be genuinely algebraic in the sense of Chapter 5. The natural expectation is that vanilla transformers produce distributions whose generating functions are approximately rational, "WFA-like" modulo approximation error at any fixed sequence length. This is the conceptual foundation for the WFA extraction program taken up in Chapter 12.

11.3 Hahn's Impossibility Theorem

Before the TC^0 machinery became standard, [22] proved a clean impossibility result by a soft-attention smoothing argument. The result remains the most intuitive lower-bound warning in the expressivity literature.

Theorem 11.3 (Hahn 2020)**.** *A fixed-depth, fixed-width transformer with soft attention cannot learn* PARITY *uniformly. More precisely, the cross-entropy loss of any such model on uniformly random binary strings of length n is bounded away from zero by a constant independent of n. The same conclusion holds for* 2DYCK*, the language of well-matched parentheses over two bracket types, at arbitrary nesting depths.*

The argument turns on a Lipschitz property of soft attention. Softmax attention weights are smooth functions of the input embeddings; if the keys and queries have bounded norm, a small perturbation of any single input token produces a small perturbation of every attention weight. By a telescoping argument across depth, the transformer's output is a Lipschitz function of each input token, with Lipschitz constant that does not grow with sequence length for fixed depth and width. But PARITY is maximally sensitive: flipping any single input bit flips the correct label, regardless of where in the string that bit appears. No Lipschitz function can track such global sensitivity uniformly over all lengths with constant Lipschitz constant. The same argument applies to 2DYCK because the matching structure at arbitrary depth requires unbounded hierarchical bookkeeping; a soft-attention model's ability to track bracket nesting collapses as nesting depth grows.

The moral for analytic combinatorics is precise. PARITY is the canonical language whose generating function has a singularity at $z = -1$: the string $0^k 1^k 0^k \cdots$ patterns that witness parity are exactly what produce alternating-sign contributions in the coefficient expansion. A model that cannot represent mod 2 counting cannot put mass consistently on such patterns, and therefore its generating function cannot have a pole at $z = -1$ arising from the underlying formal structure. More generally, periodic finite-state structure (controlled by roots of unity in the GF) and unbounded hierarchical nesting (controlled by branch points in the algebraic GF) are precisely the phenomena a vanilla transformer cannot faithfully reproduce.

11.4 Svete and Cotterell: RNN Language Models as Probabilistic Finite-State Automata

[55] establish an exact correspondence on the recurrent side of the architecture landscape. They prove that simple Elman-style RNN language models—networks where the hidden state $h_t = \sigma(W_h h_{t-1} + W_x x_t + b)$ updates recurrently—are equivalent, as distributions over Σ^*, to a specific subclass of probabilistic finite-state automata (PFSAs). The equivalence holds in both directions: every Elman RNN with a fixed activation function defines a PFSA, and every PFSA in the relevant subclass can be realized by some Elman RNN.

Theorem 11.4 (Svete, Cotterell 2023)**.** *An Elman RNN language model over alphabet Σ with hidden dimension d defines a distribution over Σ^* that is equivalent to a probabilistic finite-state automaton. Furthermore, representing an arbitrary N-state deterministic PFSA over Σ requires $\Omega(N|\Sigma|)$ hidden units, and this lower bound is asymptotically tight.*

The tightness proof is a direct construction: one can embed the transition table of an N-state PFSA into the weight matrices of an RNN using exactly $N|\Sigma|$ hidden units, with each unit tracking one state-symbol pair. The lower bound follows from a rank argument on the weight matrices.

This correspondence identifies a swath of recurrent language models with rational formal power series. For the corresponding subclass, the length-indexed generating function $L(z) = \sum_{n \geq 0} p_n z^n$, where p_n is the total probability mass on strings of length n, is a rational function of z. The Flajolet–Sedgewick toolkit then applies verbatim: one performs a partial-fraction decomposition of $L(z)$ over $\mathbb{C}$, identifies the dominant pole ρ (the smallest-modulus pole), and reads off the exponential decay rate $[z^n]L(z) \sim C\rho^{-n} n^k$ where k is the order of the pole minus one. The entire Chapter 7 program is available, and the translation from RNN architecture to analytic GF is explicit rather than approximate.

11.5 Rizvi et al.: Transformers Simulating Weighted Automata

[46] carry the equivalence across the architectural gap from recurrent networks to transformers, giving constructive simulation theorems.

Theorem 11.5 (Rizvi et al. 2024, Theorem 1)**.** *A transformer with bilinear attention layers and hard attention can exactly simulate any n-state weighted finite automaton (WFA) over inputs of length T, using depth $O(\log T)$, embedding dimension $O(n^2)$, and $O(1)$ attention heads per layer.*

Theorem 11.6 (Rizvi et al. 2024, Theorem 2)**.** *A standard transformer with softmax attention and feed-forward MLP sublayers can approximately simulate any n-state WFA over inputs of length T, at depth $O(\log T)$, with approximation error controlled by the MLP approximation quality.*

The construction underlying both results is a parallel-prefix scan of the WFA's matrix semiring. For a WFA with transition matrices $\{M_\sigma\}_{\sigma\in\Sigma}$, the weight assigned to a string $x_1x_2\cdots x_T$ is $\alpha^\top M_{x_1}M_{x_2}\cdots M_{x_T}\beta$, where α and β are initial and final weight vectors. Computing this product in a transformer requires aggregating information from all positions, which cannot be done with constant depth and a linear scan. Instead, Rizvi et al. implement the product via a divide-and-conquer binary tree: the transformer at depth ℓ holds partial products over intervals of length 2^ℓ, combining them layer by layer. This is precisely the kind of computation that TC^0 at logarithmic depth (which does reach NC^1) admits, and the $O(\log T)$ depth is not an accident—it matches the depth of the binary tree over the input positions. Rizvi et al. also extend the construction to weighted tree automata over balanced binary trees, again at logarithmic depth.

Combined with the universal approximation result of [60]—which establishes that transformers with positional encodings are dense in the space of continuous sequence-to-sequence functions on compact domains—this gives a quantitative picture: transformers are, in a precise sense, WFAs plus controlled error at each fixed input length, with the approximation quality improving as depth grows logarithmically in sequence length. The gap between "exact simulation" and "actual vanilla transformer" is an engineering matter of depth and precision, not a fundamental structural one.

11.6 Chain of Thought Breaks the TC^0 Ceiling

Everything in the previous sections applies to transformers that commit to an answer in a single forward pass. The architecture reads the input, computes through its layers, and outputs a distribution over the next token (or a label) in one shot. But modern LLMs are often deployed in a mode where they emit intermediate tokens—reasoning steps, calculations, scratch work—before committing to a final answer. This is chain-of-thought (CoT) prompting, and it fundamentally changes the expressivity analysis.

[34] give an exact characterization of what CoT enables.

Theorem 11.7 (Merrill, Sabharwal 2024)**.** *A transformer with generalized pre-norm, when allowed to emit and re-read a polynomial number of intermediate chain-of-thought tokens before producing its final output, recognizes exactly the class of languages in* P*—the class of problems decidable in polynomial time by a deterministic Turing machine.*

This is the first exact identification of a transformer variant with a standard complexity class, and it is a decisive break: $\mathsf{TC}^0 \subseteq \mathsf{NC}^1 \subseteq \mathsf{P}$ by classical inclusions, and P is believed to be strictly larger than NC^1 (this is the $\mathsf{NC} \neq \mathsf{P}$ conjecture, closely related to $\mathsf{P} \neq \mathsf{NC}$). The mechanism is that each CoT token allows the transformer to use its full TC^0 machinery to compute one step of a Turing machine simulation; stringing together polynomially many such steps gives full polynomial-time computation.

The analytic-combinatorics consequence is radical. CoT is the mechanism by which a transformer stops being (approximately) a WFA and starts producing distributions whose generating functions need not be rational at all. If the transformer uses CoT to simulate a pushdown automaton, the resulting string distribution is context-free, and its generating function satisfies an algebraic equation—yielding the $n^{-3/2}$ coefficient decay and square-root branch-point structure described in Chapter 5. If the CoT implements polynomial-time algorithms of greater complexity, even holonomic (D-finite) structure is possible, with richer singular geometry beyond the scope of this book.

Concretely: if you are computing $[z^n]L(z)$ for a CoT-enabled LLM and you expect rational asymptotics, you may be making an architectural error. The model you are analyzing may be in P, not in TC^0, and the generating function may have branch points rather than poles as its dominant singularities. The mathematical objects are different, not merely quantitatively but qualitatively.

11.7 Synthesis for the Remainder of the Book

The theoretical picture assembled in this chapter—drawing on [53], [22], [35], [55], [46], and [34]—yields three precise conclusions that organize the rest of the book.

(i) Without chain-of-thought, a transformer LM's distribution is constrained to live near the WFA-approximable corner of the TC^0 landscape. Its length generating function $L(z)$ is expected to be well-approximated by a rational function up to controlled error, with approximation quality improving with model depth and the logarithm of sequence length. The dominant pole of a WFA approximation is the natural first-order description of the exponential decay rate of the length tail, and the finite spectrum of subdominant poles controls subleading corrections. The partial-fraction decomposition framework of Chapter 7 applies, at least approximately.

(ii) With chain-of-thought, the same model can produce distributions whose analytic structure escapes the rational regime entirely. Algebraic generating functions with square-root branch points are achievable—which, by Chapter 5's transfer theorems, would produce $n^{-3/2}$ asymptotics in the length distribution rather than the pure exponential decay characteristic of poles. Whether actual deployed LLMs in CoT mode exhibit this algebraic signature is an open empirical question; it is listed as Gap 5 in Part V. The conjecture is plausible for models that have been trained to reason about balanced structures (such as mathematical proofs with matched quantifiers, or code with balanced brackets), but it has not been verified by a direct singularity analysis.

(iii) Any study of LLM output distributions via analytic combinatorics must declare which regime it operates in. The mathematics differs not by degree but by kind. Rational generating functions have finitely many singularities, all poles, with coefficient asymptotics that are linear combinations of terms of the form $C_k \rho_k^{-n} n^{m_k}$. Algebraic generating functions have branch points whose local Puiseux expansions control more complex coefficient behavior. D-finite generating functions satisfy linear differential equations with polynomial coefficients, and their asymptotics are governed by the local monodromy of those equations. These are not the same mathematical universe, and applying the wrong toolkit produces wrong answers.

Remark 11.8. The TC^0 upper bound does not rule out the possibility that a transformer in single-pass mode can approximate any rational GF arbitrarily well in the total-variation sense on strings of length up to T. What it rules out is *uniform* recognition across all lengths with a fixed circuit. In practice, the approximation will be excellent for lengths well within the training distribution and will degrade near the boundary of that distribution. This distributional mismatch is itself a singularity-like phenomenon: the model's effective GF changes character as n grows beyond the training horizon. Detecting and characterizing this transition is one of the goals of the empirical program in Part IV.

The reader who has internalized this chapter is equipped to open any 2024–2026 paper on transformer expressivity and situate its claims precisely: which architecture, which precision, which mode of operation (single-pass or CoT), and which complexity class is being invoked. Chapter 12 turns to the operational question: given a trained LLM operating in single-pass mode, can one actually extract a WFA approximation, and if so, how faithfully does the extracted automaton capture the original distribution's generating function?

Further Reading

- [53] — A comprehensive survey of transformer expressivity results, systematizing the TC^0 upper bounds across model variants (attention type, precision, positional encoding) and clarifying which architectural choices matter for the complexity classification.

- [35] — Introduces saturated transformers and proves the foundational TC^0 upper bound via an explicit circuit simulation; the paper that established the main complexity ceiling

discussed throughout this chapter.

- [22] — Proves that fixed-depth soft-attention transformers cannot represent PARITY or 2DYCK uniformly across lengths; the cleanest impossibility result in the expressivity literature and essential context for understanding the WFA approximation program.
- [55] — Establishes the exact equivalence between Elman RNN language models and probabilistic finite-state automata, with tight bounds on the hidden-dimension cost of representing a PFSA; provides the recurrent-architecture counterpart to the transformer results.
- [46] — Gives constructive simulation theorems showing that transformers at logarithmic depth can exactly (or approximately) simulate weighted finite automata via parallel-prefix scans of the WFA transition matrices.
- [34] — Proves that transformers with polynomial-length chain-of-thought exactly recognize P, establishing the decisive break beyond the TC^0 ceiling and its implications for the analytic structure of CoT-enabled LLM distributions.

Chapter 12

Approximating Language Models by Weighted Finite Automata

Chapter 11 established a hard ceiling: Transformers with fixed precision lie in TC^0, cannot compute PARITY exactly, and therefore cannot be rational power series in any faithful sense. Yet Chapter 7 gave us a clean toolkit — weighted finite automata (WFA) and their rational generating functions $F(z) = \alpha(I - zA)^{-1}\beta$ — whose asymptotic behavior we control completely via the singularity analysis of Chapter 4. The gap between these two statements is the subject of this chapter.

The proposal is pragmatic: if we cannot describe a Transformer exactly as a WFA, we can still try to *approximate* it by one. A good approximation is one that is faithful enough for the statistics we care about (per-token probabilities, conditional distributions, string-level scores) and small enough that its rational generating function is tractable. Once such an approximation is in hand, every theorem of Parts I and II applies to it: entropy rate, coefficient asymptotics, limit laws, phase transitions.

This chapter surveys four lines of attack on the approximation problem. They are not mutually exclusive; a mature pipeline would combine them.

12.1 Line 1: Black-Box Distillation

The oldest and simplest approach treats the language model as an opaque scoring function $s : \Sigma^* \to \mathbb{R}$, $s(w) = \log \mu(w)$, and fits a WFA to minimize the KL divergence between the WFA's induced distribution and the model's on a corpus.

Concretely: fix a target number of states k, a target alphabet (possibly a coarsening of the model's tokenizer), and a parameterization of $(A_\sigma)_{\sigma\in\Sigma}$ with α, β. Write the WFA-induced score as $\hat{s}(w) = \log \alpha A_{w_1} \cdots A_{w_n} \beta$. Minimize

$$\mathcal{L}(A, \alpha, \beta) = \frac{1}{|\mathcal{D}|} \sum_{w\in\mathcal{D}} \big(s(w) - \hat{s}(w)\big)^2$$

or the corresponding KL objective on a held-out sample from the LLM.

Historically this idea predates the Transformer era. Lecorvé and Motlicek 2012 [31] compile a neural recurrent language model into a weighted finite-state transducer for speech-recognition rescoring; the motivation was latency and integration with existing finite-state decoders. Suresh, Roark, Riley, Schogol *et al.* 2021 [54] extend the programme to modern neural LMs, distilling them into n-gram and WFST surrogates that fit on-device and meet real-time constraints.

The limitations are real: black-box distillation treats the LLM as a teacher whose internal structure gives no guidance to the student. The student's expressive power is bounded by its state count, and its error depends on how well the corpus $\mathcal{D}$ covers the parts of Σ^* that

matter. What it buys you: a genuinely rational $\hat{s}$, hence a genuinely rational generating function $\hat{F}(z) = \alpha(I - zA)^{-1}\beta$ where $A = \sum_\sigma A_\sigma$, to which all of Parts I and II applies.

12.2 Line 2: Architectural Equivalence

A complementary observation is that several popular recurrent architectures *are already* weighted finite automata — or rather, their computation graph is literally that of a WFA with a particular semiring and a particular transition operator. Recognising this turns an empirical fit into a structural identity.

Schwartz, Thomson and Smith 2018 [49] ("Bridging CNNs, RNNs, and Weighted Finite-State Machines") show that certain RNN variants with elementwise activations and specific gating patterns compute exactly the function

$$h_t = \bigoplus_{\sigma \in \Sigma} A_\sigma \odot h_{t-1}$$

with $(\oplus, \odot)$ a semiring. Different choices of semiring recover different RNN variants. Peng, Schwartz, Thomson and Smith 2018 [43] ("Rational Recurrences") make the equivalence precise: an RNN is a *rational recurrence* when its hidden state evolves as a WFA over an appropriate semiring. They exhibit several contemporary architectures as rational recurrences in disguise.

The payoff is conceptual. If a model is a rational recurrence, then its log-score is a log-rational function and the counting generating function of its high-probability language is governed by the same machinery as classical rational power series. Transformers are *not* rational recurrences — Chapter 11 ruled this out — but some of their submodules and many of their competitors are.

12.3 Line 3: Query Learning and L^*

A more ambitious approach refuses to pick the WFA architecture in advance. Angluin's L^* algorithm (1987) learns a minimal DFA from a teacher who answers two kinds of queries:

- *Membership queries*: "is $w \in L$?"
- *Equivalence queries*: "is my current hypothesis H equal to L? If not, return a counterexample."

L^* maintains an observation table indexed by candidate prefixes and candidate suffixes; membership queries fill in the table, and when the table is closed and consistent, the implied DFA is the current hypothesis. An equivalence query either confirms or returns a string on which H and L disagree, which the algorithm uses to refine the table. The running time is polynomial in the size of the minimal DFA and the length of the longest counterexample.

Weiss, Goldberg and Yahav 2018 [59] adapt L^* to extract automata from trained RNNs. The RNN plays the role of the teacher. Membership queries are cheap: just run the RNN on w and threshold its score. Equivalence queries are the hard part — there is no oracle that can certify two infinite-state machines are equivalent — so they are approximated by sampling from a space of potential counterexamples and by using state-partitioning heuristics on the RNN's hidden activations.

The WFA extension is straightforward in principle: replace the DFA observation table with a Hankel matrix (below) and replace minimality with rank-minimality. The hard part remains the equivalence query. For LLMs, practical L^* variants usually stop at a prescribed fidelity level on a validation set rather than at exact equivalence.

Remark 12.1. The L^* route has a feature that distillation lacks: it is *adaptive*. Rather than scoring a fixed corpus, the algorithm chooses which strings to query based on what it currently does not know. This makes it well-suited to exposing rare but structurally important strings — counter-examples — that blind corpus sampling would miss.

12.4 Line 4: Spectral Learning and the AAK Approach

The fourth line is the most mathematically substantial and connects directly to the generating-function viewpoint. It is built on the *Hankel matrix* of a string function.

Definition 12.2 (Hankel matrix)**.** Let $f : \Sigma^* \to \mathbb{R}$. The *Hankel matrix* of f is the bi-infinite matrix H_f with rows indexed by prefixes $u \in \Sigma^*$, columns indexed by suffixes $v \in \Sigma^*$, and entries

$$(H_f)_{u,v} = f(uv).$$

The fundamental theorem of rational power series (Fliess 1974; see also Berstel–Reutenauer 1988) states that f is rational (computable by a WFA with finitely many states) if and only if H_f has finite rank, and the minimal rank equals the minimal number of WFA states. This identifies the problem of finding a small WFA with the problem of finding a low-rank approximation of a Hankel matrix.

Proposition 12.3 (Spectral realization)**.** *Suppose f has Hankel matrix of rank k. Let $H = U\Sigma V^\top$ be a compact SVD truncated to rank k. Then there exist explicit formulas for $\alpha \in \mathbb{R}^k$, $\beta \in \mathbb{R}^k$, and matrices $A_\sigma \in \mathbb{R}^{k\times k}$ such that $f(w) = \alpha^\top A_{w_1} \cdots A_{w_n}\beta$ for every w.*

The explicit formulas are:

$$A_\sigma = (U^\top H)^+ U^\top H^{(\sigma)} (V^\top)^+, \qquad \alpha = (U^\top)^+ h_\varepsilon^{\text{row}}, \qquad \beta = V^+ h_\varepsilon^{\text{col}},$$

where $H^{(\sigma)}$ is the shifted Hankel matrix $H^{(\sigma)}_{u,v} = f(u\sigma v)$, and $h_\varepsilon^{\text{row}}, h_\varepsilon^{\text{col}}$ are the prefix/suffix rows at the empty string. This is the *spectral learning* construction of Hsu–Kakade–Zhang 2009, Bailly–Denis–Ralaivola 2009, and Balle–Quattoni–Carreras 2014, among others.

Rabusseau, Li and Precup 2019 [45] apply spectral learning directly to RNNs: sample enough values of $f(w) = s_{\text{RNN}}(w)$ to populate a sufficiently large finite submatrix of H_f, take its SVD, read off a WFA.

The problem is that the Hankel matrix of a real LLM is essentially full rank. Spectral learning returns a WFA whose rank you must choose, and a straightforward low-rank truncation of the SVD gives only a Frobenius-optimal approximation, not the optimal WFA in any natural norm on the string function.

The operator-theoretic remedy is the *Adamyan–Arov–Krein (AAK) theorem.* In its classical form (one-letter alphabet, Hardy space H^2), AAK states that for a bounded Hankel operator Γ, the best rank-k approximation Γ_k in operator norm satisfies $\|\Gamma - \Gamma_k\| = \sigma_{k+1}(\Gamma)$, and Γ_k is itself Hankel. In plain terms: the best low-rank approximation of a Hankel matrix that is *still Hankel* (and hence still corresponds to a genuine WFA, not to an arbitrary low-rank matrix) has error exactly the $(k+1)$st singular value.

Lacroce, Panangaden and Rabusseau 2021 [30] apply constrained AAK theory to WFA extraction from RNNs, and Lacroce, Balle, Panangaden and Rabusseau 2024 [29] (MSCS) give a clean treatment of the one-letter case. Balle and collaborators [2] extend the machinery to multi-letter alphabets, which is technically much harder because the right notion of Hankel operator is non-commutative.

12.5 One-Letter Is a Feature, Not a Bug

A running theme in lines 3 and 4 is that the cleanest theory is the one-letter theory. When $|\Sigma| = 1$, a rational power series $\sum_n f(1^n)z^n$ is an ordinary scalar power series in z, and its singularities sit in $\mathbb{C}$ where all the tools of Part I apply directly. The Hankel operator Γ is a scalar Hankel matrix in Hardy space, where AAK is classical. Spectral learning becomes linear algebra over the reals, without the free-semigroup combinatorics.

This is not a pathology to be overcome. It is the setting in which the analytic-combinatorics dictionary translates most literally, and it is the setting in which the entropy-rate identity $h = \log(1/R)$ of Chapter 9 takes its textbook form. A multi-letter LLM can be coarsened to a one-letter surrogate simply by collapsing the alphabet: define $\tilde{f}(1^n) = \sum_{w:\,|w|=n} \mu(w)$ and work with $\tilde{f}$. The resulting scalar power series is the length probability generating function of Chapter 10, and its singularity structure governs length-based statistics exactly.

The multi-letter theory then supplies refined joint information once the univariate picture is understood. The hierarchy is: start with the one-letter projection, compute its radius of convergence and singular exponent, then refine.

12.6 Scale Realities

All four lines have been validated on small recurrent models — LSTMs and GRUs with a few hundred to a few thousand hidden units, trained on Penn Treebank or similar corpora. Scaling to contemporary Transformers is an open problem with multiple obstructions.

First, the empirical Hankel matrix of a multi-billion-parameter LLM has effective rank that is certainly much larger than anything a spectral method can realize. Whether there is a meaningful low-rank approximation at all is not obvious; empirically, the singular-value decay of Hankel submatrices sampled from Transformers is slow.

Second, the equivalence-query problem in L^* becomes essentially intractable: no sample-based approximation plausibly certifies agreement with a trillion-token model.

Third, the distillation objective is well-defined but the target class of WFAs is small compared to the signal you are trying to fit. Any distilled WFA will systematically lose information at long range.

None of this argues against the programme. It argues that the programme must begin on small, controllable surrogates — for instance, Transformers trained on PCFG-generated data (Chapter 8) where the ground-truth rational structure is known — and graduate to realistic models only once the surrogate theory is in place.

12.7 What WFA Approximation Buys

Suppose we have a WFA approximation $\widehat{M}$ of an LLM with transition matrix $\hat{A} = \sum_\sigma \hat{A}_\sigma$ and rational generating function $\hat{F}(z) = \hat{\alpha}(I - z\hat{A})^{-1}\hat{\beta}$. What can we then say about the original LLM?

Proposition 12.4 (Entropy rate of the approximation)**.** *If $\widehat{M}$ is a WFA whose transition matrix $\hat{A}$ is irreducible with spectral radius $\hat{\rho} = \rho(\hat{A})$, then the entropy rate of the stationary distribution on $\widehat{M}$-emitted strings equals $\log \hat{\rho}$, and the radius of convergence of $\hat{F}(z)$ equals $1/\hat{\rho}$. In particular, the universal identity $\hat{h} = \log(1/R_{\hat{F}})$ of Chapter 9 holds for $\widehat{M}$.*

Proof sketch. The entries of $\hat{A}^n$ grow as $\hat{\rho}^n$ by Perron–Frobenius, so $[z^n]\hat{F}(z) \sim C\hat{\rho}^n$ for some constant C. The radius of convergence of $\hat{F}$ is therefore $1/\hat{\rho}$, and the exponential growth rate of the counting generating function is $\log \hat{\rho}$, which is the entropy rate under the symbolic-dynamics interpretation of Chapter 9. □

The proposition transfers immediately if you accept $\widehat{M}$ as a faithful proxy for the original. The caveat is the fidelity gap: $\hat{h}$ may differ from the true h by something controlled by the approximation error of the WFA. Quantifying this gap analytically (rather than empirically) is one of the open problems listed in Chapter 18.

12.8 Notes and Further Reading

The weighted-automata / spectral-learning literature is larger than any single chapter can cover. Berstel and Reutenauer's *Noncommutative Rational Series with Applications* (Cambridge, 2011) is the authoritative algebraic reference. Balle's survey on spectral learning of WFAs (JMLR MLOSS special issue) collects the practical algorithms. The distillation line of work is covered in the references cited in Section 12.1. For the AAK theorem in its classical Hardy-space form, Peller's *Hankel Operators and Their Applications* (Springer, 2003) is the standard monograph; Lacroce's thesis and the papers cited in Section 12.4 adapt it to the WFA setting.

- [31] — Early instantiation of the distillation program: compiles a neural recurrent language model into a weighted finite-state transducer for speech-recognition rescoring, motivating the black-box approximation approach of Section 12.1.
- [54] — Extends neural LM distillation to modern architectures, producing n-gram and WFST surrogates suitable for on-device deployment; demonstrates the practical viability of rational approximations at scale.
- [59] — Adapts Angluin's L^* algorithm to extract automata from trained RNNs, with the RNN serving as the membership oracle; the foundational paper for the query-learning line of WFA extraction.
- [45] — Applies spectral learning (Hankel-matrix SVD) directly to RNN state representations, providing explicit algorithms for reading off WFA transition matrices from the singular vectors.
- [30] — Applies constrained AAK theory to WFA extraction, showing that the best rank-k approximation that is *still* a genuine WFA (i.e., still Hankel) has operator-norm error equal to σ_{k+1} of the Hankel operator.
- [7] — Proves the ARM–EBM bijection via soft Bellman equations, establishing that autoregressive and energy-based parameterizations are equiexpressive; directly relevant to understanding the limits of WFA approximation for globally normalized models.

Exercise 12.5. Consider a 2-state WFA over $\{a\}$ with

$$A_a = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad \alpha = (1, 0), \quad \beta = (0, 1)^\top.$$

Compute the scalar generating function $F(z) = \alpha(I - zA_a)^{-1}\beta$ and identify its poles. Verify that the coefficients $[z^n]F(z)$ are Fibonacci numbers and that the dominant pole corresponds to the golden ratio.

Exercise 12.6. Let $f(1^n) = n + 1$. Show that the scalar Hankel matrix $H_{i,j} = f(1^{i+j})$ has rank 2 and exhibit an explicit 2-state WFA realizing f. (Hint: $f(1^n)$ is rational with a double pole at $z = 1$.)

Part IV

Statistical Mechanics Meets Generating Functions

Chapter 13

Boltzmann Sampling

13.1 Motivation

Suppose we want to draw a uniformly random element of $\mathcal{A}_n$, the set of all structures in a combinatorial class $\mathcal{A}$ of size exactly n. If $\mathcal{A}$ is described by a symbolic specification, the classical approach is the *recursive method* of Nijenhuis and Wilf [41], later systematized by Flajolet, Zimmermann and Van Cutsem [19]. The idea is to precompute a table of values $[z^k]A(z)$ for $k \leq n$, then walk down the recursive structure of the specification, at each node making a random choice with probabilities proportional to those table entries. This is exact and elegant, but it has costs: storing the table requires $O(n^2)$ space (one entry per size per subclass), and generating a single sample takes $O(n \log n)$ arithmetic operations once the table is built.

For large n, or for classes whose specifications are complex, this becomes unwieldy. Duchon, Flajolet, Louchard and Schaeffer [16] proposed an alternative: rather than insisting on *exact* size n, allow the size to be random, governed by a parameter x. By tuning x one can concentrate the size distribution near a target n, and by combining this with rejection sampling one recovers a uniform sampler for exact size n at a dramatically reduced cost — $\Theta(n)$ expected time for well-behaved classes, with no precomputed tables.

The approach is called *Boltzmann sampling*, and as we shall see, the name is not accidental. The probability weights assigned to structures are precisely Boltzmann weights from statistical physics. This identification — generating function equals partition function — is the central observation of this chapter, and it will be the bridge to Chapter 14, where LLM temperature decoding is read in exactly the same terms.

13.2 Boltzmann Samplers

13.2.1 Definition

Let $\mathcal{A}$ be a combinatorial class with generating function $A(x) = \sum_{n \geq 0} a_n x^n$, where $a_n = |\mathcal{A}_n|$, and let ρ_A denote the radius of convergence of A.

Definition 13.1. Fix $x \in (0, \rho_A)$. A *(free) Boltzmann sampler* $\Gamma A(x)$ is a random procedure that produces an object $\gamma \in \mathcal{A}$ with probability

$$\Pr_x(\gamma) = \frac{x^{|\gamma|}}{A(x)}.$$

Several observations are immediate. First, this is a valid probability distribution: summing over all $\gamma \in \mathcal{A}$,

$$\sum_{\gamma \in \mathcal{A}} \frac{x^{|\gamma|}}{A(x)} = \frac{1}{A(x)} \sum_{n \geq 0} a_n x^n = 1.$$

Second, all objects of the *same size* n receive equal probability $x^n/A(x)$. So conditioning on size, $\Gamma A(x)$ yields a uniform element of $\mathcal{A}_n$. This is the key uniformity property that makes Boltzmann samplers useful for random generation.

Third, the size $N = |\gamma|$ is a random variable. Its distribution is

$$\Pr_x(N = n) = \frac{a_n x^n}{A(x)},$$

which is the probability generating function (PGF) model with weight $a_n x^n$ normalized by $A(x)$. The expected size is

$$\mathbb{E}_x[N] = \sum_{n \geq 0} n \cdot \frac{a_n x^n}{A(x)} = \frac{xA'(x)}{A(x)}.$$

This last formula is important: as x increases toward ρ_A, the expected size grows (often without bound), so we can tune the expected size of our random sample by adjusting x.

13.2.2 Construction from Symbolic Specifications

The power of the Boltzmann framework lies in how naturally it interacts with the symbolic method. If $\mathcal{A}$ is defined by a combinatorial specification, the corresponding Boltzmann sampler is built recursively from the same structure.

We record the four main constructors; proofs are straightforward from the definition.

Disjoint union. Suppose $\mathcal{A} = \mathcal{B} + \mathcal{C}$, so $A(x) = B(x) + C(x)$. Then $\Gamma A(x)$ proceeds as follows: flip a biased coin that lands on $\mathcal{B}$ with probability $B(x)/A(x)$ and on $\mathcal{C}$ with probability $C(x)/A(x)$; if $\mathcal{B}$ is chosen, return $\Gamma B(x)$; otherwise return $\Gamma C(x)$.

To verify: an object $\gamma \in \mathcal{B}$ is produced with probability

$$\frac{B(x)}{A(x)} \cdot \frac{x^{|\gamma|}}{B(x)} = \frac{x^{|\gamma|}}{A(x)},$$

and similarly for $\mathcal{C}$, so the definition is satisfied.

Cartesian product. Suppose $\mathcal{A} = \mathcal{B} \times \mathcal{C}$, so $A(x) = B(x) \cdot C(x)$. Then $\Gamma A(x)$ independently draws $\beta \sim \Gamma B(x)$ and $\gamma \sim \Gamma C(x)$, and returns the pair (β, γ).

Indeed, $\Pr_x((\beta, \gamma)) = \frac{x^{|\beta|}}{B(x)} \cdot \frac{x^{|\gamma|}}{C(x)} = \frac{x^{|\beta|+|\gamma|}}{A(x)}$, as required.

Sequence. Suppose $\mathcal{A} = \mathrm{SEQ}(\mathcal{B})$, so $A(x) = 1/(1 - B(x))$, valid when $B(x) < 1$. Draw a length K from the geometric distribution $\Pr(K = k) = (1 - B(x))B(x)^k$ for $k \geq 0$, then independently draw K objects from $\Gamma B(x)$, and return the sequence $(\beta_1, \ldots, \beta_K)$.

The probability of producing a specific sequence $(\beta_1, \ldots, \beta_k)$ of total size $n = |\beta_1| + \cdots + |\beta_k|$ is

$$(1 - B(x))B(x)^k \cdot \prod_{i=1}^{k} \frac{x^{|\beta_i|}}{B(x)} = (1 - B(x)) \cdot \frac{x^n}{B(x)^k} \cdot B(x)^k = (1 - B(x))x^n = \frac{x^n}{A(x)},$$

confirming the Boltzmann property.

Set and cycle. For $\mathcal{A} = \mathrm{SET}(\mathcal{B})$ and $\mathcal{A} = \mathrm{CYC}(\mathcal{B})$, the constructions involve Poisson distributions and logarithmic-series distributions respectively. Specifically, the number of copies of a given unlabeled component of size k in a random set structure follows a Poisson distribution with mean $(x^k/k!)$, and analogously for cycles. We do not spell out the full surgery here; the reader should consult [18], Chapter VII, for the complete treatment.

13.2.3 Tuning the Size and Rejection Sampling

For a fixed target size n, one uses $\Gamma A(x)$ as a subroutine in a rejection scheme: run $\Gamma A(x)$ repeatedly and keep only outputs of size n. Since all such outputs are equally likely (by the uniformity property), each kept sample is uniform over $\mathcal{A}_n$.

The question is efficiency. The probability of getting an object of size exactly n is $a_n x^n/A(x)$. To maximize this, one should choose x to maximize $a_n x^n/A(x)$ over $x \in (0, \rho)$, which by a saddle-point argument gives $x^* = x^*(n)$ satisfying $x^* A'(x^*)/A(x^*) = n$, i.e., the expected size equals n.

Proposition 13.2. *For combinatorial classes satisfying the smoothness conditions of [16] (roughly: $a_n \sim C\rho^{-n} n^{-\alpha}$ for some $\alpha > 0$), rejection sampling with the optimal parameter $x^*(n)$ produces a uniform element of $\mathcal{A}_n$ in $\Theta(\sqrt{n})$ expected trials if $\alpha = 3/2$, and $\Theta(1)$ if $\alpha > 1$. In either case the total expected work is $\Theta(n)$.*

This should be contrasted with the recursive method's $O(n^2)$ preprocessing: for large classes and large n, the Boltzmann approach wins decisively.

13.3 The Gibbs Measure Interpretation

We now come to the central observation of this chapter.

13.3.1 Boltzmann Weights as Gibbs Weights

Make the substitution $x = e^{-\beta}$, where $\beta \in \mathbb{R}$ is to be thought of as an *inverse temperature*. The Boltzmann sampler probability becomes

$$\Pr_\beta(\gamma) = \frac{e^{-\beta|\gamma|}}{A(e^{-\beta})}.$$

Define the *energy* of a structure γ to be its size: $E(\gamma) = |\gamma|$. Then

$$\Pr_\beta(\gamma) = \frac{e^{-\beta E(\gamma)}}{Z(\beta)}, \qquad Z(\beta) = A(e^{-\beta}).$$

This is exactly the Gibbs (Boltzmann) distribution from statistical mechanics. The normalization $Z(\beta)$ is the *partition function*, and here it is simply the generating function of the class, evaluated at $e^{-\beta}$.

Remark 13.3. **The generating function is the partition function.** For any combinatorial class $\mathcal{A}$ with generating function $A(z)$ and energy function $E(\gamma) = |\gamma|$, the Gibbs distribution at inverse temperature β is

$$\Pr_\beta(\gamma) = \frac{e^{-\beta|\gamma|}}{A(e^{-\beta})},$$

and the partition function is $Z(\beta) = A(e^{-\beta})$. This identification is not a formal analogy — it is an equality of formulas. The analytic-combinatorics side (generating functions, singularities, coefficients) and the statistical-physics side (partition functions, free energy, phase transitions) are the same mathematical object viewed through different lenses. This is the single most important observation of this chapter.

13.3.2 The Critical Temperature

The radius of convergence ρ_A of $A(z)$ corresponds to a *critical inverse temperature*

$$\beta_c = -\log \rho_A.$$

For $\beta > \beta_c$ (i.e., $e^{-\beta} < \rho_A$), the partition function $Z(\beta) = A(e^{-\beta})$ is finite, and the Gibbs measure is a well-defined probability distribution on $\bigcup_{n\geq 0} \mathcal{A}_n$.

For $\beta < \beta_c$ (i.e., $e^{-\beta} > \rho_A$), the series $A(e^{-\beta})$ diverges, so no normalizable Gibbs measure exists on the whole class. One can still define a uniform distribution on each fixed $\mathcal{A}_n$ separately (the recursive method does exactly this), but there is no consistent extension to a single measure across all sizes.

At $\beta = \beta_c$, whether the partition function converges depends on the growth rate of a_n. For many natural classes, $a_n \sim C\rho^{-n}n^{-\alpha}$ with $\alpha = 3/2$ (trees, planar maps, etc.), in which case $Z(\beta_c) = A(\rho) = \sum a_n\rho^n \sim \sum Cn^{-3/2}$, which *converges* (since $3/2 > 1$). Thus the partition function is finite at criticality, and the Gibbs measure at β_c is well-defined, though the resulting size distribution $\Pr_{\beta_c}(N = n) \propto n^{-3/2}$ is heavy-tailed. This heavy-tailed sampler at $x = \rho$ feeds into the rejection scheme of Proposition 13.2.

13.4 Free Energy and the Thermodynamic Limit

For finite-system statistical mechanics, it is standard to study the partition function restricted to systems of size n:

$$Z_n(\beta) = a_n\, e^{-\beta n} = [z^n]A(z) \cdot e^{-\beta n}.$$

This is simply a_n rescaled, and it is the unnormalized probability of drawing a structure of size n from the Gibbs distribution.

The *free energy per unit size* is

$$f(\beta) = -\lim_{n\to\infty} \frac{1}{n} \log Z_n(\beta) = \beta - \lim_{n\to\infty} \frac{1}{n} \log a_n.$$

By the standard theory of exponential growth rates, if $a_n \sim C\rho^{-n}n^{-\alpha}$, then $\lim_{n\to\infty} n^{-1} \log a_n = \log(1/\rho)$. Hence

$$f(\beta) = \beta - \log(1/\rho) = \beta + \log \rho.$$

At the critical temperature $\beta_c = -\log \rho$, the free energy vanishes: $f(\beta_c) = 0$.

This connects directly to the entropy rate of Chapter 9. There we saw that the exponential growth rate $\log(1/R)$ of the coefficient sequence of a generating function is the entropy rate of the associated random process (with R the radius of convergence). Here the same quantity $\log(1/\rho)$ appears as the thermodynamic driving force: the free energy is zero when the inverse temperature is tuned to the entropy rate.

At high temperature (β small, $x = e^{-\beta}$ close to 1), the Gibbs measure strongly favors large structures; the expected size $xA'(x)/A(x)$ diverges as $x \to \rho^-$. At low temperature (β large), the measure concentrates on small structures, with $\Pr_\beta(N = 0) \to 1$ as $\beta \to \infty$ if $a_0 > 0$.

13.5 Worked Example: Binary Strings

Example 13.4. Let $\mathcal{A} = \mathrm{SEQ}(\mathcal{Z}_0 + \mathcal{Z}_1)$ be the class of binary strings, where $\mathcal{Z}_0$ and $\mathcal{Z}_1$ are atoms of size 1. The generating function is

$$A(z) = \frac{1}{1-2z}, \qquad \rho_A = \frac{1}{2}.$$

The sampler. To run $\Gamma A(x)$, we use the sequence recipe: draw $K \sim \mathrm{Geom}(1-2x)$ (so $\Pr(K=k) = (1-2x)(2x)^k$), then draw each of the K bits independently and uniformly from $\{0,1\}$. The output is a uniformly random binary string of length K.

Size distribution. The length K has distribution

$$\Pr_x(K=n) = \frac{2^n x^n}{A(x)} = (1-2x)(2x)^n,$$

which is geometric with success parameter $1-2x$. The expected length is

$$\mathbb{E}_x[K] = \frac{xA'(x)}{A(x)} = \frac{x \cdot 2/(1-2x)^2}{1/(1-2x)} = \frac{2x}{1-2x}.$$

Setting this equal to a target n and solving gives $x^*(n) = n/(2(n+1))$, which approaches $\rho = 1/2$ as $n \to \infty$.

Partition function. Substituting $x = e^{-\beta}$:

$$Z(\beta) = A(e^{-\beta}) = \frac{1}{1-2e^{-\beta}}.$$

This is finite for $\beta > \beta_c = \log 2$. The partition function restricted to size n is $Z_n(\beta) = 2^n e^{-\beta n}$, and the free energy per bit is $f(\beta) = \beta - \log 2$.

Uniformity check. Among strings of length n, each specific string has probability $\Pr_x(\text{specific string}) = x^n/A(x)$. Since all 2^n strings of length n receive the same weight, they are equiprobable, as required.

Example 13.5. For comparison, consider plane binary trees (Catalan structures), with generating function

$$C(z) = \frac{1-\sqrt{1-4z}}{2z}, \qquad \rho_C = \frac{1}{4},$$

so $[z^n]C(z) = \binom{2n}{n}/(n+1) \sim 4^n/(n^{3/2}\sqrt{\pi})$. The partition function at inverse temperature β is $Z(\beta) = C(e^{-\beta})$, finite for $\beta > \beta_c = \log 4$.

The free energy per node is $f(\beta) = \beta - \log 4$. From the statistical-physics side, the $n^{-3/2}$ factor in the coefficient asymptotics appears as a sub-exponential correction to $Z_n(\beta)$: $Z_n(\beta_c) = [z^n]C(z) \sim Cn^{-3/2}$. This is precisely the $n^{-3/2}$ singularity exponent that Chapter 5 derived via singularity analysis of $C(z)$ at $z = 1/4$. The same fact — viewed there as a transfer lemma for square-root singularities, viewed here as a sub-leading correction to the free energy — is the same computation.

13.6 Preview of Chapter 14

The machinery developed here admits a direct generalization that connects to language model temperature decoding.

Replace the combinatorial class $\mathcal{A}$ with the set $\mathcal{W}^*$ of finite strings over a vocabulary $\mathcal{V}$, and replace the uniform-within-size distribution of the Boltzmann sampler with the distribution μ defined by an autoregressive LLM (as in Chapter 10). Define the energy of a string w by

$$E(w) = -\log \mu(w),$$

the negative log-probability under the model. Then temperature-T sampling produces w with probability proportional to

$$\mu(w)^{1/T} = e^{-E(w)/T},$$

which is a Gibbs distribution at inverse temperature $\beta = 1/T$ with energy $E(w)$.

The partition function is $Z(\beta) = \sum_w \mu(w)^\beta$, a Rényi-type sum over all strings. At $\beta = 1$ (temperature 1), this is just $\sum_w \mu(w) = 1$, always finite. At $\beta > 1$ (low temperature), the measure sharpens; at $\beta < 1$ (high temperature), it flattens toward uniform. At $\beta \to 0$ (infinite temperature), all strings of a given length become equally likely, which is the analogue of the Boltzmann sampler at $x \to 1$.

The analogy is exact: the LLM is a partition function machine. Chapter 14 will develop this correspondence carefully, using the WFA approximation of Chapter 12 to connect the LLM partition function to a rational generating function, and thereby apply the singularity analysis of Chapters 3–5 to the thermodynamics of language model decoding.

13.7 Notes and Further Reading

The Boltzmann sampler framework originates in [16], which gives full proofs of all the constructions in Section 13.2.2 and the complexity results of Proposition 13.2. The connection between generating functions and partition functions is classical in combinatorics (it is implicit in Polya's enumeration theorem and the transfer matrix method), but the Boltzmann sampling paper makes it algorithmic.

The recursive method, against which Boltzmann sampling is compared in Section 13.1, is developed in [41] and [19]. For sets and cycles (sketched in Section 13.2.2), the detailed Boltzmann constructions appear in [18], Chapter VII.

The identification of the radius of convergence with the critical inverse temperature, and the free energy computation of Section 13.4, are standard in the statistical mechanics of polymers and random combinatorial structures; see the monograph of Velenik for a textbook treatment in the physics tradition.

- [16] — The foundational paper on Boltzmann samplers: introduces the framework, proves the recursive construction rules for all symbolic-method operators, and establishes the $\Theta(n)$ expected-time complexity via saddle-point tuning and rejection sampling.
- [18] — Chapter VII gives the authoritative treatment of Boltzmann samplers in the context of the full analytic combinatorics toolkit, including the set and cycle constructions omitted in Section 13.2.2 and the connection to complex analysis of generating functions.
- [4] — Analyzes composition schemes and their singularity types systematically; directly relevant to understanding how the Gibbs measure's critical behavior (Section 13.3.2) depends on whether the generating function has a square-root or higher-order branch point.
- [7] — Establishes the ARM–EBM bijection for autoregressive language models via soft Bellman equations, grounding the "LLM as partition function machine" analogy previewed in Section 13.6 in a rigorous equiexpressiveness theorem.
- [27] — Quantifies the gap between local (token-wise) and global (Gibbs) temperature sampling, providing bounds that make precise the approximation implicit in the preview of Chapter 14 at the end of this chapter.

13.8 Exercises

Exercise 13.6. Let $\mathcal{T}$ be the class of rooted plane trees, satisfying the specification $\mathcal{T} = \mathcal{Z} \times \mathrm{SEQ}(\mathcal{T})$, so $T(z) = z/(1 - T(z))$, giving $T(z) = (1 - \sqrt{1 - 4z})/2$ and $\rho_T = 1/4$.

(a) Describe the Boltzmann sampler $\Gamma T(x)$ as an explicit random procedure using the recursive specification.

(b) Compute $\mathbb{E}_x[N]$ and find the value $x^*(n)$ that makes the expected tree size equal to n.

(c) Compute the critical inverse temperature β_c and the free energy $f(\beta)$ for $\beta > \beta_c$.

(d) Explain why the singularity exponent $-3/2$ in $[z^n]T(z) \sim C \cdot 4^n n^{-3/2}$ implies that the Gibbs measure at $\beta = \beta_c$ assigns probability $\Pr_{\beta_c}(N = n) \sim C' n^{-3/2}$ to size-n trees, and find the normalizing constant (up to a multiplicative constant).

Exercise 13.7. Let $\mathcal{A}$ be a combinatorial class with generating function $A(z) = \sum_n a_n z^n$ and radius of convergence ρ.

(a) Show that the variance of the size under $\Gamma A(x)$ is $\mathrm{Var}_x(N) = x \frac{d}{dx}\left(\frac{xA'(x)}{A(x)}\right)$.

(b) For $A(z) = e^z$ (the class of all sets of labeled atoms), compute $\mathbb{E}_x[N]$ and $\mathrm{Var}_x(N)$. What distribution does N follow?

(c) In the Gibbs interpretation with $x = e^{-\beta}$, express the variance $\mathrm{Var}_\beta(E)$ of the energy (here $E(\gamma) = |\gamma|$) in terms of β and A. This quantity is the *heat capacity* $C_V(\beta) = -\partial^2 \log Z / \partial \beta^2$. Verify your formula against the result of part (b).

Chapter 14

Gibbs Measures and Temperature Decoding

14.1 Recap: The Gibbs–Generating Function Bridge

Chapter 13 introduced Boltzmann sampling on combinatorial structures via the substitution $x = e^{-\beta}$: if $A(x)$ is the ordinary generating function of a combinatorial class, the Boltzmann distribution at inverse temperature β assigns probability $e^{-\beta|a|}/A(e^{-\beta})$ to structure a, turning size into energy and the generating function evaluated at $e^{-\beta}$ into a partition function $Z(\beta) = A(e^{-\beta})$. This chapter transfers that framework to language models. The combinatorial class becomes the set of finite strings Σ^*, the "size" of a string is replaced by the negative log-probability under a learned model, and temperature becomes a control knob that interpolates between greedy decoding ($\beta \to \infty$) and uniform sampling ($\beta \to 0$). The mathematics is the same; what changes is what the energy functional means and how it is computed.

14.2 The LLM Energy Functional

Let Σ be a finite vocabulary including a distinguished end-of-sequence token EOS. An autoregressive language model, as developed in Chapter 10, defines a probability distribution μ over $\Sigma^* = \bigcup_{n\geq 0} \Sigma^n$ by the factorization

$$\mu(w) = \prod_{t=1}^{|w|} p(w_t \mid w_1, \ldots, w_{t-1}),$$

where the product runs through the tokens of w including the terminal EOS, and each conditional $p(\cdot \mid w_{<t})$ is a probability distribution over Σ produced by the model (e.g., the output softmax of a Transformer).

Definition 14.1 (LLM Energy)**.** For a string $w \in \Sigma^*$, define the *energy* of w under μ as

$$E(w) = -\log \mu(w).$$

Because μ factorizes, the energy decomposes as a sum over token positions:

$$E(w) = \sum_{t=1}^{|w|} \big(-\log p(w_t \mid w_{<t})\big).$$

Each term $-\log p(w_t \mid w_{<t})$ is the per-token *surprisal*. This additive structure is precisely what makes autoregressive sampling computationally tractable: the energy of a string can be

accumulated token by token as the string is generated, without looking ahead. If the model is tight (i.e., $\sum_{w \in \Sigma^*} \mu(w) = 1$, as discussed in Chapter 8), then μ is a genuine probability measure, and E is its negative log-likelihood.

Remark 14.2. The energy functional is not an externally imposed Hamiltonian; it is derived entirely from the model's own distribution. In statistical physics one starts from an energy and derives a distribution. Here the direction is reversed: the distribution comes first, and the energy is defined from it. The Gibbs formalism nevertheless applies, because all that the formalism requires is an energy function on the sample space.

14.3 Temperature Sampling

14.3.1 The Global Gibbs Distribution

Given the energy $E(w) = -\log \mu(w)$, the *global Gibbs distribution at inverse temperature* $\beta > 0$ is

$$\mu_\beta(w) = \frac{e^{-\beta E(w)}}{Z(\beta)} = \frac{\mu(w)^\beta}{Z(\beta)},$$

where the partition function is

$$Z(\beta) = \sum_{w \in \Sigma^*} \mu(w)^\beta.$$

At $\beta = 1$ we recover $\mu_1 = \mu$ (assuming μ is tight). As $\beta \to \infty$, μ_β concentrates on the strings that maximize $\mu(w)$, i.e., greedy (or beam-search) decoding. As $\beta \to 0$, μ_β approaches the uniform distribution over strings (modulo the issues of infinite support). The temperature $T = 1/\beta$ is the parameter commonly exposed in LLM APIs.

14.3.2 Token-Wise Temperature Sampling

In practice, temperature is not applied globally to μ. Instead, at each decoding step t, the model produces logits $\ell_i = \log p(w_t = i \mid w_{<t})$ for $i \in \Sigma$, and these are divided by temperature T before taking the softmax:

$$p_i^{(T)} = \frac{\exp(\ell_i/T)}{\sum_{j \in \Sigma} \exp(\ell_j/T)}.$$

Equivalently, writing $\beta = 1/T$, the per-step distribution is proportional to $p(i \mid w_{<t})^\beta$. The string distribution induced by iterating this local procedure will be denoted $\widetilde{\mu}_\beta$. The question is whether $\widetilde{\mu}_\beta = \mu_\beta$.

14.3.3 Local vs. Global: The Markov Approximation

The local procedure replaces each conditional $p(\cdot \mid w_{<t})$ with its β-tilted version, renormalizing at each step. The resulting string probability is

$$\widetilde{\mu}_\beta(w) = \prod_{t=1}^{|w|} \frac{p(w_t \mid w_{<t})^\beta}{\widetilde{Z}_t(\beta, w_{<t})},$$

where $\widetilde{Z}_t(\beta, w_{<t}) = \sum_{a \in \Sigma} p(a \mid w_{<t})^\beta$ is the local per-step partition function. On the other hand, the global Gibbs distribution is

$$\mu_\beta(w) = \frac{\prod_{t=1}^{|w|} p(w_t \mid w_{<t})^\beta}{Z(\beta)}.$$

These are equal if and only if the product $\prod_t \widetilde{Z}_t(\beta, w_{<t})$ equals $Z(\beta)$ for all w, which is generically false. The global partition function $Z(\beta)$ accounts for the entire future subtree of completions from every prefix, while $\widetilde{Z}_t$ accounts only for the single next token. We record this precisely.

Proposition 14.3 (Local vs. Global Temperature)**.** *Let μ be a tight autoregressive language model. For $\beta \neq 1$, the local temperature distribution $\widetilde{\mu}_\beta$ and the global Gibbs distribution μ_β satisfy*

$$\frac{\widetilde{\mu}_\beta(w)}{\mu_\beta(w)} = \frac{Z(\beta)}{\prod_{t=1}^{|w|} \widetilde{Z}_t(\beta, w_{<t})},$$

and this ratio is not constant over w unless all per-step partition functions $\widetilde{Z}_t(\beta, w_{<t})$ are equal (independent of the prefix). In particular, $\widetilde{\mu}_\beta = \mu_\beta$ when $\beta = 1$.

Proof sketch. The formula for the ratio follows immediately from the two expressions for $\widetilde{\mu}_\beta(w)$ and $\mu_\beta(w)$ written above. At $\beta = 1$, both $\widetilde{Z}_t = 1$ for all t (since $\sum_a p(a \mid w_{<t}) = 1$) and $Z(1) = \sum_{w'} \mu(w') = 1$ by tightness, so the ratio is $1/1 = 1$. For general β, fix two completions w and w' of a common prefix $w_{<t^*}$ that branch at step t^* into two subtrees with unequal total μ^β-masses. Then $\widetilde{Z}_{t^*}$ is the same for both (it depends only on the shared prefix), but $Z(\beta)$ weights each path by the entire subtree mass; the resulting ratio differs for w and w', so the correction factor is non-constant. □

14.3.4 Concrete 2-Token Example

Example 14.4 (Two tokens, two branches)**.** Let $\Sigma = \{a, b, \text{EOS}\}$, and consider strings of length exactly 2 (plus EOS). Suppose the model gives:

$$p(a \mid \varepsilon) = 0.8, \quad p(b \mid \varepsilon) = 0.2,$$

$$p(\text{EOS} \mid a) = 0.9, \quad p(\text{EOS} \mid b) = 0.1,$$

so the two complete strings are $w_1 = a\,\text{EOS}$ with $\mu(w_1) = 0.8 \times 0.9 = 0.72$ and $w_2 = b\,\text{EOS}$ with $\mu(w_2) = 0.2 \times 0.1 = 0.02$ (ignoring the small residual probability; normalize if you wish). Take $\beta = 0.5$ (temperature $T = 2$).

Global Gibbs. $Z(0.5) = 0.72^{0.5} + 0.02^{0.5} \approx 0.8485 + 0.1414 = 0.9899$. So

$$\mu_{0.5}(w_1) \approx 0.8485/0.9899 \approx 0.857, \quad \mu_{0.5}(w_2) \approx 0.143.$$

Local temperature. At step 1: $\widetilde{Z}_1 = 0.8^{0.5} + 0.2^{0.5} \approx 0.8944 + 0.4472 = 1.3416$. Local probability of choosing a: $0.8944/1.3416 \approx 0.667$; of b: $0.4472/1.3416 \approx 0.333$. At step 2 given a: $\widetilde{Z}_2(a) = 0.9^{0.5} \approx 0.9487$ (only one token continues); probability of EOS $\mid a$ is 1 (it is the only continuation here). Given b: similarly probability of EOS $\mid b$ is 1. Thus:

$$\widetilde{\mu}_{0.5}(w_1) \approx 0.667, \quad \widetilde{\mu}_{0.5}(w_2) \approx 0.333.$$

The global Gibbs distribution gives roughly 0.857 vs. 0.143, while local temperature gives 0.667 vs. 0.333. The local procedure over-promotes the low-probability branch $b\,\text{EOS}$ because it does not account for the fact that b's subtree has much lower overall mass.

[27] study this gap systematically and provide bounds on how far $\widetilde{\mu}_\beta$ can deviate from μ_β in terms of the spread of per-step partition functions across the tree of continuations.

14.4 The ARM–EBM Bijection

A natural question is whether autoregressive models (ARMs), which are locally normalized, and energy-based models (EBMs), which are globally normalized, differ in expressive power. [7] answer this precisely via a soft Bellman equation.

Theorem 14.5 (ARM–EBM Bijection, Blondel et al. 2025)**.** *Every tight autoregressive language model μ corresponds uniquely to an energy-based model via the following construction. Define the* value function

$$V(w_{<t}) = \log \sum_{w' \in \Sigma^* : w_{<t} \sqsubset w'} \mu(w'),$$

i.e., the log-probability of all completions of prefix $w_{<t}$. Define the soft Q-value

$$Q(w_{<t}, a) = -\log p(a \mid w_{<t}) + V(w_{<t} \cdot a).$$

Then μ is the Boltzmann policy *with respect to Q:*

$$p(a \mid w_{<t}) = \frac{e^{-Q(w_{<t},a)}}{\sum_b e^{-Q(w_{<t},b)}},$$

and the value function satisfies the soft Bellman equation $V(w_{<t}) = -\log \sum_a e^{-Q(w_{<t},a)}$.

The proof follows from the observation that the completions mass $\exp(V(w_{<t}))$ is exactly the probability the model assigns to all continuations of $w_{<t}$, which satisfies a Bellman-like recursion: $\exp(V(w_{<t})) = \sum_a p(a \mid w_{<t}) \exp(V(w_{<t} \cdot a))$.

The significance of this result is the following. An EBM on Σ^* specifies a string energy $E(w)$ and samples via $e^{-E(w)}/Z$; it is globally normalized and in general intractable. An ARM samples greedily by factorizing the distribution. Blondel et al. show that when the underlying function class is rich enough to represent arbitrary V and Q functions, the two parameterizations have identical expressive power: every EBM distribution on Σ^* can be realized as the path distribution of an ARM, and vice versa. The difference is not what distributions they can represent, but how they are learned and sampled. The ARM parameterization makes forward sampling trivial (left-to-right) but makes inference about global properties (e.g., computing $Z(\beta)$) hard. The EBM parameterization specifies global energies naturally but requires approximate sampling. This is the fundamental tension that temperature decoding must navigate.

14.5 Kempton and Burrell 2025: Local Normalization Distortion

The mismatch identified in Section 14.3.3 has a clear structural cause: local renormalization at each step throws away information about the future subtree. When you divide logits by T and re-softmax at step t, you are computing the β-tilted version of the one-step conditional $p(\cdot \mid w_{<t})$, but the one-step conditional is not the same as the marginal of the β-tilted global distribution at that position. The marginal of μ_β at step t is:

$$p_\beta(a \mid w_{<t}) \propto p(a \mid w_{<t})^\beta \cdot \sum_{w' \text{ extends } w_{<t} \cdot a} \mu(w')^\beta / \mu(a \mid w_{<t})^\beta,$$

which depends on the subtree mass of all continuations beyond a. Token-wise temperature sampling ignores this subtree factor, as the example in Section 14.3.4 illustrates.

[27] quantify the distortion in terms of the variance of the log-partition functions $\log \widetilde{Z}_t(\beta, w_{<t})$ across prefixes. When this variance is small (i.e., the per-step partition functions do not vary much), the local and global procedures are close. This occurs approximately when the model's token distributions are nearly uniform at every step — a situation far from typical language model behavior where distributions are peaky.

The conclusion for practitioners is sharp: temperature decoding as universally implemented in LLM APIs is not a principled sample from the global Gibbs distribution μ_β. It is a Markov approximation of μ_β, and it systematically distorts the relative weights of high- and low-probability branches in ways that depend on the subtree structure of the model's distribution.

14.6 Structural Constraints on LLM Energy

[33] provide a perspective on why the Gibbs framework is nonetheless well-behaved for language model distributions arising from physical or natural data. Their argument, originally cast for neural networks applied to physics, implies that for data generated by processes with limited long-range interactions (e.g., text, images), the log-probability $\log \mu(w)$ is approximately polynomial of low degree in a suitable representation of w. Concretely, the energy $E(w) = -\log \mu(w)$ has an approximate expansion of bounded polynomial degree, which rules out the pathological Gibbs phenomena (e.g., non-analytic phase transitions driven by infinite-degree interactions) that can occur in worst-case energy landscapes.

For language models specifically, this translates to the observation that Transformer attention has bounded context and the energy is a sum of contributions that have diminishing interaction range. This does not mean E is polynomial in the token indices — it means the energy landscape on Σ^* does not have structure that would require exponentially many parameters to describe. The implication for temperature decoding is that the partition function $Z(\beta)$ is likely analytic (as a function of β) over a useful range, so the Gibbs distribution varies smoothly with temperature rather than exhibiting sharp discontinuities. Chapter 15 will revisit this claim in the generating-function setting.

14.7 Beyond Full Gibbs: Top-k and Nucleus Sampling

In practice, temperature sampling is often combined with truncation heuristics. Two common variants are:

- **Top-k sampling**: at each step, restrict the next-token distribution to the k highest-probability tokens, re-normalize, then sample. This is a truncation of the local distribution $p(\cdot \mid w_{<t})$ before applying temperature.
- **Nucleus (top-p) sampling**: at each step, retain the smallest set of tokens whose cumulative probability exceeds p, re-normalize, then sample.

From the Gibbs perspective, neither of these is a principled operation on μ_β. Top-k truncation removes tokens whose contribution to $Z(\beta)$ may be non-negligible — the discarded tail can carry significant probability mass at high temperatures ($\beta < 1$). Nucleus sampling adapts the truncation to the shape of the distribution at each step, which is heuristically more sensible but still does not correspond to any well-defined global distribution on Σ^*. Analytically, both operations introduce a step-dependent truncation that breaks the Markov structure of the generation process and makes the resulting string distribution very difficult to characterize. They are better understood as variance-reduction heuristics for sampling than as approximations to a Gibbs measure.

14.8 Creative vs. Greedy Decoding

The temperature parameter $T = 1/\beta$ has a natural interpretation in the generating-function sense. As $\beta \to \infty$ ($T \to 0$), the Gibbs distribution concentrates on the mode: μ_β assigns essentially all mass to the string $w^* = \arg\max_w \mu(w)$, and sampling from μ_β is equivalent to greedy decoding. As $\beta \to 0$ ($T \to \infty$), μ_β approaches the distribution that assigns equal mass to all strings — pure noise. The regime $\beta < 1$ ($T > 1$) flattens the distribution relative to μ, increasing entropy and diversity of samples; this is colloquially called "creative" decoding. The regime $\beta > 1$ ($T < 1$) sharpens the distribution, reducing diversity and pushing toward the most probable completions.

In the generating-function language of Chapter 13, varying β corresponds to moving along the real axis in the $x = e^{-\beta}$ substitution. The radius of convergence of $A(x) = \sum_w \mu(w) x^{-\log \mu(w)/\log x}$ (with suitable interpretation) sets the range of β over which the partition function is finite. Chapter 15 will show that the boundary of this convergence region is where phase transitions occur: qualitative changes in the character of typical samples from μ_β.

Remark 14.6. It is a common misconception that high temperature always makes a model "more creative" in a meaningful sense. What it does is increase entropy and reduce the probability gap between the most likely and least likely continuations. Whether this produces outputs that humans judge as creative depends on the structure of μ; for a model trained on diverse text, the high-entropy region of μ may correspond to semantically coherent but unusual continuations, or to incoherent sequences, depending on β.

14.9 Notes and Further Reading

The measure-theoretic framework for language model tightness is developed in [15], which also discusses the energy-based perspective on autoregressive models. The ARM–EBM bijection of [7] (arXiv:2512.15605) sharpens this picture by showing the two model classes are equiexpressive under suitable function-class assumptions, with the equivalence mediated by soft Bellman equations from reinforcement learning theory.

[27] provide the most careful mathematical treatment of the local-vs.-global temperature gap. Their analysis is closely related to the theory of weighted automata (Chapter 7) and to the notion of "backward" vs. "forward" partition functions in graphical models.

The observation of [33] that physical data distributions have low-degree energy functions connects to the renormalization-group perspective on deep learning, which is beyond the scope of this text. The key point for our purposes is that it gives a heuristic justification for treating $Z(\beta)$ as analytic in β over a wide range.

For the statistical physics background, the standard reference is Georgii's *Gibbs Measures and Phase Transitions*. The connection between Boltzmann sampling and analytic combinatorics is treated in Flajolet and Sedgewick, Chapter VII.

- [7] — Proves the ARM–EBM bijection via soft Bellman equations (Theorem 14.5): every tight autoregressive model has a unique energy-based representation, and the two classes are equiexpressive under rich function approximators; the paper that grounds the Gibbs interpretation of temperature decoding in a structural theorem.

- [27] — Provides tight bounds on the divergence between local token-wise temperature sampling $\widetilde{\mu}_\beta$ and the global Gibbs distribution μ_β, quantifying the distortion in terms of the variance of per-step log-partition functions across the continuation tree.

- [18] — Chapter VII develops the Gibbs/Boltzmann correspondence for combinatorial structures that underpins Chapter 13 and this chapter; the primary reference for the generating-function-as-partition-function identification.

- [39] — Applies statistical-mechanics methodology (susceptibility, finite-size scaling) to GPT-2 temperature decoding, providing the first systematic evidence for a temperature-driven phase transition near $T_c \approx 1$; previews the empirical program of Chapter 15.

- [1] — Extends the phase-transition analysis to Pythia, Mistral, and Llama across parameter scales, establishing that the critical temperature behavior is robust across architectures and training procedures.

Exercises

Exercise 14.7. Let $\Sigma = \{0, 1, \text{EOS}\}$ and define a model by $p(0 \mid \varepsilon) = p(1 \mid \varepsilon) = 0.5$, $p(\text{EOS} \mid 0) = 0.8$, $p(\text{EOS} \mid 1) = 0.2$ (so strings are of length at most 2). Compute $Z(\beta)$ for general $\beta > 0$. Show that the global Gibbs probabilities of the two length-2 strings at $\beta = 2$ differ from the probabilities produced by local temperature sampling with $T = 0.5$. What is the ratio $\widetilde{\mu}_\beta(w)/\mu_\beta(w)$ for each string?

Exercise 14.8. Suppose μ is a tight autoregressive model on Σ^* such that the per-step conditional distributions $p(\cdot \mid w_{<t})$ are all equal to the same fixed distribution q on Σ (i.e., the model is memoryless and time-homogeneous). Show that in this case the local temperature distribution $\widetilde{\mu}_\beta$ equals the global Gibbs distribution μ_β for all $\beta > 0$. Explain why the result in Proposition 14.3 is consistent with this: what happens to the per-step partition functions?

Chapter 15

Phase Transitions in Language Model Output

The temperature parameter of a language model is not merely a knob for controlling output randomness. At certain critical values it can trigger qualitative, system-wide changes in the structure of the output distribution — changes that physicists call *phase transitions*. This chapter makes that claim precise, connects it to the singularity theory developed in Chapters 1–5, and reviews the growing empirical evidence from large-scale experiments on GPT-2, Pythia, Mistral, and Llama.

15.1 Phase Transitions: The Analytic Definition

Throughout Chapters 13–14 we associated to any probability measure μ on Σ^* the *partition function*

$$Z(\beta) = \sum_{w \in \Sigma^*} \mu(w)^{\beta}, \qquad \beta > 0,$$

and the *free energy*

$$f(\beta) = -\frac{1}{\beta} \log Z(\beta).$$

The Gibbs measure at inverse temperature β assigns to each word w the probability $\mu_\beta(w) = \mu(w)^\beta / Z(\beta)$. Temperature in the sense of Chapter 14 is $T = 1/\beta$.

Definition 15.1 (Phase transition)**.** A *phase transition* of the system (μ, β) is a point $\beta_c \in (0, \infty)$ at which the free energy $f(\beta)$ fails to be real-analytic. The transition is *first-order* if f' has a jump discontinuity at β_c; it is *second-order* (or *continuous*) if f' is continuous at β_c but f'' diverges there.

In statistical mechanics the second derivative $-\beta^2 f''(\beta)$ is proportional to the *susceptibility*, measuring how sensitively the mean energy responds to a change in temperature. A diverging susceptibility is therefore the signature of a second-order transition.

15.1.1 Connection to Singularities of Generating Functions

Write $z = e^{-\beta}$, so $Z(\beta) = A(z)$ where $A(z) = \sum_w \mu(w)^\beta$ is interpreted as a generating function evaluated at $z = e^{-\beta}$. Then β_c is a phase-transition point if and only if $z_c = e^{-\beta_c}$ is a singularity of $A(z)$ on the boundary of its disk of convergence.

This is precisely the setting of Chapter 4's singularity classification:

- A *simple pole* of $A(z)$ at z_c gives a first-order transition. The dominant coefficient grows as z_c^{-n} with no polynomial correction, and the free energy has a kink.

- A *square-root branch point*, $A(z) \sim C(1 - z/z_c)^{1/2}$, gives a second-order transition: $Z(\beta)$ is continuous through β_c but $Z''(\beta)$ diverges. The associated coefficient asymptotics carry the $n^{-3/2}$ factor familiar from Chapter 5.
- An *essential singularity* can produce more exotic behavior (infinitely many derivatives discontinuous) that lies outside the standard first/second-order classification.

Proposition 15.2. *Suppose $A(z)$ has a unique dominant singularity at z_c of the square-root algebraic type,*

$$A(z) = g(z) - h(z)\sqrt{1 - z/z_c},$$

where g, h are analytic and nonzero at z_c. Then the free energy $f(\beta) = -\beta^{-1} \log A(e^{-\beta})$ is C^1 at $\beta_c = -\log z_c$ but f'' has an integrable singularity there, giving a second-order phase transition.

The proof is a direct computation: one writes $A(e^{-\beta}) = g(e^{-\beta}) - h(e^{-\beta})\sqrt{1 - e^{-(\beta-\beta_c)}}$ and expands near $\beta = \beta_c$. The square root produces a branch cut in β, so all higher derivatives of f diverge at β_c, while f and f' remain finite.

The upshot: *the order of the phase transition is determined by the type of the dominant singularity of $A(z)$*. This is the bridge between Chapter 4's taxonomy and the thermodynamics of Chapters 13–14.

15.2 Composition Schemes and Universal Transition Signatures

A natural mechanism by which a smooth generating function can develop a singularity is the *composition scheme*: take an outer function $A(u)$, analytic in a neighborhood of $u = 0$, and an inner function $B(z)$ with radius of convergence ρ_B. The composed function $F(z) = A(B(z))$ has its dominant singularity at $z = \rho_B$, but the *type* of that singularity depends on whether ρ_B is an ordinary point or a critical point of A.

[4] carried out a systematic analysis of such composition schemes and classified the possible singularity types that arise. The central finding is:

- If $B(\rho_B) < \rho_A$ (where ρ_A is the radius of convergence of A), the singularity of F at ρ_B is of the same type as that of B.
- If $B(\rho_B) = \rho_A$ — the *critical* or *confluent* case — the singularity type changes, typically from rational to square-root algebraic. The coefficient asymptotics transition from exponential-only to exponential times $n^{-3/2}$, with an Airy-function limit law governing the fluctuations.
- More degenerate confluences (higher-order critical points) produce $n^{-5/3}$ and related exponents tied to higher Airy-type kernels.

In the language of this chapter, the passage from the sub-critical to the critical composition scheme is itself a second-order phase transition. As a control parameter (say, the mean branching number in a context-free grammar) crosses a threshold, the dominant singularity of $F(z)$ changes character, and the free energy acquires a new non-analyticity.

Remark 15.3. The Airy-function limit law is *universal* in the sense that the same law emerges for a wide variety of combinatorial structures (binary trees, planar maps, context-free languages) whenever the composition scheme is at criticality. This universality is the analytic-combinatorics analog of universality classes in physics.

The composition-scheme framework gives a *catalogue* of possible phase-transition signatures for systems whose partition function is (or is approximated by) a composed generating function. Chapter 18 will revisit which entries of this catalogue are plausible candidates for real language models.

15.3 A Worked Example: The Catalan Partition Function

Before turning to the empirical literature, we work through the simplest non-trivial case.

Example 15.4 (Catalan partition function)**.** Let $A(z) = \frac{1-\sqrt{1-4z}}{2z}$, the generating function of the Catalan numbers $C_n = \binom{2n}{n}/(n+1)$. Its radius of convergence is $\rho = 1/4$, with a square-root branch point there.

Define the partition function $Z(\beta) = A(e^{-\beta})$ for $\beta > 0$. The critical inverse temperature is

$$\beta_c = -\log \rho = -\log(1/4) = \log 4.$$

For $\beta > \beta_c$ we have $e^{-\beta} < 1/4$ and $Z(\beta)$ is analytic; for $\beta < \beta_c$ the argument exits the radius of convergence and $Z(\beta)$ is defined by analytic continuation through the cut.

Near β_c write $\epsilon = \beta - \beta_c$ and expand:

$$1 - 4e^{-\beta} = 1 - 4e^{-\beta_c - \epsilon} = 1 - e^{-\epsilon} \approx \epsilon - \tfrac{\epsilon^2}{2} + O(\epsilon^3).$$

Thus

$$Z(\beta) \approx \frac{1-\sqrt{\epsilon}}{2e^{-\beta_c}} + O(\epsilon) = 2(1 - \sqrt{\epsilon} + O(\epsilon))$$

(where we use $e^{\beta_c} = 4$ and the leading term of the Catalan GF at the singularity). The free energy is

$$f(\beta) = -\frac{1}{\beta}\log Z(\beta) \approx -\frac{1}{\beta_c}\big(\log 2 - \tfrac{1}{2}\epsilon + O(\epsilon)\big).$$

Computing: $f'(\beta)$ is continuous at β_c (the $\sqrt{\epsilon}$ term in Z contributes a $\epsilon^{-1/2}$ term to Z'/Z, which enters f' as $(\log Z)'/\beta + (\log Z)/\beta^2$; the leading $\epsilon^{-1/2}$ singularity is integrable). However, $f''(\beta)$ diverges as $\epsilon \to 0$ like $\epsilon^{-1/2}$, confirming a second-order transition.

The coefficient asymptotics of $A(z)$, namely $[z^n]A(z) = C_n \sim 4^n n^{-3/2}/\sqrt{\pi}$, appear on the Gibbs side as the probability mass that the Boltzmann sampler assigns to strings of length n at the critical temperature: it decays as $n^{-3/2}$, the signature of the algebraic singularity noted in Chapter 5.

This example is the canonical prototype. Every algebraic generating function with a square-root dominant singularity gives the same exponents, which is why the $n^{-3/2}$ law and its associated Airy limit are called *universal*.

15.4 Empirical Evidence from Language Models

We now turn to recent experimental work. The experiments are independent and use different models, but they converge on a consistent picture: GPT-family language models exhibit temperature-driven phase transitions whose signatures are quantitatively consistent with second-order universality.

15.4.1 GPT-2: Nakaishi, Nishikawa, and Hukushima (2024)

[39] (arXiv:2406.05335) studied GPT-2 by treating the temperature $T = 1/\beta$ as a tunable control parameter and measuring standard finite-size-scaling observables on the output distribution. Their methodology is borrowed from computational statistical mechanics: they measure a quantity analogous to the magnetic susceptibility, locate its peak as a function of T, and track how the peak height and location shift as the model size (or the prompt length used as a proxy for system size) varies.

Their main findings are:

- There is a temperature $T_c \approx 1$ (i.e., $\beta_c \approx 1$) at which the susceptibility-analog peaks sharply.
- The peak height scales with the system-size proxy in a manner consistent with a second-order transition, with fitted critical exponents.
- The transition separates a low-temperature *ordered* phase (where the output concentrates on a small number of high-probability completions) from a high-temperature *disordered* phase (where the output approaches uniform).

These are empirical observations with fitted exponents. The paper does not prove that $Z(\beta)$ is non-analytic at β_c; it demonstrates scaling behavior that is consistent with such non-analyticity. The distinction matters: finite models always have analytic $Z(\beta)$ (the sum is finite), so what one observes in practice is a rapid crossover that sharpens as model size grows, analogous to the finite-size rounding of phase transitions in statistical mechanics.

15.4.2 Pythia, Mistral, Llama: Arnold et al. (2024)

[1] (arXiv:2405.17088) extended the investigation to three open-weight model families: Pythia (70M to 12B parameters), Mistral (7B), and Llama (7B to 70B). They scanned temperature and measured several observables including token-rank distributions, entropy, and a susceptibility proxy across all these models.

The key findings of [1] are:

- Phase-transition signatures are *robust across model families and parameter scales.* The critical temperature T_c varies somewhat, but all models show qualitatively identical behavior.
- The scaling of observables near T_c is consistent with second-order criticality.
- There is a *temperature window* around $T = 1$ where the output distribution is neither too concentrated nor too flat, and within this window the Zipf-like rank-frequency exponent is approximately constant across models.

The robustness across model families is significant: it suggests the phenomenon is not an artifact of a particular training procedure or architecture, but a structural property of the output distributions produced by autoregressive models trained on natural-language corpora.

Again, this is empirical evidence. The universality class is inferred from exponent fits, not derived from first principles.

15.4.3 Universality and the Temperature Window: Mikhaylovskiy (2025)

[37] goes a step further and argues that the observed transitions across models sit in a *common universality class* that is consistent with the predictions of analytic-combinatorics composition schemes. Specifically, the fitted critical exponents are compatible with those arising from square-root singularities of composed generating functions — the same universality class as the Catalan example of Section 15.3.

[37] also identifies and analyzes the *temperature window* regime: a range of temperatures near T_c in which the rank-frequency (Zipf) exponent α satisfies $\alpha \approx 1$ and is approximately constant as a function of T. This is consistent with the plateau in the free energy's second derivative that one would expect if the singularity is approached slowly from a non-analytic direction. Chapter 16 will examine the temperature window in detail; here we note only that its existence is further evidence of an underlying analytic structure.

The universality claim of [37] is stronger than those of [39] and [1] — it makes a claim about *which* universality class, not merely that one exists. It remains to be verified whether the exponents are truly universal or whether finite-size effects or model-family-specific features contaminate the fits.

15.4.4 Additional Corroboration: Li et al. (2024)

[32] provide additional empirical corroboration from a somewhat different angle. They carry out a finite-size-scaling analysis of LLM output distributions and show that an $O(N)$ estimation procedure for the partition function (where N is vocabulary size or sequence length) reproduces the transition signatures found by the other groups. Their work is notable for its computational efficiency: rather than sampling extensively, they use analytic approximations to $Z(\beta)$ that are tractable for large models, making the phase-transition analysis practical at scale.

15.5 The Bridging Proposal

We can now state the central proposal of Part V of this book.

Remark 15.5 (The analytic-combinatorics explanation). The empirical phase transitions observed by [39], [1], and [37] are proposed to be *analytic phase transitions* of the partition function

$$Z(\beta) = \sum_{w \in \Sigma^*} \mu(w)^{\beta},$$

where μ is the LLM's output distribution, viewed through the Boltzmann correspondence of Chapter 13 and the singularity classification of Chapter 4.

To make this concrete, consider what happens if μ were *exactly* a PCFG distribution in the sense of Chapter 8. Then by the Chomsky–Schützenberger theorem (Chapter 5), the generating function $A(z) = Z(\beta)\big|_{z=e^{-\beta}}$ would be algebraic over $\mathbb{Q}(z)$. Its dominant singularity would be a branch point of algebraic type, typically square-root. The partition function $Z(\beta) = A(e^{-\beta})$ would then exhibit a second-order phase transition at $\beta_c = -\log \rho$, with universal $n^{-3/2}$ Boltzmann weights at criticality. The Nakaishi–Arnold–Mikhaylovskiy scaling would be a theorem rather than an empirical observation.

The LLM case is not provably of this form. A trained Transformer is not a PCFG; its output distribution has no known closed-form algebraic structure. But the empirical scaling is consistent with the algebraic scenario, and the robustness across model families suggests that the phenomenon is structural rather than incidental.

The working hypothesis, which Chapter 18 will list as Gap 2 among the ten open problems of Part V, is:

> *The output distribution μ of a large autoregressive language model trained on natural language admits an effective algebraic approximation in the sense that $Z(\beta) = \sum_w \mu(w)^{\beta}$ has a dominant singularity of square-root type at some β_c, and the resulting second-order phase transition is what the empirical experiments are detecting.*

What would it take to verify this? One would need either (a) a theoretical derivation of the algebraic structure of the LLM's output distribution — which seems currently out of reach — or (b) a more refined experimental protocol that distinguishes square-root from other singularity types by measuring not just the location but the *shape* of the transition in detail. Chapter 16 takes a step toward (b) by analyzing the temperature window, and Chapter 19 discusses what theoretical progress on (a) might look like.

15.6 Summary

The main points of this chapter:

1. A phase transition of a temperature-parameterized system is a non-analyticity of the free energy $f(\beta)$. The order of the transition is determined by which derivative of f first diverges.

2. Via the substitution $z = e^{-\beta}$, the type of the dominant singularity of $A(z) = Z(\beta)$ determines the order of the transition. Square-root singularities give second-order transitions; poles give first-order.

3. Composition schemes [4] provide a catalogue of universal transition signatures, with the Airy-function universality class (associated with $n^{-3/2}$ coefficient asymptotics) being the most prominent.

4. Empirical experiments on GPT-2 [39], and on Pythia, Mistral, and Llama [1], provide evidence for temperature-driven phase transitions consistent with second-order criticality. [37] identifies the likely universality class, and [32] provide additional corroboration.

5. The bridging proposal: these empirical transitions may be explained as analytic phase transitions of the LLM partition function, consistent with the PCFG/algebraic scenario but not yet proved to be so.

15.7 Notes and Further Reading

The connection between singularities of generating functions and thermodynamic phase transitions is classical in the theory of exactly solvable models; see [18], Chapter VII, for the combinatorics perspective. Composition schemes and their singularity-confluence phenomena are developed in detail in [4]; the Airy limit law they discuss has a long history in map enumeration going back to Tutte.

The statistical-mechanics methodology applied to language models — finite-size scaling, susceptibility peaks, critical exponent fitting — is standard in the physics of disordered systems. Readers unfamiliar with this methodology should consult any graduate statistical-mechanics text before trying to interpret the exponent fits in [39] and [1].

The temperature window and its relation to Zipf's law in LLM output is discussed further in Chapter 16. The relationship between the ARM–EBM bijection (Chapter 14) and the phase-transition picture deserves further study: briefly, the EBM formulation makes it natural to define an order parameter for the transition, which the ARM formulation does not provide directly.

- [4] — Systematic analysis of composition schemes $F(z) = A(B(z))$ and their singularity types; establishes the catalogue of universal phase-transition signatures (including the Airy universality class with $n^{-3/2}$ exponents) that provides the theoretical backbone for interpreting the empirical LLM results.

- [18] — Chapter VII develops the connection between generating-function singularities and thermodynamic phase transitions in the combinatorial setting, including the Gibbs/Boltzmann correspondence; the primary analytic-combinatorics reference for the framework of this chapter.

- [39] — Studies GPT-2 using finite-size-scaling observables borrowed from computational statistical mechanics, finding a temperature-driven phase transition near $T_c \approx 1$ with scaling exponents consistent with second-order criticality.

- [1] — Extends the phase-transition analysis to Pythia, Mistral, and Llama at multiple parameter scales, demonstrating that the critical behavior is robust across architectures and establishing the existence of a temperature window with stable Zipf exponent.
- [37] — Argues that the LLM phase transitions sit in a specific universality class consistent with square-root singularities of composed generating functions, providing the strongest current link between the empirical scaling and the analytic-combinatorics prediction.
- [7] — The ARM–EBM bijection underpins the energy-functional view of LLM output; relevant here because the EBM formulation naturally furnishes an order parameter for the phase transition that the ARM formulation does not directly provide.

Exercise 15.6. Let $A(z) = (1-z)^{-2}$, which has a double pole at $z = 1$. Compute $Z(\beta) = A(e^{-\beta})$ for $\beta > 0$, identify β_c, and compute $f(\beta) = -\beta^{-1} \log Z(\beta)$ near β_c. Show that f' has a jump discontinuity at β_c, confirming a first-order transition. Compare with the Catalan (square-root) case of Section 15.3.

Exercise 15.7. The generating function of Motzkin numbers satisfies $M(z) = 1 + zM(z) + z^2 M(z)^2$, giving $M(z) = (1 - z - \sqrt{(1-z)^2 - 4z^2})/(2z^2)$ with radius of convergence $\rho = 1/3$. Let $Z_M(\beta) = M(e^{-\beta})$ and $\beta_c = \log 3$. (a) Show that Z_M has a square-root singularity at β_c, giving a second-order transition. (b) Determine the coefficient asymptotics $[z^n]M(z)$ and verify that the Boltzmann weight at β_c decays as $n^{-3/2}$. (c) Explain why the Catalan and Motzkin systems are in the same universality class despite having different ρ and different β_c.

Part V

Applications, Open Problems, and the Road Ahead

Chapter 16

Zipf's Law and Token Distributions

Empirical linguistics has known for nearly a century that word frequencies in large corpora are not spread evenly across the vocabulary. A handful of words dominate, while the vast majority appear rarely. This chapter makes that observation precise, reviews the main theoretical explanations (Pitman–Yor processes, combinatorial growth arguments, and the quantization model), and proposes how the bivariate generating function machinery developed in Chapters 4–5 might be brought to bear on the problem.

16.1 Zipf's Law

Definition 16.1 (Zipf's Law)**.** Let a corpus of tokens be given. Rank the distinct token types by decreasing frequency, so type ranked r is the r-th most common. Zipf's law asserts that the frequency of the r-th type is approximately

$$f(r) \approx \frac{C}{r^s},$$

where $C > 0$ is a corpus-dependent constant and the exponent $s \approx 1$ for natural language corpora.

This is an *empirical observation*, not a theorem. The exponent s is close to 1 across many languages and many corpora, which is striking, but it is not a mathematical law with a proof. Measuring s requires fitting a power law to noisy rank-frequency data, which introduces its own methodological complications (see Piantadosi's review below).

The connection to the complementary cumulative distribution is direct: if $f(r) \propto r^{-s}$, then the probability that a randomly drawn token has rank at most r grows as $\sum_{k=1}^{r} k^{-s}$, and by Euler–Maclaurin the probability that a token's frequency exceeds some threshold f satisfies $\Pr[\text{freq} \geq f] \propto f^{-1/s}$. For $s = 1$ this gives a power-law tail with exponent 1, sometimes written as saying the frequency distribution has a Pareto tail of index 1.

16.1.1 Mandelbrot's Refinement

Mandelbrot observed that the simple formula C/r^s fits poorly for small ranks (the very top of the frequency list) and proposed

$$f(r) \approx \frac{C}{(r+b)^s},$$

where $b \geq 0$ is a shift parameter. This one-parameter extension gives noticeably better empirical fits, particularly for high-frequency function words. Mandelbrot connected this formula to information-theoretic arguments about optimal coding, though those arguments remain contested. For our purposes, the Mandelbrot form is a useful phenomenological model; it does not by itself explain *why* such a distribution arises.

16.2 Piantadosi's Critical Review

The apparent universality of Zipf's law has attracted an enormous variety of proposed explanations: preferential attachment (Simon 1955), optimal coding (Mandelbrot), maximum-entropy arguments, and more. Piantadosi [44] provides a systematic critical evaluation of these explanations.

The central point of [44] is methodological: virtually every proposed generative model for language—from random text generated by a monkey typing uniformly on a keyboard, to preferential-attachment processes, to context-free grammars—can be tuned to produce a power-law rank–frequency distribution. Matching Zipf's law is therefore *weak evidence* for any particular explanation. A model that predicts Zipf's law is not confirmed; it has merely failed to be ruled out by this one summary statistic.

Piantadosi argues that a more informative test would involve the *joint* distribution of token frequency with other observables—word length, syntactic category, contextual predictability—rather than frequency alone. For example, there is a well-documented negative correlation between word frequency and word length (high-frequency words tend to be short), and any serious explanation of Zipf's law should also account for this. Models that reproduce the rank–frequency plot while getting the length distribution wrong are incomplete.

This critical stance is useful for us: it tells us that finding a generating function model whose singularity analysis yields a $1/r$ rank distribution is not, by itself, evidence that the model is correct. The bivariate generating function proposal in Section 16.9 is motivated partly by the desire to track *both* rank and length simultaneously, as Piantadosi recommends.

16.3 Pitman–Yor Processes and Power Laws

The most mathematically clean explanation of Zipf's law in the language modeling literature is via the Pitman–Yor (PY) process [20].

16.3.1 The Pitman–Yor Process

The Pitman–Yor process with discount parameter $a \in (0, 1)$ and concentration parameter $\theta > -a$ is a distribution over infinite sequences of tokens. It can be described by the *Chinese restaurant process*: tokens arrive one at a time. When the n-th token arrives and k distinct types have appeared so far, with type i having count c_i:

- with probability $(c_i - a)/(n - 1 + \theta)$, emit type i (reinforcement),
- with probability $(\theta + ak)/(n - 1 + \theta)$, emit a brand-new type (innovation).

The key property is that the marginal frequency distribution of types, in the limit of a long sequence, has a power-law tail. Specifically [20]:

$$\Pr[r\text{-th most frequent type after } n \text{ tokens}] \sim C\, n^{-a} \cdot r^{-(1+1/a)},$$

so the rank–frequency exponent is $s = 1 + 1/a$. For $a = 0.5$, we get $s = 3$; for a close to 1, s approaches 2 from above; for a close to 0, s grows without bound (the distribution becomes more concentrated). To recover empirical Zipf with $s \approx 1$, one would need a close to 1.

Remark 16.2. The exponent $s = 1 + 1/a$ is greater than 1 for all $a \in (0, 1)$. So a single-level Pitman–Yor process cannot directly produce the empirical $s \approx 1$ unless $a \to 1$, which is a degenerate limit. The two-stage model below addresses this.

16.3.2 The Two-Stage Model

Goldwater, Griffiths, and Johnson [20] propose a hierarchical (two-stage) construction: a *base* Pitman–Yor process generates an abstract "cached vocabulary," and a second-level process selects from this vocabulary or creates novel items. This composition of two PY processes allows the effective exponent to be lower and more flexibly tuned to match empirical data. The two-stage structure is natural from a linguistic standpoint: speakers draw on a mental lexicon (the cache), and the lexicon itself grows over time via a process with power-law dynamics.

16.4 Berman's Combinatorial Derivation

A different perspective comes from purely combinatorial reasoning. Berman [5] (arXiv:2511.17575) derives Zipf-type exponents without Bayesian priors or preferential attachment, working instead from the growth rate of the number of distinct types as a function of corpus size.

The argument proceeds roughly as follows. Consider a random text generated by a memoryless source. Track the *type count* $V(n)$, the number of distinct token types seen after n tokens. If $V(n) \sim n^\beta$ for some $\beta \in (0, 1)$, a combinatorial counting argument on the expected multiplicity of a type at rank r yields a Zipf exponent that is determined by β. The key insight is that the Zipf exponent is not a free parameter to be fit; it is constrained by the vocabulary growth exponent, which is in principle observable independently. This gives a combinatorial prediction: if you know how fast new words appear in a corpus, you can predict the shape of the rank–frequency plot.

This is qualitatively different from the PY explanation: Berman's derivation is deterministic in the sense that the exponent emerges from a combinatorial growth law, not from the random branching structure of a stochastic process. Whether the two explanations are ultimately equivalent (perhaps the PY vocabulary growth has $V(n) \sim n^a$, matching the PY discount parameter) is an open question worth investigating.

16.5 Mikhaylovskiy's Temperature Window

Chapter 15 established that autoregressive language models with temperature T sampling undergo phase transitions in their output statistics as T varies. Mikhaylovskiy [38] connects this directly to Zipf's law.

The finding is: for transformer language models, there exists a finite interval $[T_{\min}, T_{\max}]$ of temperatures within which the empirical rank–frequency exponent of model outputs is approximately 1 (matching natural language). Outside this interval, the exponent drifts—below $T_{\min}$, the model becomes repetitive and the distribution concentrates on a few high-frequency types, steepening the rank–frequency curve; above $T_{\max}$, the model becomes diffuse and the distribution flattens.

Mikhaylovskiy frames this as a *universality window*: the Zipf behavior is not a property of any single temperature but of a region in parameter space, and the boundaries of this region are the two phase transitions identified in Chapter 15. This is conceptually important. It means that a transformer model does not "naturally" produce Zipf distributions; it requires that the temperature parameter be within a specific range. A poorly calibrated model (or a model sampled at $T = 1$ without any tuning) may or may not fall inside this window.

Remark 16.3. This also has a practical implication. If a practitioner observes that a language model's output does not follow Zipf's law, the diagnosis may not be a model failure but a temperature miscalibration. Conversely, observing Zipf-like output is not evidence that the model is generating "natural" text; it only says the temperature is inside the window.

16.6 The Quantization Model

Michaud et al. [36] (arXiv:2303.13506) propose that the internal representations of a neural language model are organized around a discrete set of "quanta"—atomic skills or features, each associated with a subset of the vocabulary. The model's outputs are produced by composing these quanta, and the usage frequency of each quantum follows a power law.

The quantization model makes two related claims:

1. The power-law shape of neural scaling laws (loss decreases as $L \sim N^{-\alpha}$ in parameters N) follows from the discrete-quanta structure: as model capacity increases, more quanta are acquired, each contributing a roughly equal increment of capability.

2. The Zipf-like frequency distribution of output tokens follows from the same structure: tokens associated with common quanta are produced frequently, tokens associated with rare quanta are produced rarely, and if quantum usage frequencies are themselves power-law distributed, token frequencies inherit this distribution.

The connection to the Goldwater–Griffiths–Johnson two-stage model [20] is direct: the quanta of Michaud et al. play the role of the "cached vocabulary" in the Pitman–Yor two-stage construction. A token is produced by first selecting a quantum (the first stage, PY-like) and then emitting a token associated with that quantum (the second stage). If quantum selection follows a power law with exponent determined by a, and each quantum maps to a roughly fixed number of tokens, the output token distribution inherits the power-law shape. The quantization model thus gives a mechanistic neural network interpretation of the abstract PY hierarchy.

16.7 A Formal Proposition on Multinomial Models

Before turning to the bivariate generating function proposal, we record a basic proposition that clarifies what Zipf's law does and does not require.

Proposition 16.4. *Let tokens be drawn i.i.d. from a fixed vocabulary $\{w_1, w_2, \ldots\}$ with probabilities $p_1 \geq p_2 \geq \cdots > 0$. After n draws, let $\hat{f}(r, n)$ denote the empirical frequency of the type with the r-th largest count. Then, as $n \to \infty$, $\hat{f}(r, n) \to p_r$ almost surely. The empirical rank–frequency plot converges to the true parameter curve $r \mapsto p_r$. Consequently, the empirical distribution is Zipfian with exponent s if and only if $p_r \propto r^{-s}$.*

Proof. By the strong law of large numbers, the empirical frequency of each type w_i converges almost surely to p_i. Since the ranking is determined by these frequencies, and the p_r are distinct (generically), the rank of each type stabilizes almost surely after finitely many steps. Therefore $\hat{f}(r, n) \to p_r$ almost surely as $n \to \infty$. □

Remark 16.5. This proposition says that a multinomial model with Zipfian parameters produces Zipfian output, and a multinomial model with non-Zipfian parameters does not. The proposition is trivial in one direction—it simply says empirical frequencies converge—but its point is conceptual: Zipf's law in the output is equivalent to Zipf's law in the model parameters when the model is i.i.d. The interesting scientific question is not whether an i.i.d. model can produce Zipf's law (it can, trivially, if parameterized correctly), but where the Zipfian parameter values come from. The PY, combinatorial, and quantization frameworks are answers to this deeper question.

16.8 Worked Mini-Example

Example 16.6 (Pitman–Yor with $a = 0.5$)**.** Set $a = 0.5$. The Pitman–Yor prediction is $s = 1 + 1/a = 1 + 2 = 3$. The rank–frequency plot should follow $f(r) \propto r^{-3}$, which is *steeper*

than empirical Zipf ($s \approx 1$). In a log–log plot, the slope would be -3 rather than -1. To achieve $s \approx 1$, one would need $a \approx 1$, but the PY process is only defined for $a < 1$. The two-stage construction of [20] provides a way around this by composing two levels.

Example 16.7 (Memoryless unigram model)**.** Consider a memoryless (i.i.d.) model with vocabulary $\{w_1, \ldots, w_V\}$ and uniform probabilities $p_i = 1/V$. By Proposition 16.4, the empirical rank–frequency plot converges to the flat line $f(r) = 1/V$. This is *not* Zipfian; it has exponent $s = 0$. If instead $p_i \propto i^{-1}$ (a manually specified Zipfian distribution), then the model outputs Zipfian frequencies, but we have simply assumed what we wanted to explain. The conclusion is that Zipf's law cannot emerge from a simple i.i.d. model unless it is pre-loaded into the parameters. This is why the process-level explanations—PY, combinatorial growth, quantization—are necessary: they explain how Zipfian parameters arise from simpler primitives.

These two examples illustrate a general point. Zipf's law is a *non-trivial* statistical regularity that discriminates between model classes. An i.i.d. model with a fixed vocabulary will not generate it spontaneously. A model with memory, preferential attachment, or hierarchical structure can generate it, and the exponent encodes something about the structure of that memory or hierarchy.

16.9 A Bivariate Generating Function Proposal

The tools developed in Chapters 4–5 and formalized in Flajolet and Sedgewick [18], Chapter IX, handle bivariate generating functions (BGFs) of the form $F(z, u) = \sum_{n,k} f_{n,k} z^n u^k$, where the variable u marks a secondary parameter. The coefficient $[z^n u^k]F(z, u)$ counts objects of size n with secondary parameter value k.

We propose the following BGF as a tool for studying the joint distribution of token length and rank:

$$F(z, u) \;=\; \sum_{w \in \mathcal{V}} z^{|w|}\, u^{\mathrm{rank}(w)},$$

where $\mathcal{V}$ is the vocabulary, $|w|$ is the character-length of token w, and $\mathrm{rank}(w)$ is the frequency rank of w in the corpus. If the rank–length joint distribution follows Zipf's law with a length–rank correlation, then $F(z, u)$ encodes this correlation in its singular behavior.

Specifically, the singularity of $F(z, u)$ in z, as a function of u, would shift as u varies. If the dominant singularity $\rho(u)$ satisfies $\rho(u) \sim \rho_0 \cdot u^\gamma$ near $u = 1$, then by the transfer lemma of singularity analysis (Flajolet–Sedgewick Ch. VI), the coefficient $[z^n]F(z, u)$ is dominated by tokens near a characteristic rank $r^\star(n) \sim n^{1/\gamma}$. Inverting, this would give $s = 1/\gamma$.

This is a proposal, not a theorem. Several technical difficulties stand in the way:

1. The vocabulary is finite in practice, so $F(z, u)$ is a polynomial in z for each fixed u; a singularity analysis requires a suitable infinite or asymptotic model.

2. The rank function depends on the corpus, which is itself random; incorporating this randomness into the generating function requires a random analytic function framework.

3. The connection between the PY process or Berman's growth model and a specific BGF functional equation is not yet established.

This is related to the open problems of Chapter 18. The nearest solved analogues are the BGF analyses of random tries (Flajolet–Sedgewick Ch. IX), where length and depth play roles analogous to length and rank here. A systematic translation of the trie analysis to the token-rank setting would be a concrete step toward a theorem.

16.10 Notes and Further Reading

Zipf's original data and the rank–frequency observation appear in [61] (not cited as a primary source here but standard background). Mandelbrot's information-theoretic derivation is in his 1953 paper. The Simon (1955) preferential-attachment model is the discrete-time precursor to Barabási–Albert networks and reproduces Zipf's law from a simple urn process.

Piantadosi [44] is the clearest statement of why reproducing the rank–frequency plot is insufficient evidence for a mechanistic model, and should be read alongside any of the theoretical derivations. Goldwater, Griffiths, and Johnson [20] give the full Pitman–Yor derivation with empirical validation on English and Chinese corpora.

Berman [5] (arXiv:2511.17575) provides the combinatorial perspective. Mikhaylovskiy [38] provides the empirical temperature-window analysis for transformers. The quantization model of Michaud et al. [36] (arXiv:2303.13506) connects Zipf's law to neural scaling laws via a discrete-quanta structure.

For the BGF methodology, Flajolet and Sedgewick [18] Chapter IX is the standard reference. Random tries are the worked-out example closest to the token-rank problem.

Recommended Reading

- [61] — The original empirical study establishing the rank–frequency regularity in English. Read for historical grounding; the data analysis is crude by modern standards but the observation is real.
- [44] — A systematic critical review of explanations for Zipf's law, arguing that matching the rank–frequency plot is weak evidence for any model. The benchmark methodological reference before engaging any proposed mechanism.
- [20] — Derives Zipf-like frequency distributions from a hierarchical Pitman–Yor process and validates the model on English and Chinese corpora. The standard probabilistic explanation in the NLP literature.
- [5] — Derives the Zipf exponent from a combinatorial vocabulary-growth argument, without Bayesian priors. Connects the rank–frequency exponent directly to the observable rate at which new word types appear.
- [38] — Identifies a finite temperature window within which transformer outputs satisfy Zipf's law, linking the rank–frequency exponent to the phase transitions of Chapter 15.
- [36] — Proposes a discrete-quanta model of neural representations in which power-law frequency distributions and neural scaling laws share a common origin. Gives a mechanistic neural-network interpretation of the two-stage Pitman–Yor hierarchy.

Exercises

Exercise 16.8. Let $p_r = Cr^{-s}$ for $r = 1, 2, \ldots, V$ with C chosen so that $\sum_{r=1}^{V} p_r = 1$. For a corpus of n i.i.d. draws from this distribution, compute the expected number of distinct types seen after n draws as a function of n, V, and s. For $s = 1$ and $V \to \infty$, show that this expected count grows as $\Theta(C \log n)$. Compare with Berman's prediction that $V(n) \sim n^{\beta}$ for some β; for what β are the two consistent?

Exercise 16.9. Consider the Pitman–Yor Chinese restaurant process with parameters $a = 0.5$, $\theta = 1$. Simulate $n = 10000$ token draws. Plot the empirical rank–frequency distribution on log–log axes and estimate the slope. Compare your empirical estimate with the theoretical prediction $s = 1 + 1/a = 3$. How many tokens are needed before the power-law regime is clearly

visible? What does this say about the difficulty of distinguishing power-law tails from, say, a log-normal distribution with comparable data?

Chapter 17

Motif Statistics and Analytic Information Theory

This chapter is the most classical in Part V. The questions we ask — how often does a pattern appear in a random string? how deep is a random trie? what is the typical information content of a long sequence? — were answered, for Markov sources, in the 1990s and early 2000s by a school of researchers centered on Wojciech Szpankowski. The toolkit they developed is called *analytic information theory*, and it is precisely the toolkit the LLM community should reach for when it wants rigorous answers about sequence statistics.

17.1 What Is Analytic Information Theory?

Analytic information theory is the research program of applying complex-analytic methods — Mellin transforms, saddle-point integration, Cauchy's coefficient formula, singularity analysis — to classical problems in information theory: entropy rates, data compression, pattern matching, suffix trees, tries. The program is associated above all with the book of Szpankowski [56], which remains the definitive reference.

Remark 17.1. Szpankowski's book is organized by *technique*, not by problem: one chapter for poissonization, one for the Mellin transform, one for the saddle-point method, one for singularity analysis (in the sense of Flajolet–Sedgewick [18]). This is deliberate. The same complex-analytic machine applies to problems that look unrelated at the problem level. A reader who learns the techniques acquires a reusable kit.

The core observation is that many combinatorial and probabilistic quantities — expected values, variances, limit distributions — can be read off from the analytic behavior of a generating function near its dominant singularities. For sequences generated by a Markov source (or, more generally, a finite-state automaton), the relevant generating functions are rational, which makes the singularity analysis particularly clean.

The LLM connection is direct: as argued in Chapter 12, an LLM approximated by a WFA is essentially a high-dimensional Markov source. Every result in analytic information theory that holds for Markov sources therefore applies, in principle, to WFA-approximated LLMs. The remainder of this chapter makes this precise.

17.2 Motif Occurrence Generating Functions

Let Σ be a finite alphabet of size σ, and let M be a *motif* — either a fixed word or, more generally, the language recognized by some regular expression. We want to study, in a random string over Σ, the number of occurrences of M.

Definition 17.2 (Bivariate motif generating function)**.** The *bivariate generating function* for strings counted by length and motif count is

$$F(z,u) = \sum_{w \in \Sigma^*} z^{|w|}\, u^{\#M(w)},$$

where $\#M(w)$ denotes the number of (possibly overlapping) occurrences of M in w, and each string w is weighted by the probability assigned to it by the source.

For a *uniform* source over Σ, every word of length n has weight σ^{-n}, so we set $z \mapsto z/\sigma$ and count unweighted. For a Markov source the weights are products of transition probabilities.

The coefficient $[z^n u^k]F(z,u)$ then gives the (weighted) number of strings of length n containing exactly k occurrences of M. Extracting asymptotics amounts to analyzing $F(z,u)$ for u near 1 and z near its dominant singularity.

17.2.1 The Autocorrelation Polynomial

The structure of $F(z,u)$ is controlled by the *autocorrelation polynomial* of M, introduced by Guibas and Odlyzko [21].

Definition 17.3 (Autocorrelation polynomial)**.** Let $M = m_1 m_2 \cdots m_\ell$ be a word of length ℓ. The autocorrelation polynomial is

$$C(z) = \sum_{k=0}^{\ell-1} c_k\, z^k,$$

where $c_k = 1$ if the suffix of M of length k equals the prefix of M of length k, and $c_k = 0$ otherwise. Equivalently, $c_k = [m_{\ell-k+1} \cdots m_\ell = m_1 \cdots m_k]$.

The autocorrelation polynomial encodes how M can overlap with itself: $c_k = 1$ means M can begin at position $\ell-k+1$ of a previous occurrence of M and still produce a valid occurrence. This self-overlap structure determines, via the theory of Guibas–Odlyzko, the generating function for the waiting time until the first occurrence of M in a random string, and thence the bivariate GF.

For random texts modeled by a Markov source, and for M a word or a regular expression, the general bivariate-GF framework is developed in Nicodème–Salvy–Flajolet [40]. The recipe is:

1. Build a DFA $\mathcal{A}_k$ recognizing the language of strings containing *exactly k occurrences* of M (or, more efficiently, a single DFA tracking the current occurrence count).

2. Form the *transfer matrix* $T(z,u)$ of the weighted automaton, where a transition reading letter a from source state i to state j contributes a factor of $z \cdot p_{ij}$ (the source transition probability) and u if a new occurrence of M is completed.

3. The bivariate GF is the rational function $F(z,u) = \mathbf{v}^T (I - T(z,u))^{-1}\mathbf{e}$ for appropriate initial vector $\mathbf{v}$ and final-state indicator $\mathbf{e}$.

4. Singularity analysis: for fixed u near 1, the dominant singularity $z_0(u)$ of $F(z,u)$ is the zero of $\det(I - T(z,u))$ of smallest modulus. Perturbing u from 1 perturbs $z_0(u)$, and the derivatives of $z_0(u)$ at $u = 1$ give the mean and variance of $\#M$ in a string of length n.

This is entirely mechanical once the automaton is specified. It is one of the cleanest applications of the symbolic method and transfer-matrix technology developed in Chapters 4–6.

Proposition 17.4 (Gaussian limit law for motif counts; Nicodème–Salvy–Flajolet [40])**.** *Let M be a motif recognized by a finite automaton, and let the source be an ergodic Markov chain on Σ. Let X_n denote the number of occurrences of M in a random string of length n. Then:*

$$\mathbb{E}\, X_n = \mu n + O(1), \qquad \operatorname{Var} X_n = \sigma^2 n + O(1),$$

for constants $\mu > 0$ and $\sigma^2 > 0$ computable from the transfer matrix, and

$$\frac{X_n - \mu n}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1).$$

The proof uses singularity perturbation: one shows that $\log F(z, e^t)$ has a simple pole at $z_0(e^t)$ whose location is analytic in t near 0, and the cumulant generating function of X_n then satisfies the hypotheses of the quasi-power theorem (see Flajolet–Sedgewick [18], Chapter IX). The Gaussian limit follows.

17.3 Worked Example: The Motif "aba"

Example 17.5 (Autocorrelation of "aba")**.** Let $\Sigma = \{a, b\}$ with uniform probabilities $p_a = p_b = 1/2$, and let $M = \texttt{aba}$, a word of length $\ell = 3$.

We compute the autocorrelation polynomial $C(z) = c_0 + c_1 z + c_2 z^2$.

- $c_0 = 1$ always (the empty overlap).
- c_1: is the suffix of M of length 1, namely `a`, equal to the prefix of length 1, namely `a`? Yes. So $c_1 = 1$.
- c_2: is the suffix of M of length 2, namely `ba`, equal to the prefix of length 2, namely `ab`? No. So $c_2 = 0$.

Therefore $C(z) = 1 + z$.

This means `aba` can overlap with a subsequent occurrence of itself only at offset 2 (one character back), because the single-character suffix `a` of `aba` matches the single-character prefix `a`. Concretely, in the string `ababa`, the substring `aba` occurs at positions 1 and 3, overlapping in the `a` at position 3.

The bivariate generating function for strings over $\{a, b\}$ with uniform weights has the form, in the Guibas–Odlyzko–Nicodème framework:

$$F(z, u) = \frac{C(z/2)}{C(z/2) - (u-1)z^\ell/2^\ell} \cdot \frac{1}{1-z},$$

where the correction from the autocorrelation polynomial modifies the pole location.

More precisely, the expected number of occurrences of `aba` in a random string of length n is

$$\mathbb{E}\, X_n = \frac{n-2}{8} + O(1) \;\approx\; \frac{n}{8},$$

since $\mu = p_a^2 p_b = (1/2)^2 \cdot (1/2) = 1/8$. The autocorrelation affects the variance and the higher-order terms but not the leading mean.

17.4 Trie Depth and Markov Source Analysis

A *trie* (retrieval tree) built from n strings $w_1, \ldots, w_n \in \Sigma^\omega$ is the rooted tree in which each string corresponds to a root-to-leaf path, and internal nodes branch on the next character. The *depth* D_n of a random string in the trie — equivalently, the length of the longest common prefix of a random string with all others — is a fundamental quantity: it measures how long one must read a string before it becomes distinguishable.

Jacquet and Szpankowski [26] analyzed trie depth under an ergodic Markov source:

Proposition 17.6 (Trie depth under Markov source; Jacquet–Szpankowski [26])**.** *Let strings be generated by an ergodic Markov chain on Σ with entropy rate h_1 (in nats, or bits if logarithms are base 2). Then the expected depth of a random key in a trie built from n independent strings satisfies*

$$\mathbb{E}\, D_n \sim \frac{1}{h_1} \log n \quad \text{as } n \to \infty.$$

More precisely, there is a periodic fluctuation: $\mathbb{E}\, D_n = \frac{1}{h_1} \log n + \Lambda(\log n) + O(1/n)$, where Λ is a small-amplitude periodic function coming from poles of a Mellin transform off the real axis.

The constant $1/h_1$ is the reciprocal of the entropy rate. For the binary uniform source, $h_1 = \log 2$ and $\mathbb{E}\, D_n \sim \log_2 n$. For $n = 1000$, this gives $\mathbb{E}\, D_{1000} \approx \log_2 1000 \approx 9.97 \approx 10$. A random binary string requires about 10 bits before it is distinguished from 1000 competitors — exactly what information theory predicts, now with a proof via analytic methods.

The interpretation for language modeling is immediate. If we model a natural-language corpus as generated by a Markov source (or a WFA approximation of an LLM), then $D_n \sim \frac{1}{h} \log n$ says that the typical *distinguishing context length* grows logarithmically in the corpus size, with the entropy rate as the determining constant.

17.5 Connection to Chapter 9: The Entropy Rate Revisited

In Chapter 9, we established the identity

$$h = \log \frac{1}{R},$$

where R is the radius of convergence of the generating function counting strings by probability weight, and h is the (Shannon) entropy rate of the source. The dominant singularity of the counting GF is at $z = R$, and h is determined by its location.

In the language of Perron–Frobenius theory: for a primitive (irreducible, aperiodic) non-negative matrix T, the spectral radius $\rho(T)$ is the unique largest eigenvalue, with positive eigenvectors. For a Markov chain with transition matrix P, the relevant matrix in the trie-depth and motif-statistics problems is a *weighted* version of P, with weights chosen to count string length. The dominant singularity of the transfer matrix GF occurs at $z = 1/\rho(T_z)$, where T_z is P weighted by z.

The entropy rate satisfies

$$h = -\sum_{i,j} \pi_i P_{ij} \log P_{ij},$$

where π is the stationary distribution. This is exactly $\log(1/R)$ when R is the radius of convergence of the path-counting GF. The Perron–Frobenius theorem guarantees that this dominant singularity is a square-root or simple pole, not an accumulation point, which is why singularity analysis applies cleanly.

The trie-depth constant $1/h_1$ in Proposition 17.6 is the same h appearing in Chapter 9, computed by the same spectral method. The two analyses are not independent: they are two applications of the same underlying complex-analytic framework to the same spectral radius.

17.6 The Saddle-Point Method and Mellin Transforms

Szpankowski's book is organized around two complex-analytic tools that go beyond the algebraic singularity analysis of Flajolet–Sedgewick [18]. We sketch them briefly.

17.6.1 The Saddle-Point Method

Singularity analysis extracts $[z^n]f(z)$ when f has an algebraic or logarithmic singularity at a finite point z_0. But some generating functions are entire — no finite singularity at all — and grow like $e^{g(z)}$ for some g. The prototypical example is e^z, with $[z^n]e^z = 1/n!$.

For such functions, one uses the *saddle-point method*. By Cauchy's formula,

$$[z^n]f(z) = \frac{1}{2\pi i}\oint \frac{f(z)}{z^{n+1}}\,dz.$$

The integrand $h(z) = f(z)/z^{n+1}$ has a saddle point at z_0 where $h'(z_0) = 0$, equivalently where $f'(z_0)/f(z_0) = (n+1)/z_0$. One deforms the contour to pass through z_0 in the direction of steepest descent, so the integrand is concentrated near z_0 and the remaining integral is Gaussian. The result is

$$[z^n]f(z) \approx \frac{f(z_0)}{z_0^n\sqrt{2\pi\kappa_2}},$$

where κ_2 is the second cumulant of the local Gaussian. Hayman's theorem gives sufficient conditions for this approximation to be valid for entire functions of so-called *admissible* type.

In analytic information theory, the saddle-point method appears when one analyzes the distribution of the number of distinct words in a random string, or the profile of a random trie (how many nodes are at depth k). These quantities involve entire GFs and require saddle-point analysis for their asymptotics.

17.6.2 The Mellin Transform

The Mellin transform of a function $F : (0, \infty) \to \mathbb{R}$ is

$$\mathcal{M}[F](s) = \int_0^\infty x^{s-1}F(x)\,dx,$$

defined for s in a vertical strip of the complex plane. The Mellin transform is particularly suited to functions of the form $F(x) = \sum_k f(a_k x)$, which arise naturally when one counts strings by combining independent contributions from each position or branch.

The key property: if $F(x) = \sum_{k\geq 1} f(kx)$, then

$$\mathcal{M}[F](s) = \mathcal{M}[f](s)\cdot\zeta(s),$$

where ζ is the Riemann zeta function (or its analog for the relevant Dirichlet series). The poles of $\mathcal{M}[F](s)$ in the complex plane then correspond to terms in the asymptotic expansion of $F(x)$ via Mellin–Perron inversion.

In trie analysis, the Mellin transform handles the sum over depths k of the contribution of strings reaching depth k. The poles of the Mellin transform at non-real values of s produce the periodic fluctuation $\Lambda(\log n)$ in Proposition 17.6.

17.7 Why This Matters for LLMs

The results of this chapter were proved for Markov sources. Chapter 12 argued that LLMs can be approximated by WFAs, and WFAs are precisely weighted finite automata — a generalization of Markov chains in which the weights need not be non-negative or normalized, but which still admit a transfer-matrix representation.

For a WFA approximation A of an LLM:

- The *motif-count* analysis of Section 17.2 applies directly. Build the product automaton of A with the motif-tracking DFA; extract the bivariate GF; apply singularity perturbation. This gives the mean, variance, and Gaussian limit law for the number of occurrences of any fixed phrase in LLM-generated text.

- The *trie-depth* result of Section 17.4 applies once the effective entropy rate h_A of the WFA is defined. For an ergodic WFA, $h_A = \log(1/R_A)$ where R_A is the spectral radius of the weighted transition matrix, exactly as in Chapter 9.

- The *typical-set* analysis — what is the set of strings an LLM assigns high probability to? — is equivalent to a trie-depth or entropy-rate computation, and is therefore solved by the same methods.

Remark 17.7. There is a practical caveat. The WFA approximation of a large Transformer is high-dimensional: the transfer matrix may have dimension in the thousands or millions. The spectral quantities are then expensive to compute, though not conceptually different. This is a computational, not a mathematical, obstacle.

The open problem, which we call Gap 9 and return to in Chapter 18, is this: *no one has yet carried out the full Szpankowski-style analysis for a WFA approximation of a production LLM.* The mathematical pipeline exists. The WFA approximation methods of Chapter 12 exist. What is missing is the engineering to connect them, and perhaps the WFA approximations of sufficient quality to make the results meaningful.

This is not a fundamental obstacle. It is a gap between two mature bodies of work that have not yet been combined.

17.8 Notes and Further Reading

The foundational reference for analytic information theory is Szpankowski [56]. The autocorrelation polynomial and the exact GF for waiting times until a pattern first appears are due to Guibas and Odlyzko [21]; their paper remains a model of elegant combinatorial argument. The extension to regular-expression motifs, including the bivariate GF and the Gaussian limit law stated in Proposition 17.4, is in Nicodème–Salvy–Flajolet [40].

Trie depth under Markov sources was analyzed by Jacquet and Szpankowski [26]; the same authors have a long series of papers on digital search trees, suffix trees, and Lempel–Ziv compression under Markov sources, all applying the Mellin-transform and saddle-point machinery. The Flajolet–Sedgewick book [18] covers singularity analysis (Chapter VI) and the saddle-point method (Chapter VIII) systematically.

For readers who want to understand the periodic fluctuations in trie depth and digital tree statistics, the key tool is the Mellin–Perron formula combined with Lindström–Gessel–Viennot for lattice paths; Szpankowski [56] Chapter 8 is the place to start.

The Hayman admissibility criterion for the saddle-point method is in Hayman's 1956 paper; see also Flajolet–Sedgewick [18] Section VIII.5 for a modern treatment.

Recommended Reading

- [56] — The definitive monograph on analytic information theory. Organized by technique (Mellin transforms, saddle-point integration, poissonization) rather than by problem; a reader who works through it acquires a reusable analytical toolkit applicable to entropy estimation, data compression, and word statistics.

- [21] — Introduces the autocorrelation polynomial and derives exact generating functions for the waiting time until a pattern first occurs in a random string. Elegant and self-contained; the starting point for all subsequent motif-statistics work.

- [40] — Extends the Guibas–Odlyzko framework to regular-expression motifs and general Markov sources, establishing the bivariate generating function recipe and proving the Gaussian limit law for motif counts stated in Proposition 17.4.
- [26] — Proves the $\mathbb{E}\, D_n \sim (\log n)/h$ law for trie depth under a Markov source using the Mellin transform, and identifies the periodic fluctuation $\Lambda(\log n)$ arising from poles off the real axis. The model paper for the Mellin-transform technique in digital-tree analysis.
- [18] — Chapters VI–VIII cover singularity analysis, algebraic functions, and the saddle-point method in systematic detail. Chapter IX on words and maps is the direct bridge to this chapter's content.
- [24] — Proves the quasi-power meta-theorem that converts quasi-power form of a bivariate generating function into a Gaussian limit law. A concise read that pays dividends across all the limit-law applications in Part V.

Exercises

Exercise 17.8. Compute the autocorrelation polynomial for the motif $M = \texttt{abab}$ over $\Sigma = \{a, b\}$. Identify all non-trivial overlaps (those with $c_k = 1$ for $k \geq 1$), and explain in terms of the string structure why each overlap exists or does not exist. How does having $c_2 = 1$ affect the expected waiting time until the first occurrence of M in a uniform random string, compared to a motif of the same length with $C(z) = 1$?

Exercise 17.9. Let $\Sigma = \{0, 1\}$ and consider the Markov chain with transition matrix

$$P = \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix}$$

for parameters $\alpha, \beta \in (0, 1)$.

(a) Compute the stationary distribution π.

(b) Compute the entropy rate $h_1 = -\sum_{i,j} \pi_i P_{ij} \log P_{ij}$ (in bits).

(c) For $n = 10^6$ strings generated by this chain, estimate the expected trie depth $\mathbb{E}\, D_n$ for the parameter values $\alpha = 0.1, \beta = 0.2$.

(d) Verify your answer makes sense by checking the limiting cases $\alpha = \beta = 1/2$ (uniform iid) and $\alpha \to 0, \beta \to 0$ (nearly deterministic).

Chapter 18

Ten Open Problems at the Frontier

This chapter collects ten precisely stated open problems at the intersection of analytic combinatorics and large language models. Each problem is accompanied by a statement of what is known, why existing work stops short, and what a solution would look like. The numbering is for reference only and does not imply an ordering by importance; Chapter 19 addresses tractability.

18.1 Gap 1a: A Generating Function for LLM Length Distributions

Let μ be the distribution over Σ^* induced by an autoregressive language model (Chapter 10). The *length probability generating function* (length PGF) is

$$L(z) = \sum_{n=0}^{\infty} \Pr[|W| = n]\, z^n = \sum_{n=0}^{\infty} \Bigg(\sum_{\substack{w \in \Sigma^* \\ |w| = n}} \mu(w) \Bigg) z^n.$$

This is a well-defined formal power series with non-negative coefficients summing to 1 (by tightness, Chapter 8). Its radius of convergence R_L satisfies $R_L \geq 1$.

What is known. Du *et al.* 2023 [15] give a necessary and sufficient condition for tightness ($\sum_t \tilde{p}_{\mathrm{EOS}}(t) = \infty$) and prove that every softmax Transformer is tight. They do not construct $L(z)$ explicitly or characterize its singularity structure.

Why it stops short. Tightness is the statement $L(1) = 1$. The analytic-combinatorics programme needs far more: the location of the dominant singularity of $L(z)$, its type (polar, algebraic, logarithmic), and the resulting coefficient asymptotics $\Pr[|W| = n] \sim ?$ These would tell us, for example, whether LLM output lengths follow a geometric tail (pole), a subexponential power-law tail ($n^{-3/2}$ from a square-root branch point), or something else entirely.

What a solution looks like. An explicit formula for $L(z)$ in terms of the model's parameters — or at least a proof that $L(z)$ belongs to a specific function class (rational, algebraic, D-finite) — together with a singularity analysis yielding the asymptotic form of $\Pr[|W| = n]$.

18.2 Gap 1b: The Entropy-Rate Identity for LLM Counting GFs

Chapter 9 established the identity $h = \log(1/R)$ connecting the entropy rate of a stochastic source to the radius of convergence of its *counting* generating function $C(z) = \sum_n |\mathcal{L}_n|\, z^n$, where $\mathcal{L}_n$ is the set of strings of length n in the source's support. This identity holds for regular languages (rational C) and for unambiguous context-free languages (algebraic C).

What is known. The identity is classical for Markov sources and PCFGs. Shannon's original work, Kontoyiannis 1998, and Takahashi and Tanaka-Ishii 2018 [57] (who estimate the entropy rate of natural language at approximately 1.12 bits per character) all operate in this regime. Scheibner, Smith and Bialek 2025 [48] (arXiv:2512.24969) revisit the question with modern LLM-based estimators.

Why it stops short. The counting GF of an LLM's support is not the same object as the length PGF of Gap 1a. The length PGF weights strings by their probability; the counting GF counts strings in the support without weighting. For PCFGs these are related by the Chomsky–Schützenberger theorem; for LLMs the relationship is unclear. No one has carefully stated, let alone proved, $h = \log(1/R)$ for a counting GF associated with a Transformer's support language.

What a solution looks like. A rigorous statement and proof of $h = \log(1/R_C)$ for an appropriate counting GF $C(z)$ of a class of autoregressive models, with explicit hypotheses on the model class.

18.3 Gap 2: Rigorous Boltzmann–Temperature Correspondence

Chapter 13 introduced the substitution $x = e^{-\beta}$ that turns a combinatorial generating function $A(x)$ into a partition function $Z(\beta) = A(e^{-\beta})$. Chapter 14 defined the LLM Gibbs measure $\mu_\beta(w) \propto \mu(w)^\beta$. The bridge between them is the *global* partition function

$$Z(\beta) = \sum_{w \in \Sigma^*} \mu(w)^\beta.$$

What is known. Blondel *et al.* 2025 [7] (arXiv:2512.15605) establish a bijection between autoregressive models and energy-based models via the soft Bellman equation. Kempton and Burrell 2025 [27] quantify the gap between local temperature sampling and the global Gibbs measure. Neither constructs $Z(\beta)$ as an analytic function of β or identifies its singularities.

What a solution looks like. For a concrete non-trivial model class (e.g. a softmax Transformer trained on a PCFG corpus), construct $Z(\beta)$ explicitly, determine its domain of analyticity in β, and locate any phase-transition points β_c where Z or its derivatives become singular.

18.4 Gap 3: ACSV for Tokenized Models

When the alphabet $|\Sigma| > 2$ and we want joint statistics — for example, the joint distribution of the counts of two specific tokens in a sample of length n — the natural tool is multivariate generating functions and the Pemantle–Wilson framework of analytic combinatorics in several variables (ACSV) [42].

What is known. ACSV provides a complete asymptotic theory for coefficients of rational and algebraic multivariate generating functions, governed by the geometry of the singular variety. The framework is mature for combinatorial applications (lattice paths, queuing models).

Why it stops short. No one has applied ACSV to LLM output distributions. The multivariate GF of token-count statistics under an autoregressive model has not been constructed, and the singular variety has not been analyzed.

What a solution looks like. A multivariate GF $F(z_1, \ldots, z_k)$ tracking token counts under a WFA-approximated LLM, with ACSV applied to extract joint asymptotics.

18.5 Gap 4: A Chomsky–Schützenberger Test for LLM Support

Chapter 5 established that unambiguous context-free languages have algebraic generating functions whose coefficients obey $[z^n]C(z) \sim c \cdot n^{-3/2}\rho^{-n}$. This is a *testable prediction*: sample

strings from an LLM, estimate the length distribution, and fit the asymptotic form.

What is known. Banderier and Drmota 2015 [3] classify the possible asymptotic forms of coefficients of algebraic generating functions. The $n^{-3/2}$ exponent is universal for strongly connected unambiguous CFGs.

Why it stops short. No one has run this test on LLM outputs. It is not known whether the length distribution of samples from GPT-4, Llama, or any open-weight model is consistent with the algebraic asymptotic class.

What a solution looks like. An empirical study fitting $\Pr[|W| = n]$ from LLM samples to the form $c \cdot n^{\alpha}\rho^{-n}$ and testing whether $\alpha = -3/2$ is consistent with the data. A positive result would be evidence that LLM output languages are "approximately context-free" in a precise analytic sense.

18.6 Gap 5: Gaussian Limit Laws via Hwang's Quasi-Power Theorem

Hwang 1998 [24] proved a meta-theorem: if a bivariate generating function $F(z, u)$ tracking a statistic has the "quasi-power" form

$$[z^n]F(z, u) = A(u) \cdot B(u)^n \cdot \big(1 + O(1/n)\big)$$

uniformly near $u = 1$, then the statistic is asymptotically Gaussian with mean $\mu_n \sim n \cdot B'(1)/B(1)$ and variance $\sigma_n^2 \sim n \cdot V$ for an explicit V.

What is known. The quasi-power theorem is a workhorse of analytic combinatorics, applied to hundreds of combinatorial statistics (inversions, descents, pattern occurrences). Chapter 17's motif-count statistics are a direct instance.

Why it stops short. Verifying the quasi-power hypothesis for a statistic measured on *LLM output* — e.g. the count of a motif in a sample of length n — requires knowing the bivariate GF $F(z, u)$ of the LLM, which is Gap 1a generalized. For WFA-approximated LLMs, $F(z, u)$ is rational in (z, u) and the hypothesis is often checkable.

What a solution looks like. For a WFA-approximated LLM, verify the quasi-power form for a concrete statistic (e.g. occurrences of a specific bigram), state the Gaussian limit, and compare with empirical histograms.

18.7 Gap 6: Thermodynamic Formalism and Generating Functions

The thermodynamic formalism of Ruelle [47], Bowen [9], and Sinai studies shift-invariant measures on symbolic sequences via transfer operators and pressure functions. Chapter 13's Gibbs/GF correspondence is a special case. The full formalism provides a theory of equilibrium states, variational principles, and phase transitions that subsumes both the combinatorial Boltzmann picture and the information-theoretic entropy picture.

What is known. The thermodynamic formalism is fully developed for subshifts of finite type (which are the symbolic-dynamics analogues of regular languages / finite-state sources). The pressure function $P(\phi)$ for a Hölder-continuous potential ϕ on a mixing subshift is real-analytic, and equilibrium states are unique.

Why it stops short. LLMs are not subshifts of finite type. Their potential functions (the log-probabilities) are computed by deep networks, not by finite-state rules. Extending the formalism to "soft" potentials arising from neural networks requires either (a) approximation by finite-state potentials (back to Chapter 12) or (b) new theorems about transfer operators with neural-network potentials.

What a solution looks like. A rigorous pressure function for a class of autoregressive models, with analyticity properties and a variational characterization, expressed in terms of the model's generating function.

18.8 Gap 7: Classification of LLM Singularity Types

Analytic combinatorics has a well-developed zoology of singular behaviors: polar (rational GFs, geometric coefficients), algebraic-logarithmic (algebraic GFs, $n^{-3/2}$ and its cousins $n^{-(2k+1)/2}$), and essential (entire functions with super-exponential growth, requiring saddle-point methods). Which of these appear in LLM counting or length generating functions? Is there a new type?

What is known. WFA approximations give rational GFs (polar singularities). PCFG approximations give algebraic GFs (branch-point singularities). Real LLMs presumably have singularity structure that is neither purely rational nor purely algebraic.

What a solution looks like. A classification theorem: "for autoregressive models in class $\mathcal{C}$, the length PGF has singularities of type X, and the coefficient asymptotics are of form Y." Even a conjectural classification backed by numerical evidence would be a contribution.

18.9 Gap 8: Phase-Transition Classification via Composition Schemes

Banderier, Kuba and Wallner 2021 [4] classified the phase transitions arising from composition-scheme generating functions $A(B(z))$ where the inner and outer singularities interact. Chapter 15 argued that empirical LLM phase transitions (Nakaishi *et al.* [39], Arnold *et al.* [1]) may be instances of this classification.

What is known. The composition-scheme classification produces Airy-type universal behavior at critical points. Mikhaylovskiy 2025 [37] argues that empirical LLM phase transitions are consistent with this universality class.

Why it stops short. The match is qualitative. No one has derived a composition-scheme structure from an LLM's architecture or from a WFA approximation of one, and then matched the predicted critical exponents to the empirical ones.

What a solution looks like. For a concrete LLM surrogate (e.g. a Transformer trained on a PCFG), derive the composition-scheme structure of $Z(\beta)$, predict the critical exponents, and verify them empirically.

18.10 Gap 9: A WFA Pipeline with Analytic Error Bounds

Chapter 12 surveyed four approaches to WFA extraction. None comes with end-to-end *analytic* error bounds relating the approximation quality (in KL divergence, total variation, or Hankel-operator norm) to the rank of the extracted WFA and the spectral properties of the original model.

What is known. AAK theory (Section 12.4) gives optimal Hankel-operator-norm bounds for the one-letter case. Spectral learning gives Frobenius-norm bounds on the Hankel matrix. L^* has polynomial query complexity in the number of states of the target DFA. These bounds exist in isolation.

What a solution looks like. A combined pipeline: extract a WFA of rank k from an LLM, with a theorem stating that the KL divergence between the WFA's distribution and the LLM's distribution is at most $\varepsilon(k, \sigma_{k+1})$ where σ_{k+1} is the $(k+1)$st singular value of the empirical Hankel matrix. This is the most tractable of the ten gaps.

18.11 Gap 10: Phase Transitions as Confluent Singularities

A confluent singularity is a point in parameter space where two or more singular surfaces of a generating function merge. In the composition-scheme setting of Gap 8, these produce Airy-function asymptotics with universal critical exponents. The conjecture, motivated by Mikhaylovskiy 2025 [37] and the empirical evidence of Chapter 15, is that the observed phase transitions in LLM output are confluent singularities of $Z(\beta)$.

What is known. Confluent singularities are well understood for algebraic generating functions arising from context-free grammars (Flajolet and Sedgewick [18], Chapter VII). The Airy function $\mathrm{Ai}(\cdot)$ appears universally.

What a solution looks like. For a concrete model, prove that $Z(\beta)$ has a confluent singularity at β_c, identify the Airy scaling, and match to the empirical critical exponents of Nakaishi *et al.* and Arnold *et al.*

18.12 Summary of Gaps

Gap 1a Construct the length PGF $L(z)$ for an LLM and analyze its singularities.

Gap 1b Prove $h = \log(1/R)$ for the counting GF of an LLM's support.

Gap 2 Construct $Z(\beta) = \sum_w \mu(w)^\beta$ as an analytic function and locate phase transitions.

Gap 3 Apply ACSV to multivariate token-count GFs from LLMs.

Gap 4 Test whether LLM length distributions exhibit the $n^{-3/2}$ algebraic signature.

Gap 5 Verify the quasi-power hypothesis and derive Gaussian limit laws for LLM statistics.

Gap 6 Extend thermodynamic formalism to neural-network potentials.

Gap 7 Classify the singularity types of LLM generating functions.

Gap 8 Match empirical LLM phase transitions to composition-scheme universality classes.

Gap 9 Build a WFA extraction pipeline with end-to-end analytic error bounds.

Gap 10 Prove that LLM phase transitions are confluent singularities with Airy scaling.

18.13 Research Suggestions

Gap 9 is the most tractable starting point: it synthesizes existing tools (AAK bounds, spectral learning, L^*) and does not require new theorems about LLMs themselves. Gap 5 is an attractive second target because Hwang's quasi-power theorem is a ready-made meta-theorem with checkable hypotheses. Gap 10 is the prize: a proof would unify the empirical phase-transition observations with the analytic-combinatorics classification programme. Chapter 19 develops a concrete research roadmap from these priorities.

18.14 Notes and Further Reading

The ten gaps are distilled from the broader research programme described in the source literature. Flajolet and Sedgewick [18] remains the central reference for the analytic side; their Chapters VI–IX cover the singularity analysis, algebraic functions, and limit-law machinery that most of these gaps require. Pemantle and Wilson [42] is the reference for Gap 3. The empirical phase-transition literature (Nakaishi *et al.* [39], Arnold *et al.* [1], Mikhaylovskiy [37]) motivates Gaps 8 and 10 directly.

Recommended Reading

- [18] — Chapters VI–IX supply the singularity analysis, algebraic-function theory, transfer-matrix machinery, and limit-law framework that underpin Gaps 1a, 1b, 4, 5, 7, and 10. Required reading before attempting any of the analytic gaps.
- [42] — The reference for analytic combinatorics in several variables (ACSV). Develops the critical-point theory and stratified-Morse-theory tools needed for Gap 3 (multivariate token-count generating functions).
- [24] — Proves the quasi-power meta-theorem central to Gap 5. Reading this paper makes clear exactly what hypotheses need to be verified for a WFA-approximated LLM, and how Gaussian limit laws follow once they are.
- [47] — The foundational reference for the thermodynamic formalism of equilibrium states, pressure functions, and transfer operators on subshifts of finite type. Essential background for Gap 6 (extending the formalism to neural-network potentials).
- [9] — Develops the Ruelle–Bowen theory of Gibbs states and the variational principle for topological pressure on Axiom A systems. Read alongside Ruelle for Gap 6; together they give the ergodic-theory substrate that Gap 6 seeks to extend.
- [38] — Empirically identifies temperature-driven phase transitions in transformer outputs and connects them to Zipf-exponent shifts. The primary empirical motivation for Gaps 8 and 10, and a concrete target for what a positive resolution would need to explain.

Chapter 19

A Research Roadmap

19.1 Where We Stand

Part I of this book developed the toolkit of analytic combinatorics: generating functions, singularity analysis, transfer matrices, and Boltzmann samplers [18]. Part II established the linguistic foundations: weighted finite automata, the Hankel characterisation, PCFGs, entropy rates, and the Chomsky–Schützenberger theorem. Part III introduced the Transformer architecture and surveyed the theoretical results bearing on its expressive power and limitations [59, 35, 46]. Part IV built the bridge: Boltzmann sampling interpreted as a Gibbs measure, the partition function $Z(\beta)$ as a generating function with temperature as a parameter, and the emergence of phase transitions at confluent singularities [4]. Part V surveyed Zipf distributions, motif statistics, and their analytic explanations [56], before Chapter 18 catalogued ten open problems — Gaps 1a, 1b, 2, 3, 4, 5, 6, 7, 8, 9, 10 — at the frontier. This final chapter answers a practical question: given that background, what should a researcher actually work on first?

19.2 Ranking the Gaps by Tractability

Not all ten gaps are equally open. Some are computation problems waiting for careful accounting; others require new mathematics that is not yet in sight. The following ranking is a judgement call, not a theorem.

Most tractable: Gap 9 (WFA-approximation pipeline with analytic error bounds). The four approximation strategies surveyed in Chapter 12 — spectral learning from moment matrices, AAK-based Hankel truncation, L^*-style automaton learning, and tensor-network contraction — are each well understood in isolation. What is missing is a unified error budget: given that each step introduces its own approximation error, what does the compound bound look like? The ingredients are already on the shelf. The AAK singular-value bounds were made explicit for weighted automata by Lacroce, Panangaden, and Rabusseau [30]; the L^* convergence guarantees are classical. The missing piece is to compose these bounds and state a single theorem of the form: "for a rank-r WFA approximation to an LLM distribution over sequences of length at most n, the total-variation error is at most $f(r, n)$." This is, in essence, a careful accounting exercise. It does not require proving anything new about LLMs themselves.

Second: Gap 5 (Gaussian limit laws via the Hwang quasi-power theorem). Hwang's 1998 meta-theorem [24] gives Gaussian limits essentially for free once the probability generating function satisfies a quasi-power hypothesis. Verifying that hypothesis for a concrete LLM surrogate — say, a small PCFG with a softmax output head — is a calculation, not an open problem in the usual sense. The difficulty is making the surrogate tractable enough that the PGF can be computed or tightly bounded. A PCFG with k nonterminals has an algebraic GF of degree at most k; the softmax perturbs it in a controlled way. Working through this

for the simplest non-trivial case would constitute a publishable result and, more importantly, would establish whether the quasi-power hypothesis is robust to the softmax perturbation.

Third: Gap 1b (the entropy-rate identity $h = \log(1/R)$ for LLM counting GFs). Once Gap 9 is resolved and a WFA approximation exists, Gap 1b is nearly a corollary. The transition matrix M of a WFA defines a linear recurrence for the sequence of partition functions Z_n (the sums of weights over all strings of length n). The radius of convergence of the associated generating function $Z(x) = \sum_n Z_n x^n$ is then $R = 1/\rho(M)$, where $\rho(M)$ is the spectral radius of M, by Perron–Frobenius. On the other hand, the entropy rate of the WFA-approximated distribution is $h = \log \rho(M)$ by standard spectral theory of Markov-type operators. So $h = \log(1/R)$ holds immediately for the WFA approximation. The substantive question is how faithfully this approximation captures the entropy rate of the original LLM distribution, which circles back to the quality of the Gap 9 bound. One must also be precise about the distinction between the length probability generating function (Gap 1a, concerning $\sum_n p_n x^n$ where $p_n = \Pr[\text{length} = n]$) and the counting GF (Gap 1b, concerning $\sum_n Z_n x^n$ where Z_n counts weighted paths). These are not the same object and the singularities can differ; the chapter's exposition must keep them separate.

Big prize: Gap 10 (phase transitions as confluent singularities). The empirical observations of Nakaishi et al. [39], Arnold et al. [1], and Mikhaylovskiy [37] all point to sharp transitions in LLM-generated text statistics at critical temperatures. The Banderier–Kuba–Wallner composition-scheme framework [4] gives a rigorous mechanism for such transitions: they arise when two dominant singularities of the partition function $Z(\beta, x)$ coalesce as β varies. A positive result connecting these two bodies of work — proving that the observed transitions correspond to singularity confluence in $Z(\beta, x)$ for an explicit surrogate model — would be a significant contribution. This is the right long-term target but not the right starting point.

Long shot in the sense of ease of entry, not in probability of success: Gap 4 (Chomsky–Schützenberger test). This is an empirical project. The idea is simple: sample strings from a medium-sized open-weight LLM at fixed temperature, fit a counting GF to the empirical length distribution, and ask whether the dominant singularity is algebraic of the type $\sim C \cdot n^{-3/2} r^{-n}$ (the signature of a non-degenerate context-free language). Anyone with a GPU cluster and access to an open-weight model can start this tomorrow. The theoretical interpretation of a positive or negative result is less straightforward, but the empirical finding would itself be informative. This gap is listed last in tractability not because it is hard to attempt but because the result is harder to interpret without the theoretical scaffolding of Phases 1–2.

19.3 A Three-Phase Programme

The following programme is designed so that each phase produces publishable intermediate results while also building the substrate for the next phase.

Phase 1: Implement and benchmark a WFA-extraction pipeline. Choose a small LLM surrogate: a 2-layer Transformer trained on a corpus generated by a known PCFG. Using the PCFG, the ground-truth distribution is computable, so errors can be measured exactly. Extract the Hankel matrix H of the surrogate's string probabilities up to length L, compute its singular-value decomposition, and truncate to rank r. Record the singular-value decay as a function of r (this is a diagnostic of how well a WFA can approximate the model). Construct the rank-r WFA by the spectral-learning procedure of Balle–Mohri or equivalently via Lacroce–Panangaden–Rabusseau [30]. Measure the empirical KL divergence between the WFA distribution and the surrogate distribution as a function of r and L. Compare to the theoretical AAK bound.

Deliverables: a codebase, a table of singular-value plots, and a graph of empirical KL divergence versus rank. The gap between the empirical error and the AAK bound, if large, is itself

an interesting finding.

Phase 2: Verify the entropy-rate identity. Using the rank-r WFA from Phase 1, compute the transition matrix M and extract $\rho(M)$. Compute $R = 1/\rho(M)$ and compare to the empirically measured entropy rate h of the surrogate (estimated by the standard cross-entropy method on a held-out corpus). If the WFA approximation is tight, the identity $h \approx \log(1/R)$ should hold up to an error controlled by the Phase 1 KL bound. Do this for several values of r and record how quickly the discrepancy decays.

This closes Gap 1b for the surrogate model and provides a template for how the argument would go in the general case. The writeup should make explicit which step fails if the WFA approximation is not tight, pointing toward what Gap 9's missing theoretical bound would need to say.

Phase 3: Scan temperature and look for singularity confluence. Fix the surrogate from Phases 1–2. Define the temperature-parameterised partition function $Z(\beta) = \sum_{\mathbf{x}} p(\mathbf{x})^\beta$, where the sum runs over all strings $\mathbf{x}$ up to length L. For each β in a grid, compute the dominant singularity of the associated WFA (the reciprocal spectral radius $R(\beta)$). Plot $R(\beta)$ as a function of β and look for non-analyticity: a kink in $R(\beta)$ signals a singularity crossing, which in the Banderier–Kuba–Wallner framework [4] corresponds to a phase transition. If a transition is found, attempt to classify the type of confluence (smooth, cusp, or higher order) by examining the local expansion of $Z(\beta, x)$ near the critical point.

A positive finding here constitutes empirical evidence for Gap 10. A negative finding is also informative: it would suggest that LLM phase transitions, when they occur, are not of the composition-scheme type and would require a different theoretical explanation.

19.4 Parallel Tracks and What to Avoid First

Gap 4 (the Chomsky–Schützenberger numerical test) is an orthogonal project that does not depend on the WFA pipeline. It can be pursued in parallel by a researcher with access to a real open-weight LLM [7, 15] and a good sampling implementation. The main risk is that the length distribution of LLM output is heavily confounded by prompt length and stopping heuristics; controlling for these requires care.

Gap 5 (Hwang quasi-power limits) can be pursued as a theoretical side-project by a combinatorics-oriented researcher who does not have access to GPU resources. The calculation for the PCFG-plus-softmax surrogate can be carried out on paper or with a computer-algebra system [24].

What not to work on first. Three gaps should be deferred.

Gap 6 (thermodynamic formalism) is where the cleanest theory lives: the Ruelle–Bowen–Sinai transfer operator gives a complete account of the free energy and its singularities. But this background is a semester course on its own, with prerequisites in ergodic theory that most combinatorics or machine-learning students will not have. Return to it after Phases 1–3 have built intuition.

Gap 7 (classification of all singularity types arising in LLM partition functions) is probably the ultimate theoretical destination of this research programme, but it presupposes Gap 10 being resolved and a rich collection of examples to classify. It is not a starting point.

Gap 3 (ACSV for tokenized models) requires the Pemantle–Wilson multivariate machinery [42] in its full generality. This is technically demanding — the critical-point analysis for multivariate GFs involves stratified Morse theory — and the payoff depends on first establishing that LLM generating functions have the right structure to make the analysis applicable. Defer until the univariate picture (Phases 1–2) is settled.

19.5 For a Student Entering the Field

The following reading order is recommended for a graduate student who has not yet read this book and wants to enter this research area efficiently.

1. **Flajolet–Sedgewick [18], Chapters I–IV.** This covers symbolic combinatorics, ordinary and exponential GFs, and the singularity-analysis framework. These chapters correspond to Chapters 1–5 of this book. This material is non-negotiable; everything else builds on it.

2. **This book, Part II** (WFAs, PCFGs, entropy rates). The primary mathematical objects are the Hankel matrix and the Perron–Frobenius theorem applied to weighted transition matrices.

3. **Szpankowski [56].** For researchers whose interest is in word statistics, motif occurrences, and large-deviation results. Chapter 17 of this book is an appetiser; Szpankowski's monograph is the full treatment.

4. **Rizvi et al. [46] and Merrill–Sabharwal–Smith [35].** For the LLM side: what Transformers can and cannot express as formal languages, and what circuit-complexity arguments say about their computational depth. Read alongside Strobl et al. [53] for the broader survey of formal language results.

5. **Duchon–Flajolet–Louchard–Schaeffer [16].** The foundational reference for Boltzmann sampling. Read this before attempting Phase 3; the connection between the Boltzmann parameter and the thermodynamic β is made explicit there.

This reading list is not comprehensive, but it covers the minimum required to engage with the three-phase programme above without repeatedly needing to pause and build background.

19.6 Closing

The purpose of this book has been to place in one document the pieces of two literatures — analytic combinatorics and large language model theory — that have developed largely independently. The mathematical community has Flajolet–Sedgewick, Szpankowski, and Pemantle–Wilson; the machine-learning community has Transformer theory, scaling laws, and empirical phase-transition observations. These are not separate subjects. The formal-language perspective forces precision about what a language model is computing; the singularity-analysis perspective gives a vocabulary for what happens at critical temperatures. If the book has done its job, a reader should be able to move between a chapter of Flajolet–Sedgewick and a survey such as Strobl et al. [53] without feeling that they have changed disciplines. The ten gaps of Chapter 18 and the programme sketched here are an invitation to close that distance further. They are not a manifesto; they are a list of things that seem doable.

Further Reading

The reading list in Section 19.5 covers the minimum required for the three-phase programme. The items below are the core references specifically for the roadmap itself.

- [18] — The analytical backbone of Phases 1–3. Chapters VI–IX on singularity analysis, algebraic functions, and the saddle-point method are used directly; Chapter IX on words and tries is the template for the WFA transition-matrix computations in Phase 2.

- [56] — The bridge between analytic combinatorics and information theory. For researchers entering from the combinatorics side, this monograph makes entropy rates, trie depths, and suffix-tree statistics into singularity-analysis problems, exactly the translation needed for Phase 2.
- [42] — Required for Gap 3 and for Phase 3 once the analysis extends beyond univariate generating functions. The critical-point and stratified-Morse-theory toolkit for multivariate coefficient asymptotics.
- [24] — The quasi-power meta-theorem that converts a bivariate generating function in quasi-power form into a Gaussian limit law. A short paper with large returns: reading it clarifies exactly what Phase 1's WFA extraction must deliver for Gap 5 to follow automatically.
- [47] — Background for Gap 6 and for understanding why Phase 3's $Z(\beta)$ behaves like a thermodynamic pressure function. The Ruelle transfer operator is the infinite-alphabet generalisation of the WFA transition matrix.
- [9] — Companion to Ruelle for the variational principle and Gibbs-state characterisation. Together these two short books give the ergodic-theoretic language that the roadmap's Phase 3 is, in spirit, trying to instantiate for neural-network potentials.

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